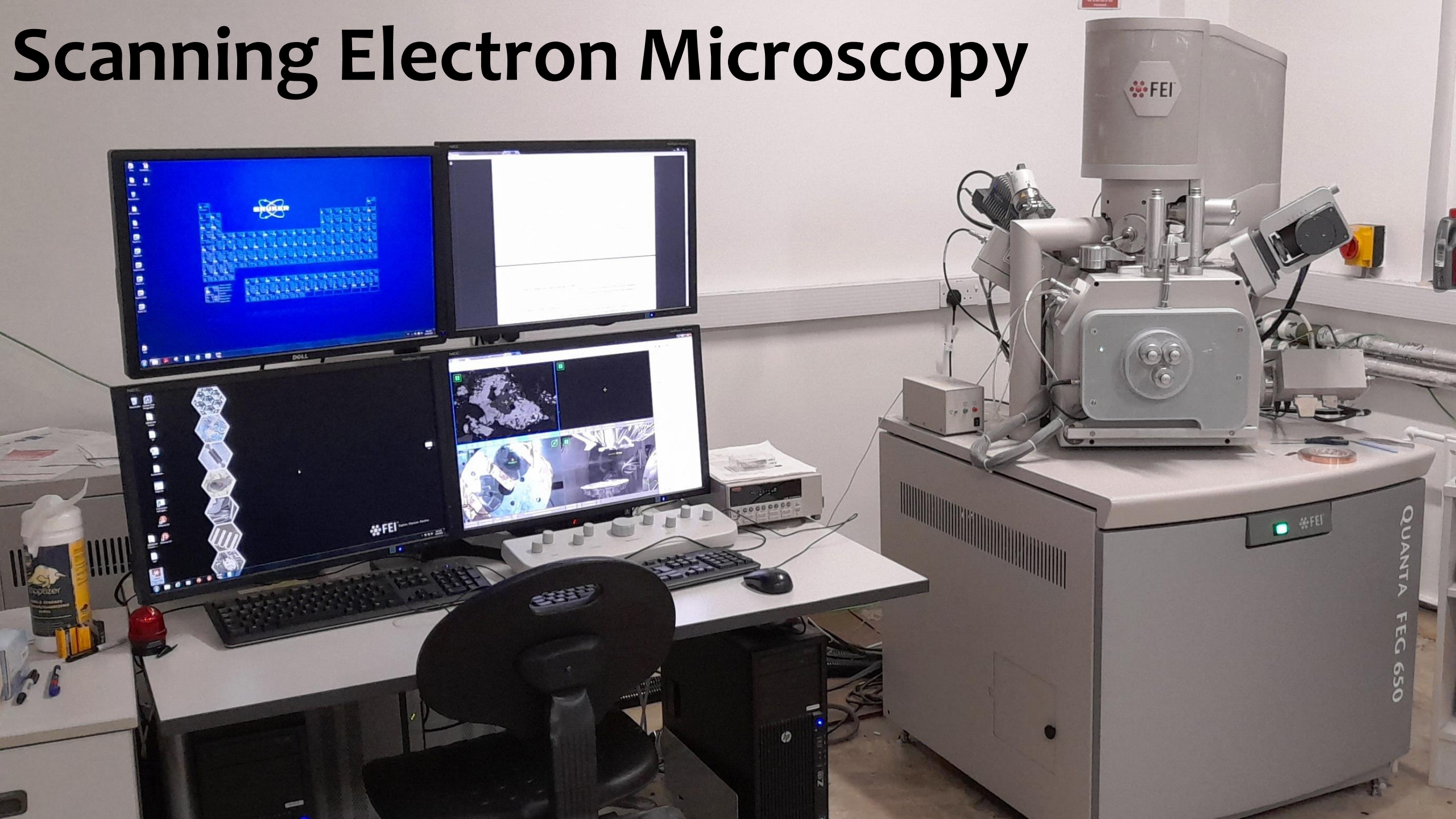
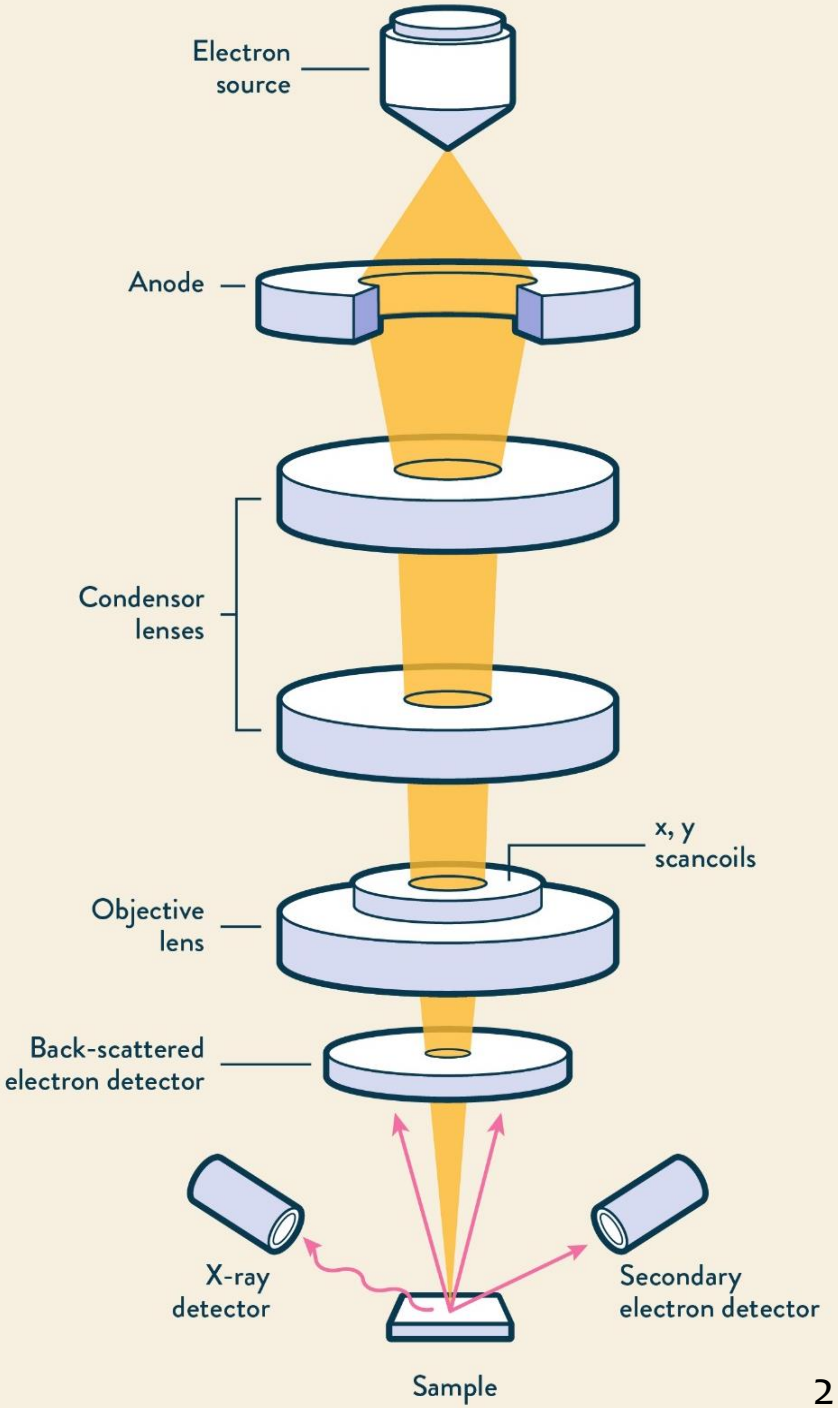
Scanning Electron Microscopy



System overview

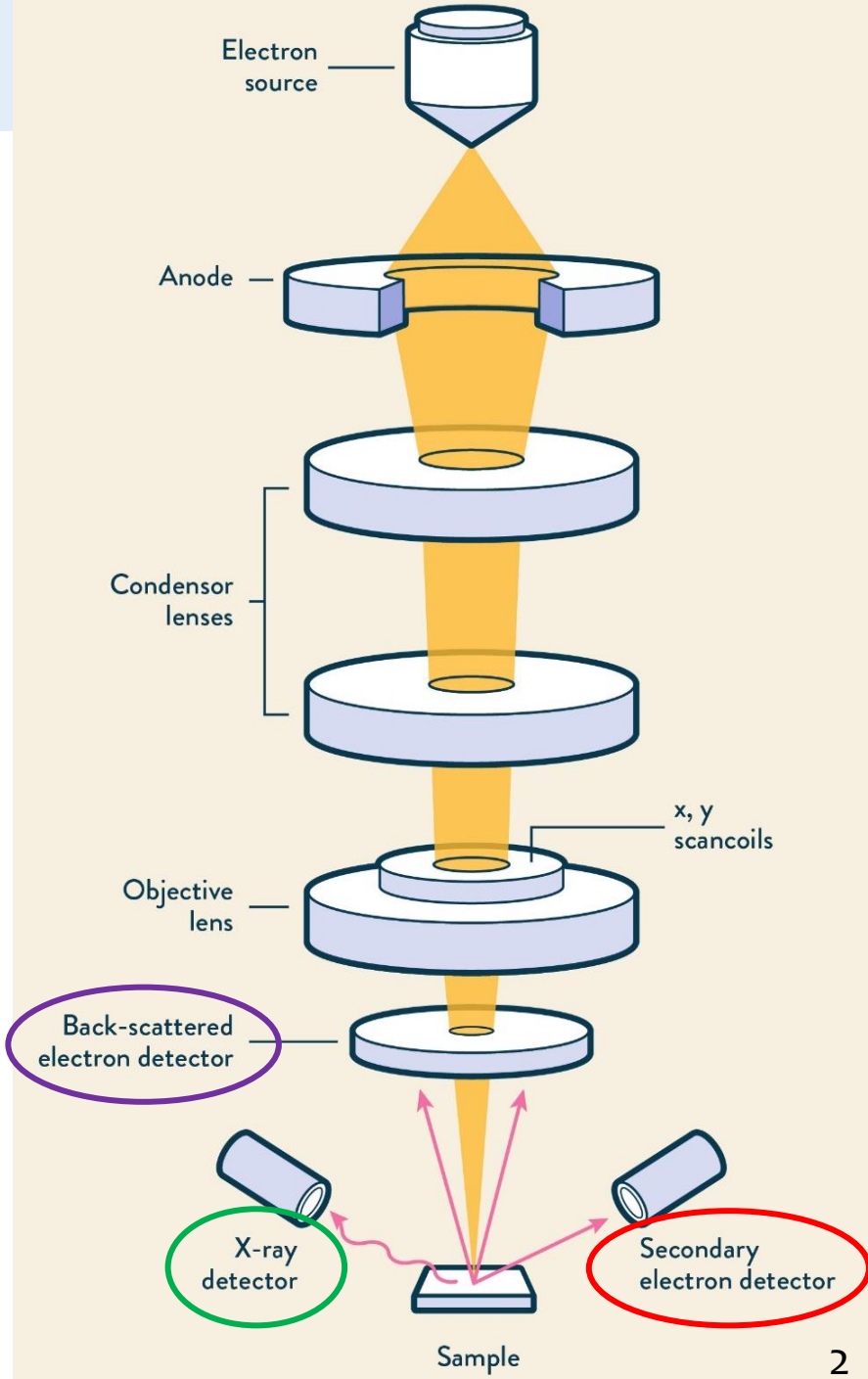
- **<u>Electron source/gun</u>** produces beam of e^-. Typically W filament, LaB_6, or field emission gun (FEG).



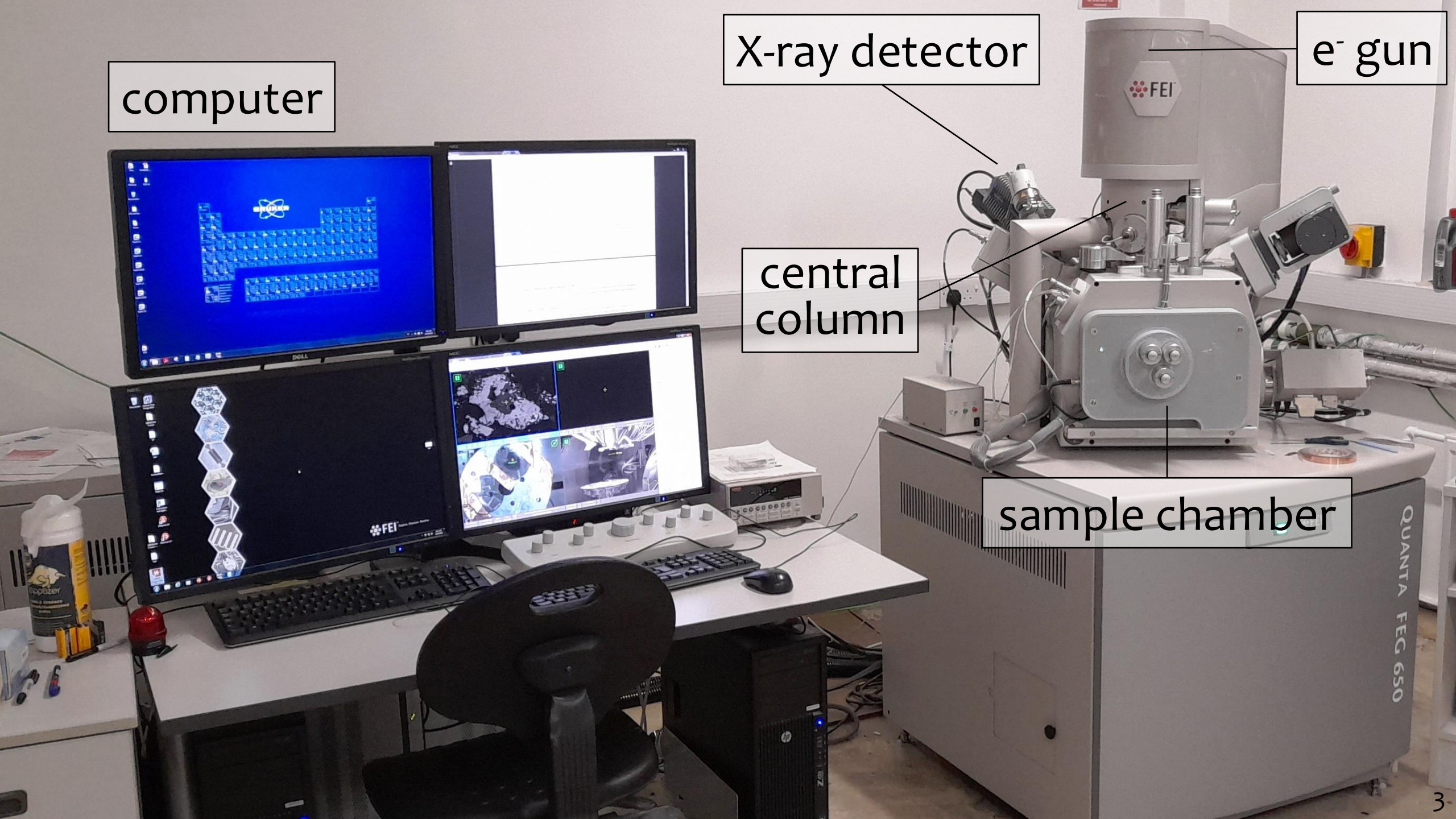
https://www.technologynetworks.com/analysis/articles/sem-vs-tem-331262

System overview

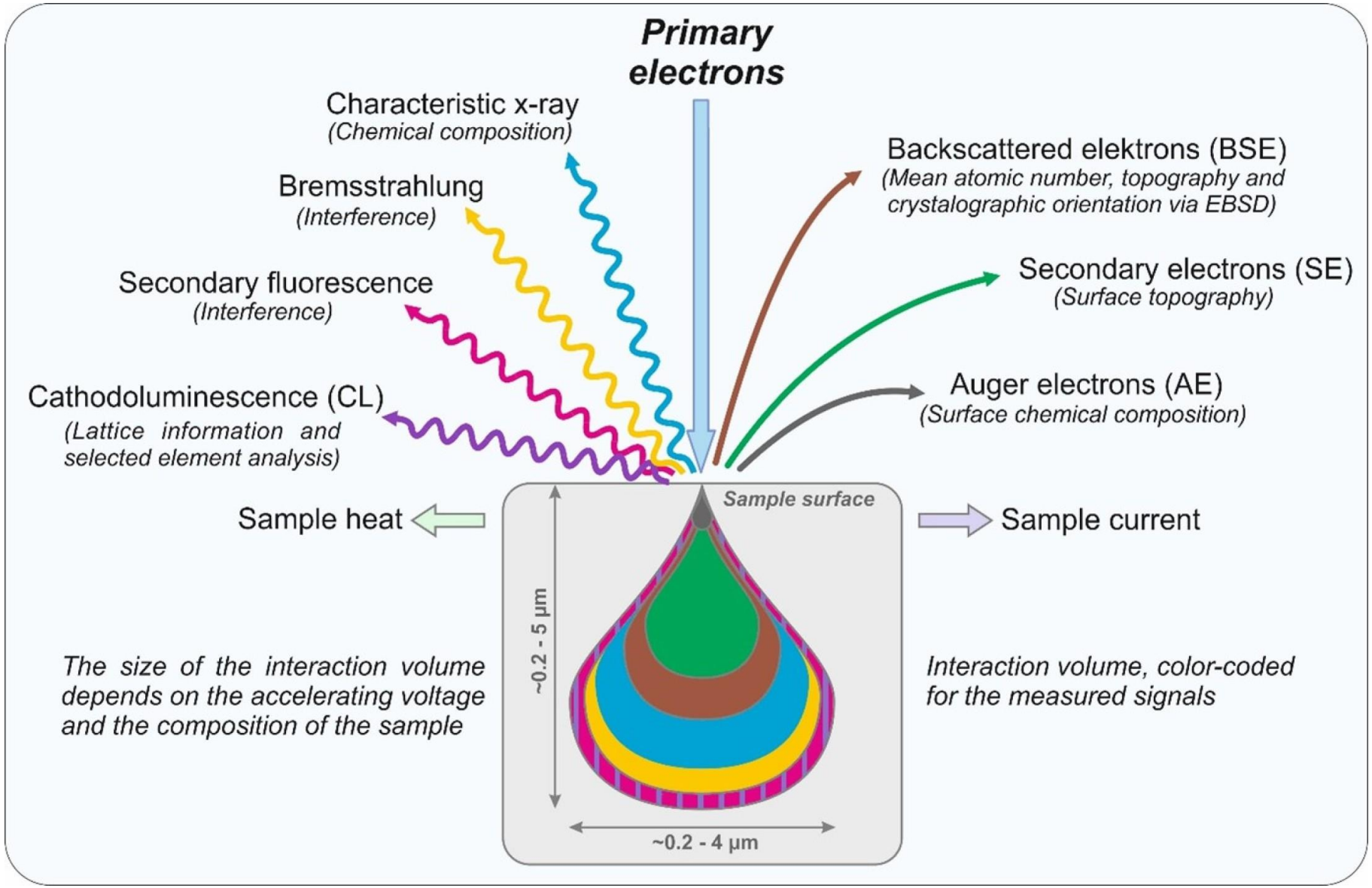
- **Electron source/gun** produces beam of e^-. Typically W filament, LaB_6, or field emission gun (FEG).
- e^- beam passes through the **central column**, which comprises **condenser lens(es)** [*determines beam current that impinges on sample*] and **objective lens** [*determines final spot size of the e^- beam*].
- **Scanning coil** used to deflect the beam in X-Y directions → raster scan over the sample surface.
- **Specimen chamber**, hosts sample as well as a range of detectors, typically for secondary electron (SE), back-scattered electron (BSE), and X-rays. Typically under vacuum conditions.



https://www.technologynetworks.com/analysis/articles/sem-vs-tem-331262

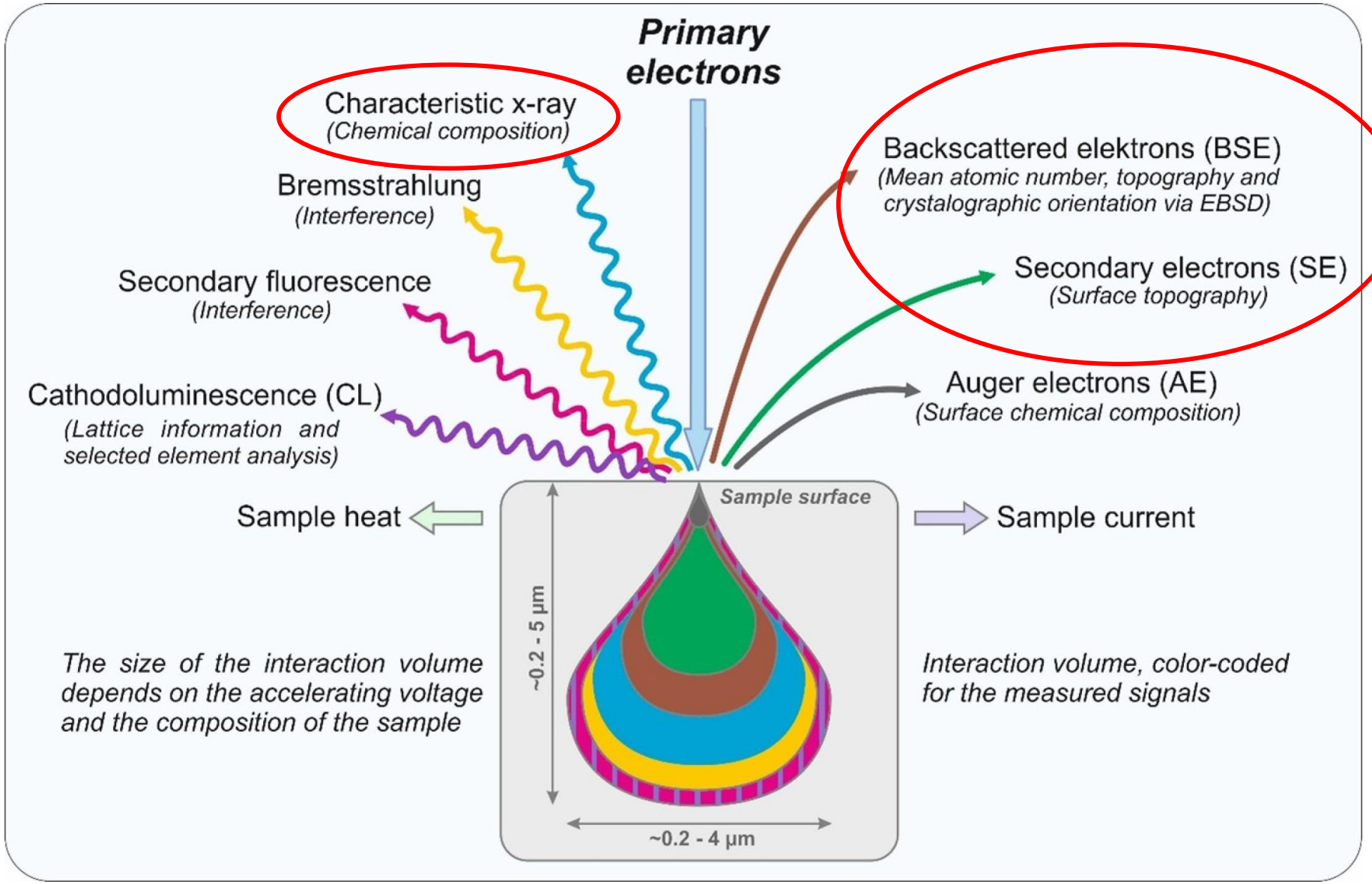

computer
X-ray detector
e- gun
central column
sample chamber
FEI
QUANTA FEG 650

Electron beam – sample interactions



https://www.nhm-wien.ac.at/en/research/central_research_laboratories/analytical_electron_microscopy/theory

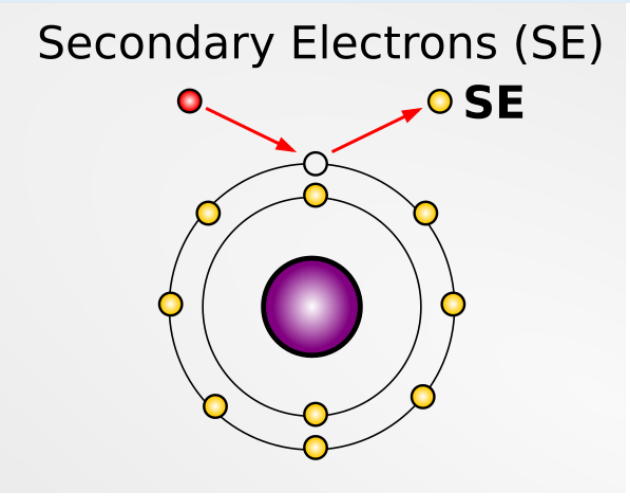
Electron beam – sample interactions



- Typically interested in SE (topography), BSE (composition), CL (composition), and characteristic X-ray (chemical composition).
- Fluorescence and Bremsstrahlung important for background characterisation.

https://www.nhm-wien.ac.at/en/research/central_research_laboratories/analytical_electron_microscopy/theory

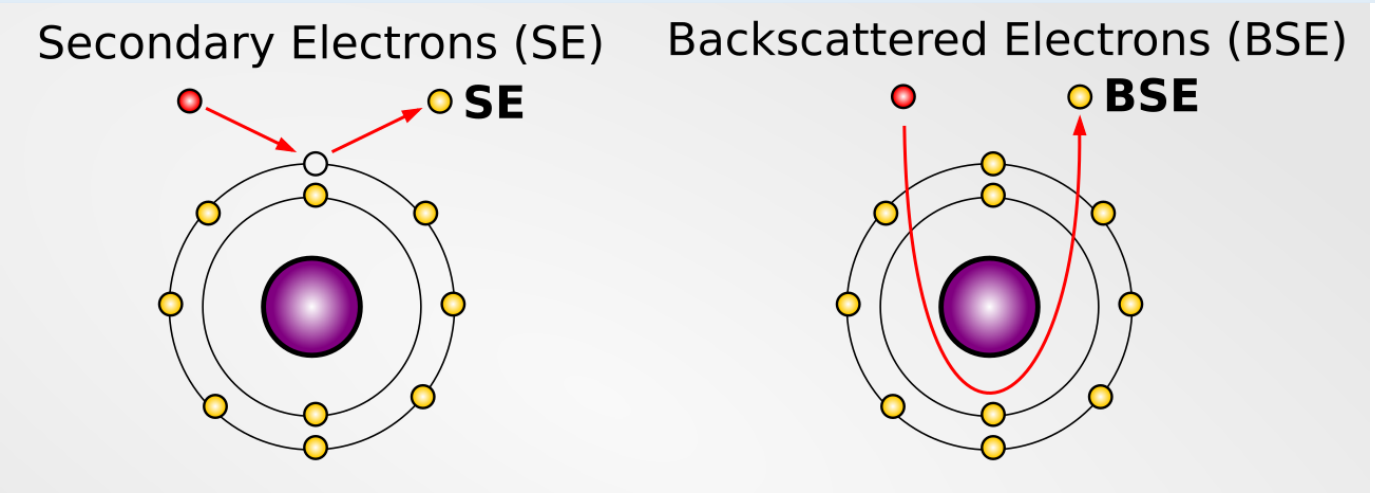
Electron beam – sample interactions



- **Secondary electrons**: inelastically scattered from the emission of valence e^-. *Low energy*.

https://msestudent.com/what-does-bse-stand-for-backscattered-electron/

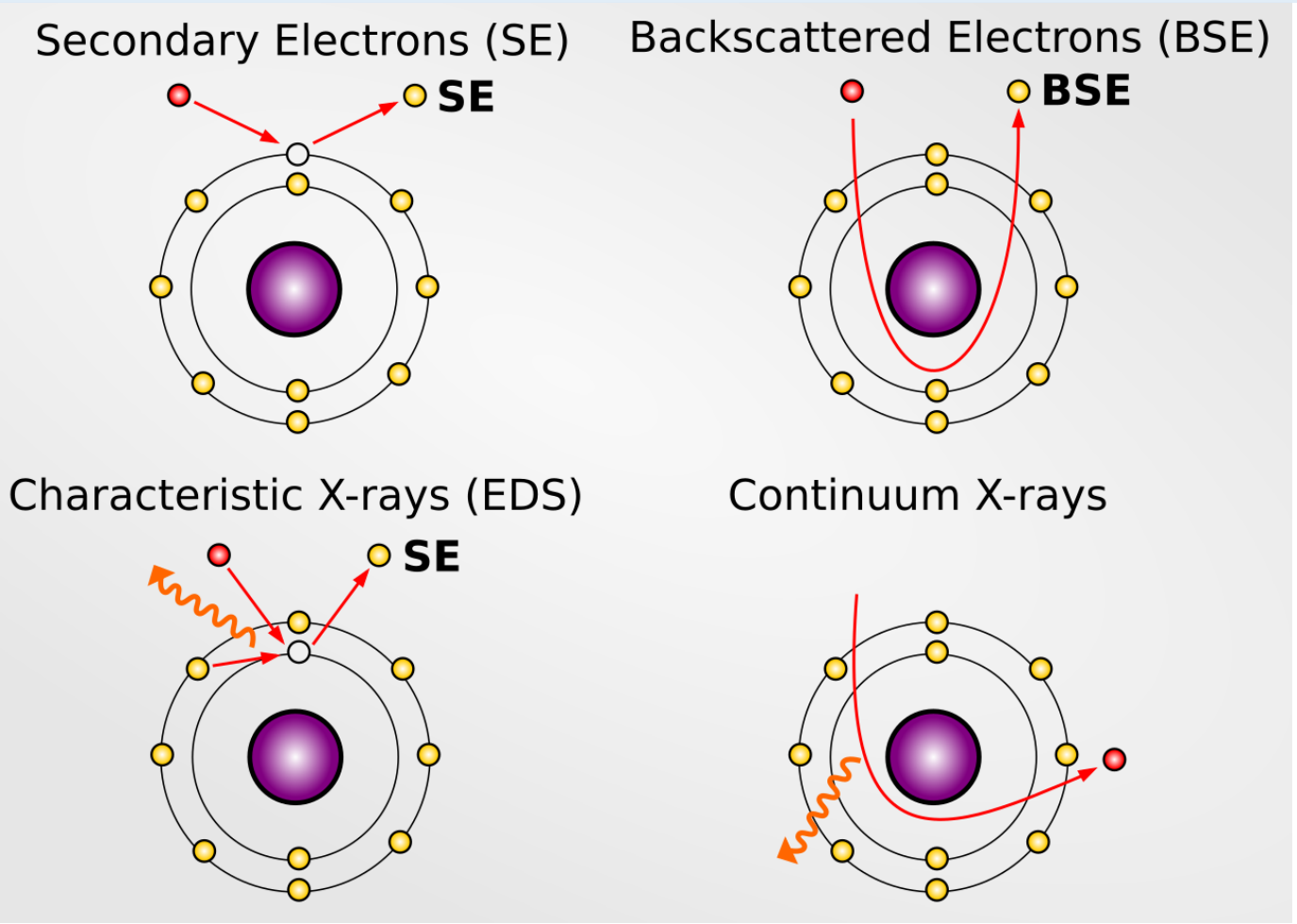
Electron beam – sample interactions



- **<u>Secondary electrons</u>**: inelastically scattered from the emission of valence e^-. *Low energy*.
- **<u>Backscattered electrons</u>**: elastically scattered; high Z elements backscatter e^- more strongly than low Z ones. *High energy*.

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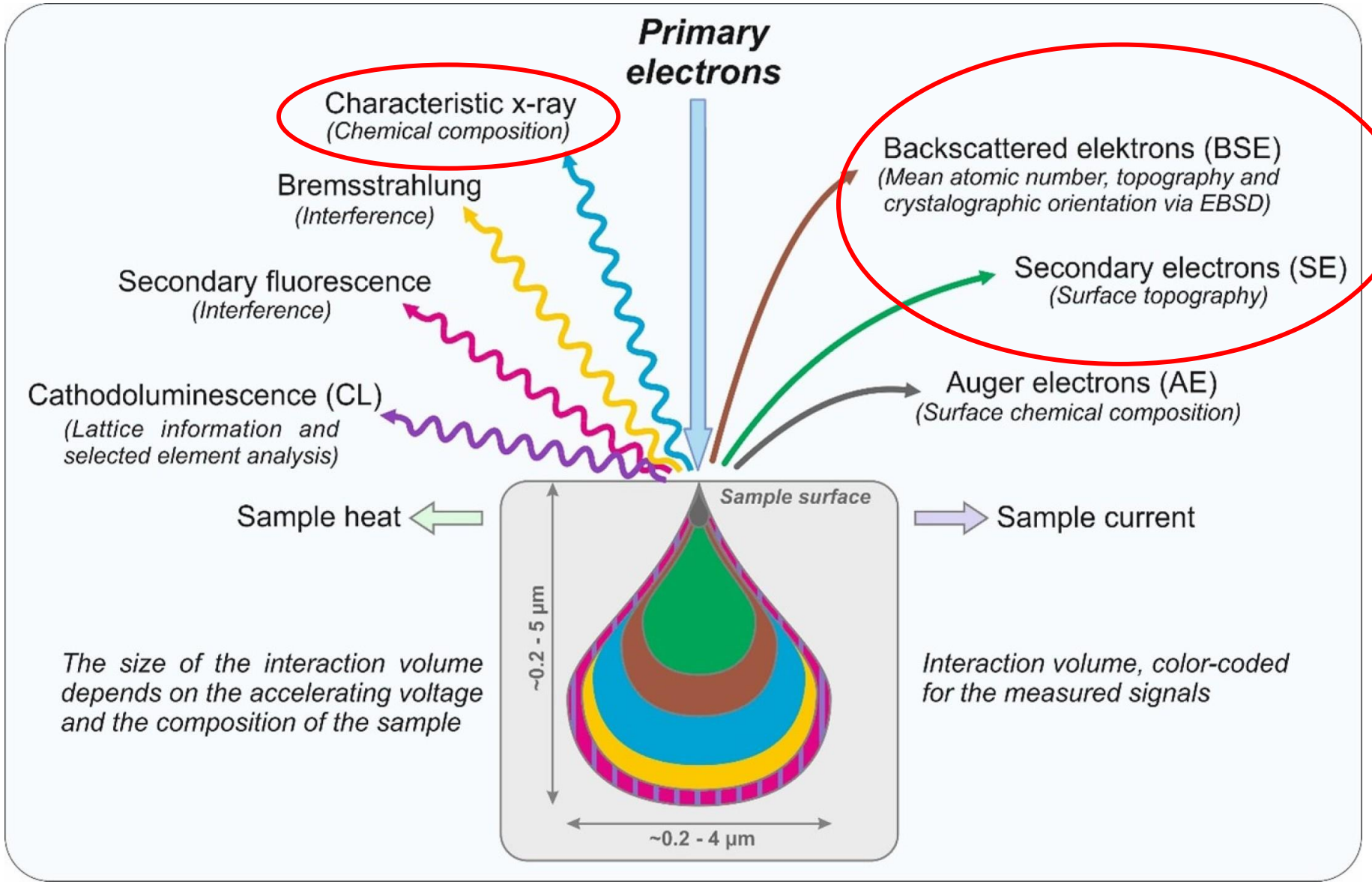
Electron beam – sample interactions



https://msestudent.com/what-does-bse-stand-for-backscattered-electron/

- **Secondary electrons**: inelastically scattered from the emission of valence e^-. *Low energy*.
- **Backscattered electrons**: elastically scattered; high Z elements backscatter e^- more strongly than low Z ones. *High energy*.
- **Characteristic X-rays**: ejection of inner shell e^- compensated for by e^- in higher orbital dropping down → excess energy given off as X-ray.
- **Continuous X-rays**: result from e^- decelerating when travelling close to target nucleus → looses energy as X-ray photon.

Electron beam – sample interactions



- Typically interested in SE (topography), BSE (composition), CL (composition), and characteristic X-ray (chemical composition).
- Fluorescence and Bremsstrahlung important for background characterisation.

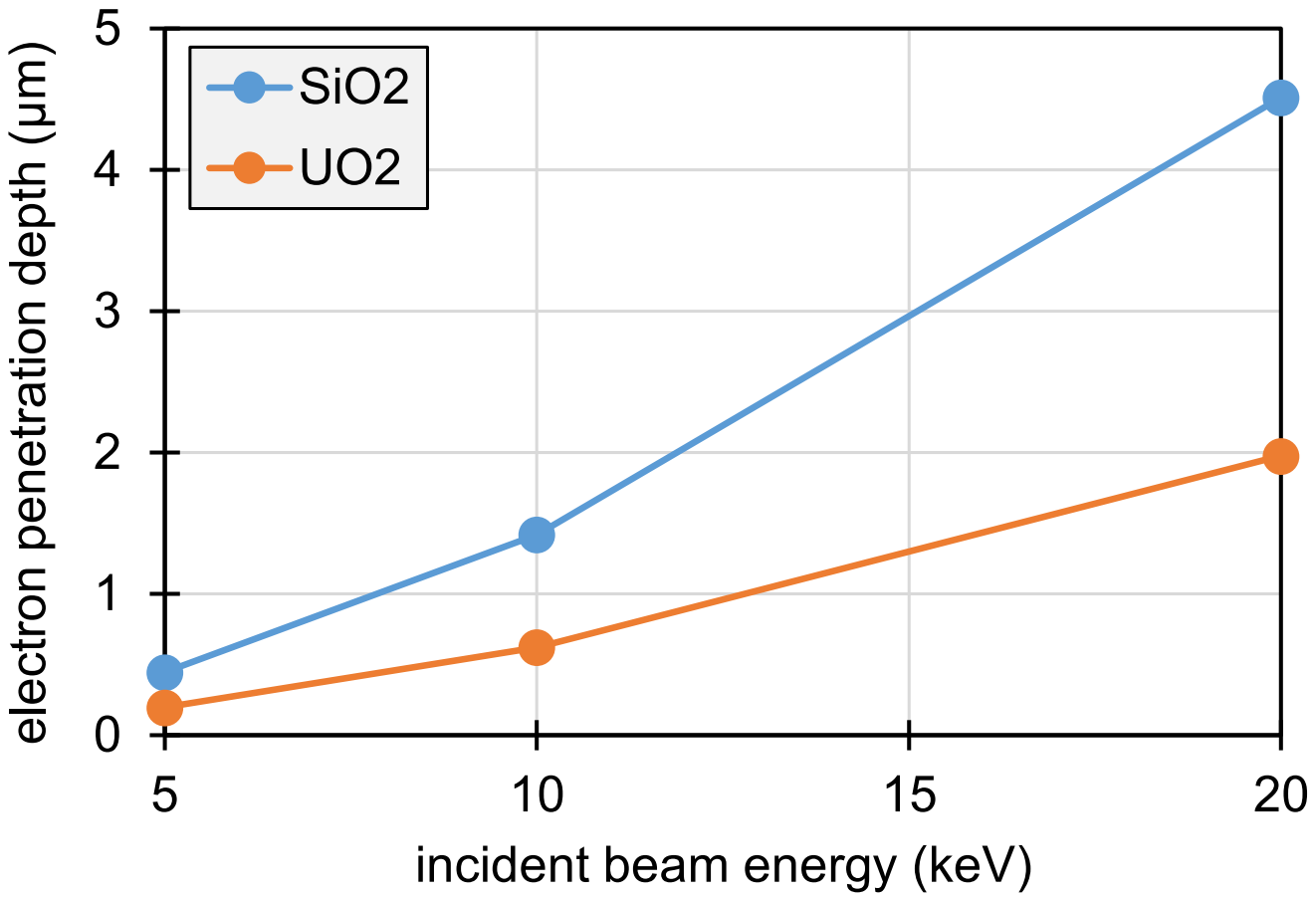
https://www.nhm-wien.ac.at/en/research/central_research_laboratories/analytical_electron_microscopy/theory

Electron interaction volume

$$R = \frac{0.0276 \times E^{1.67} \times A}{\rho \times Z^{0.89}}$$

E = electron beam energy (keV)
A = atomic weight (g/mol)
ρ = density of target (g/cm³)
Z = atomic number of target

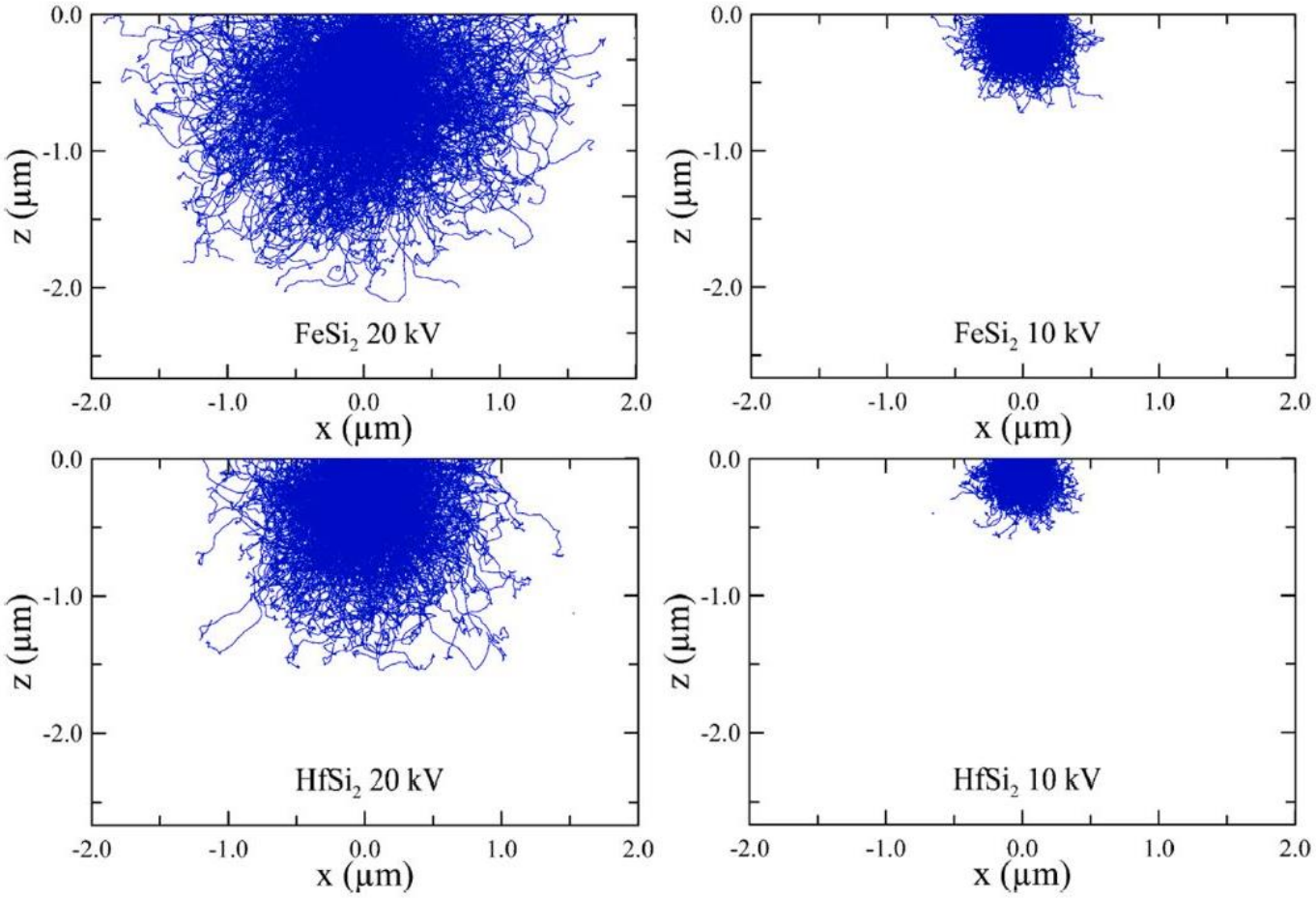
From Kanaya & Okayama (1972) Penetration and energy-loss theory of electrons in solid targets. J. Phys. D: Appl. Phys. 5, p. 43 (link).

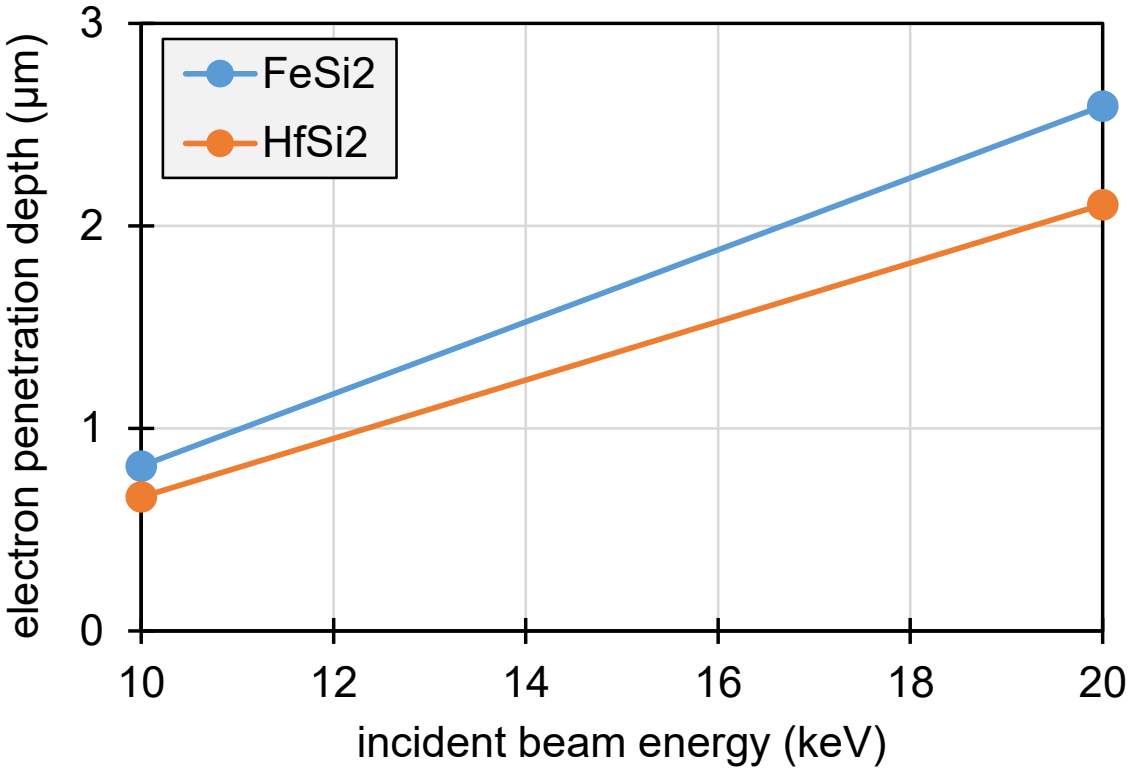


→ R increases with electron beam energy E
→ R decreases with target atomic number Z

Electron interaction volume

$$R = \frac{0.0276 \times E^{1.67} \times A}{\rho \times Z^{0.89}}$$

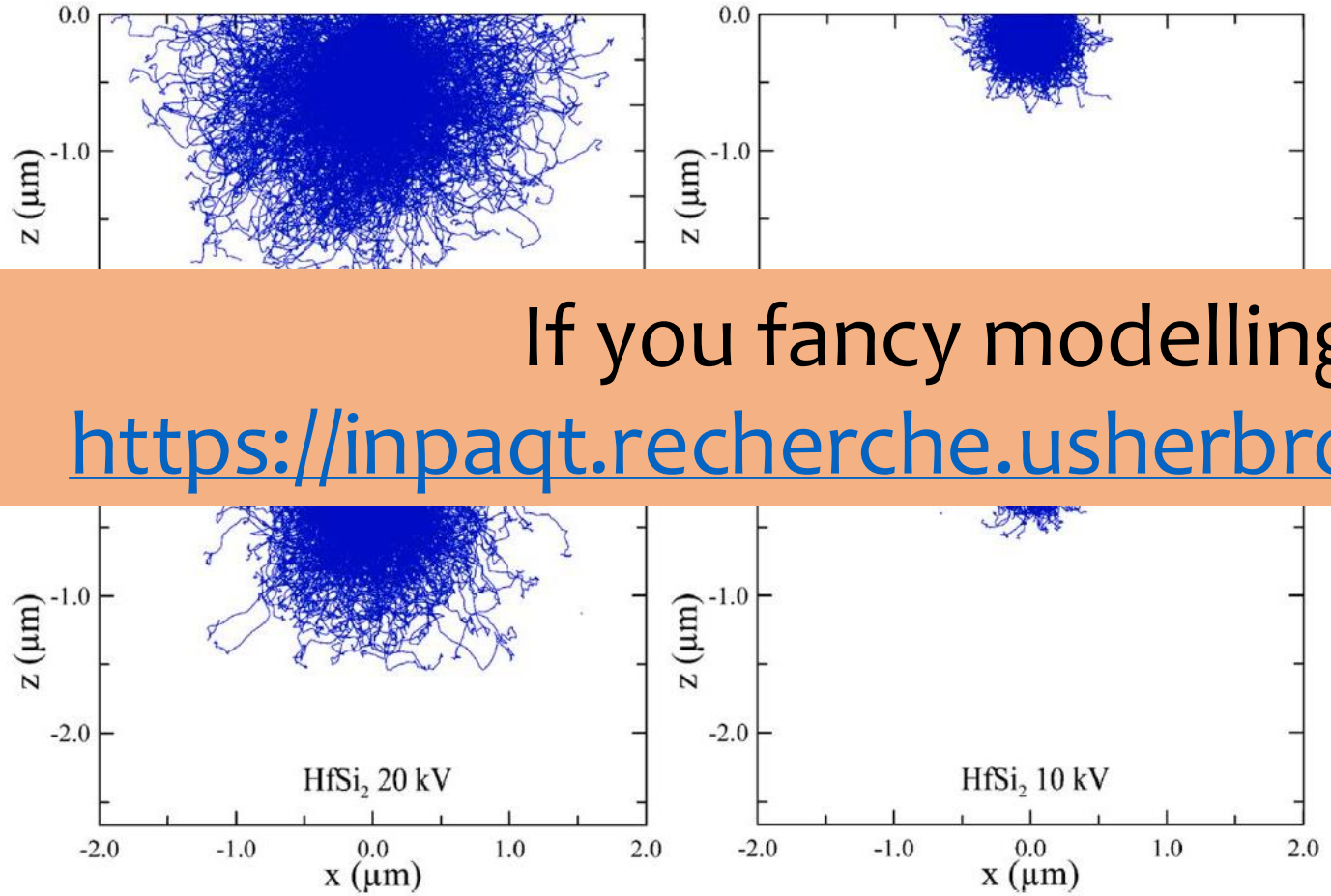


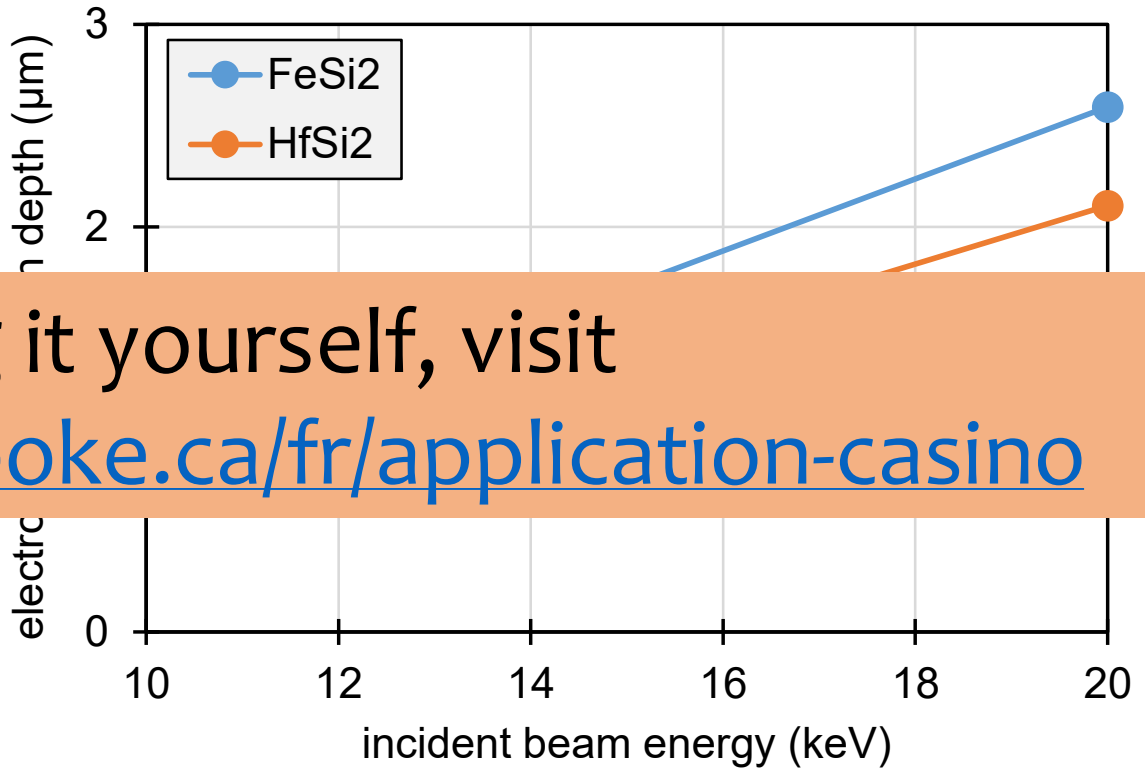


from Llovet *et al.* (2021) – [link]

Electron interaction volume

$$R = \frac{0.0276 \times E^{1.67} \times A}{\rho \times Z^{0.89}}$$





If you fancy modelling it yourself, visit
https://inpaqt.recherche.usherbrooke.ca/fr/application-casino

from Llovet *et al.* (2021) – link

SE and BSE imaging

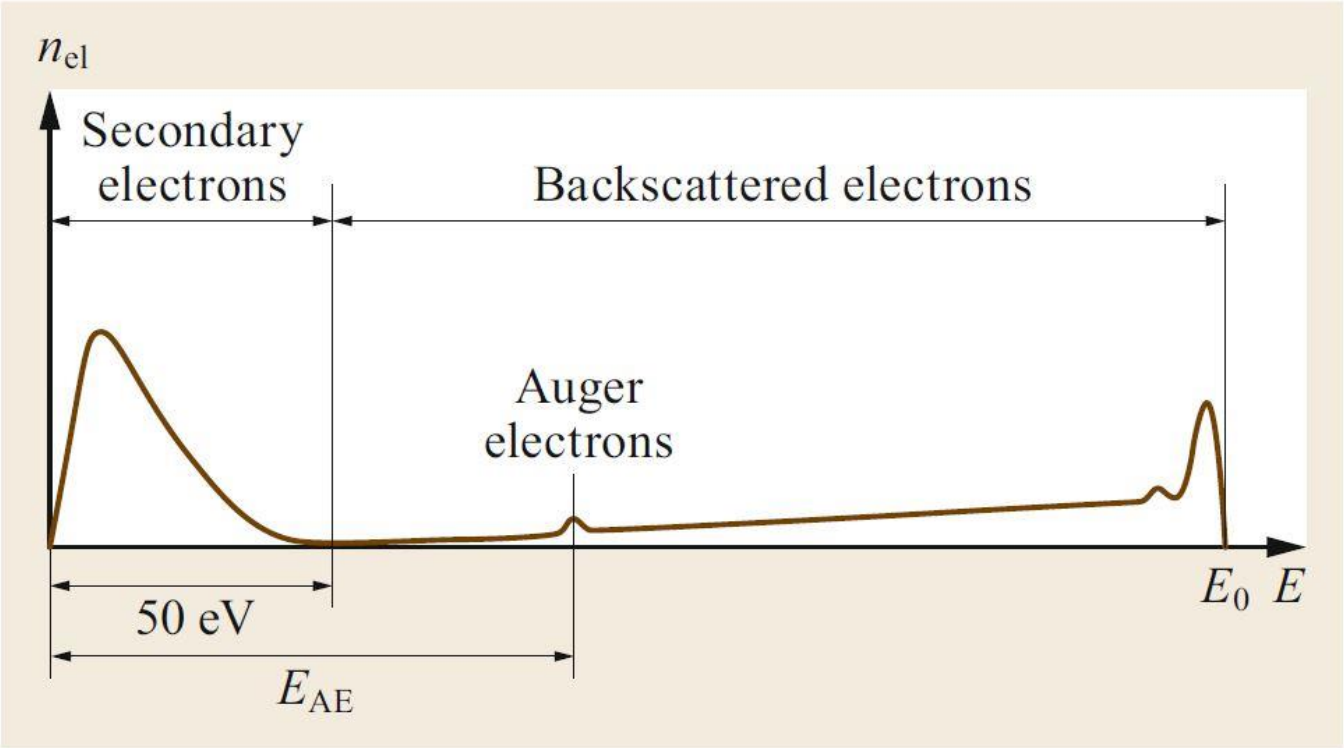


Fig. 5.12 Schematic energy distribution of electrons emitted n_{el} from a surface as a result of its bombardment with fast electrons with energy E_0

from Herdman *et al.* (2019) – link

SE and BSE imaging

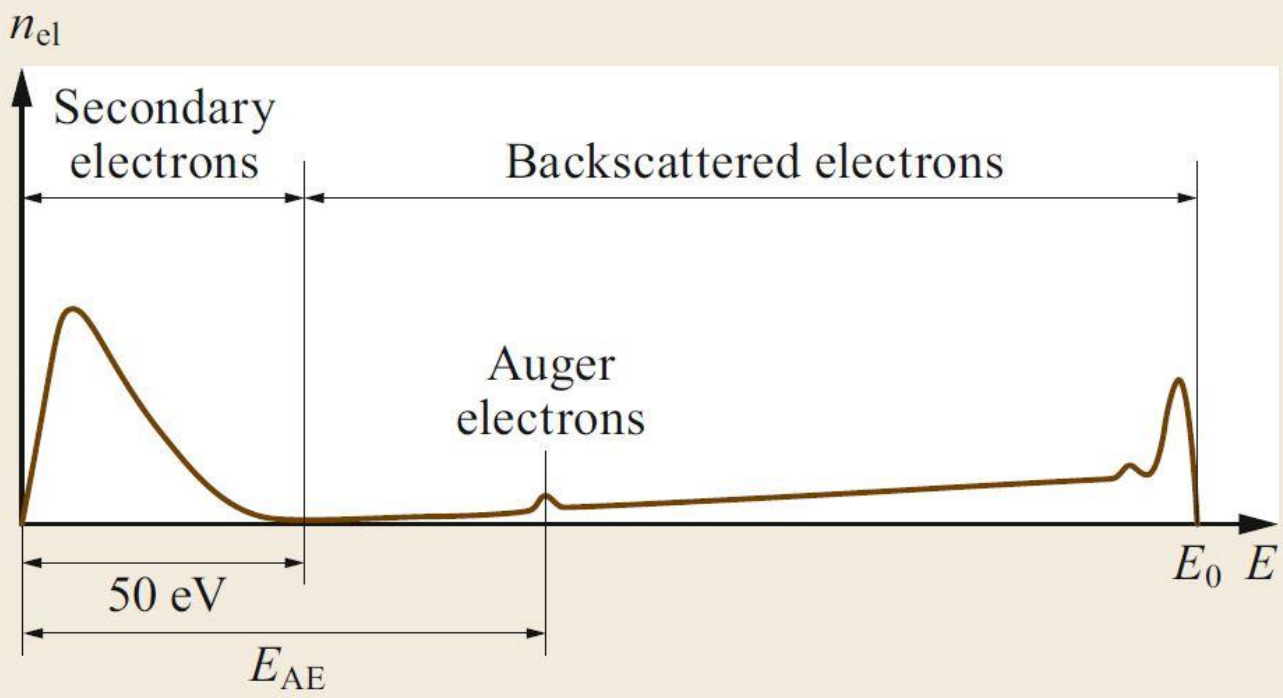


Fig. 5.12 Schematic energy distribution of electrons emitted n_{el} from a surface as a result of its bombardment with fast electrons with energy E_0

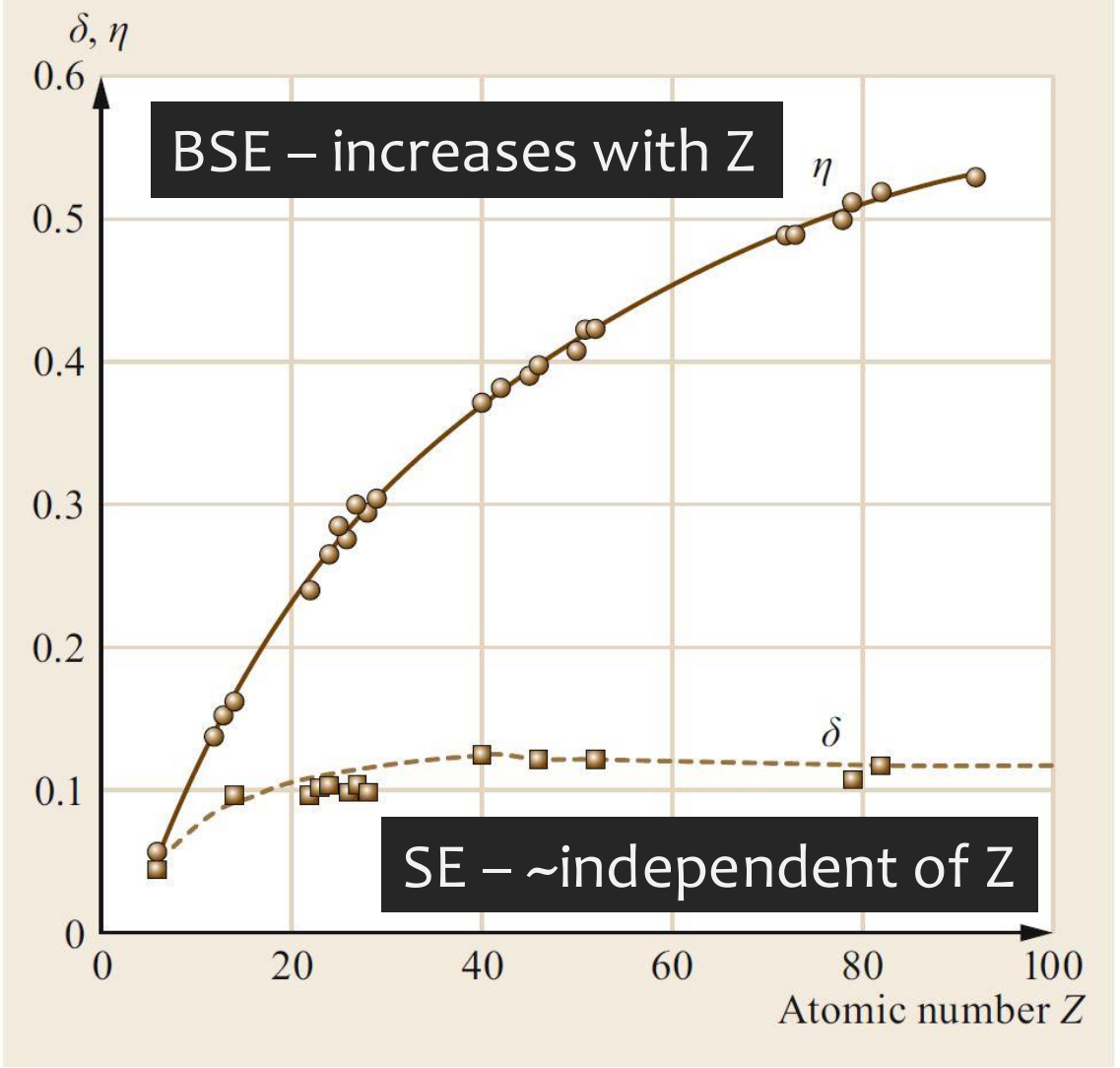
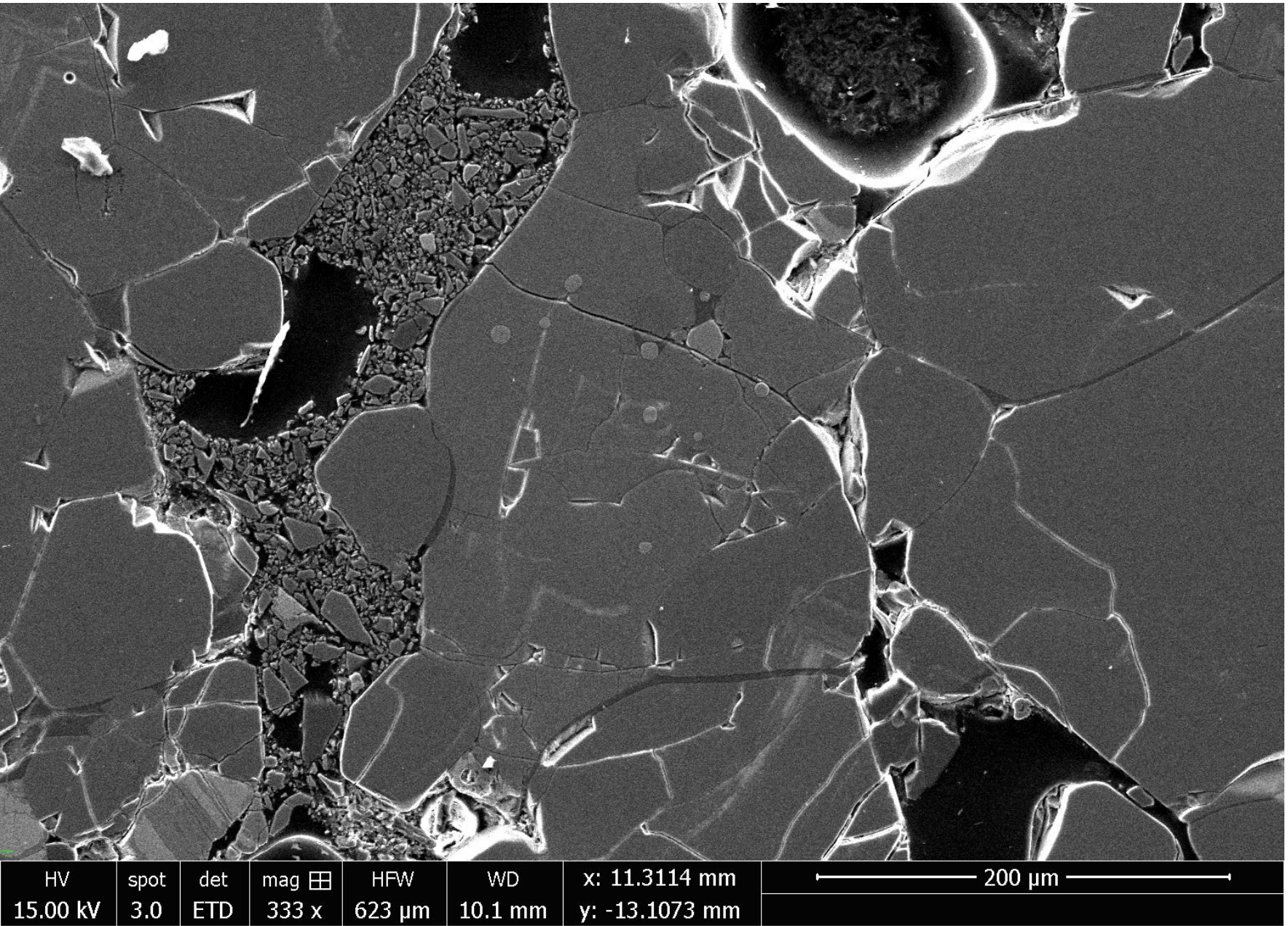


Fig. 5.17 SE yield δ and BSE yield η versus atomic number Z at $E_0 = 30\,\text{keV}$ and $\theta = 0°$. (Data from [5.146, 147])

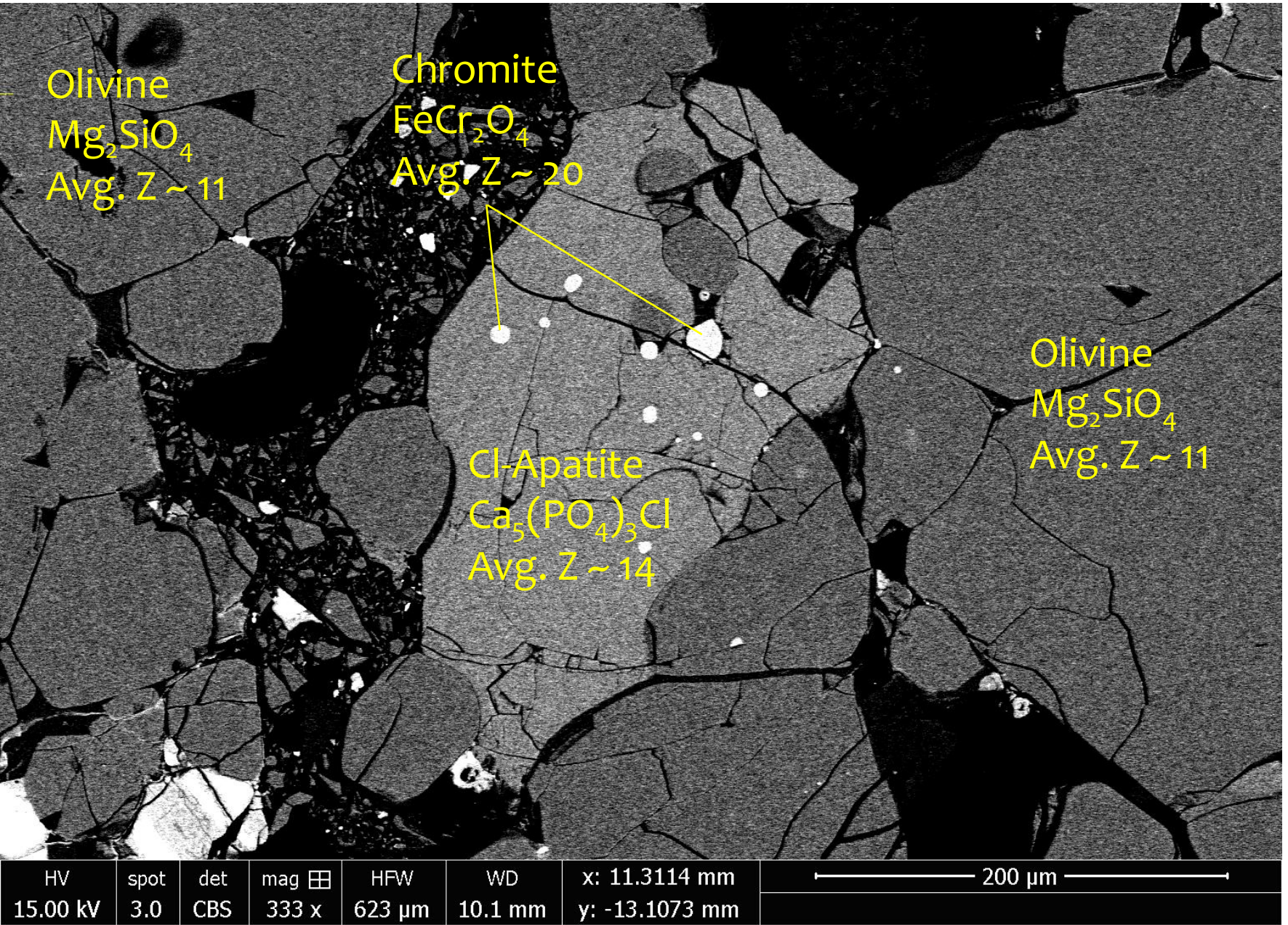
from Herdman *et al.* (2019) – link

SE and BSE imaging



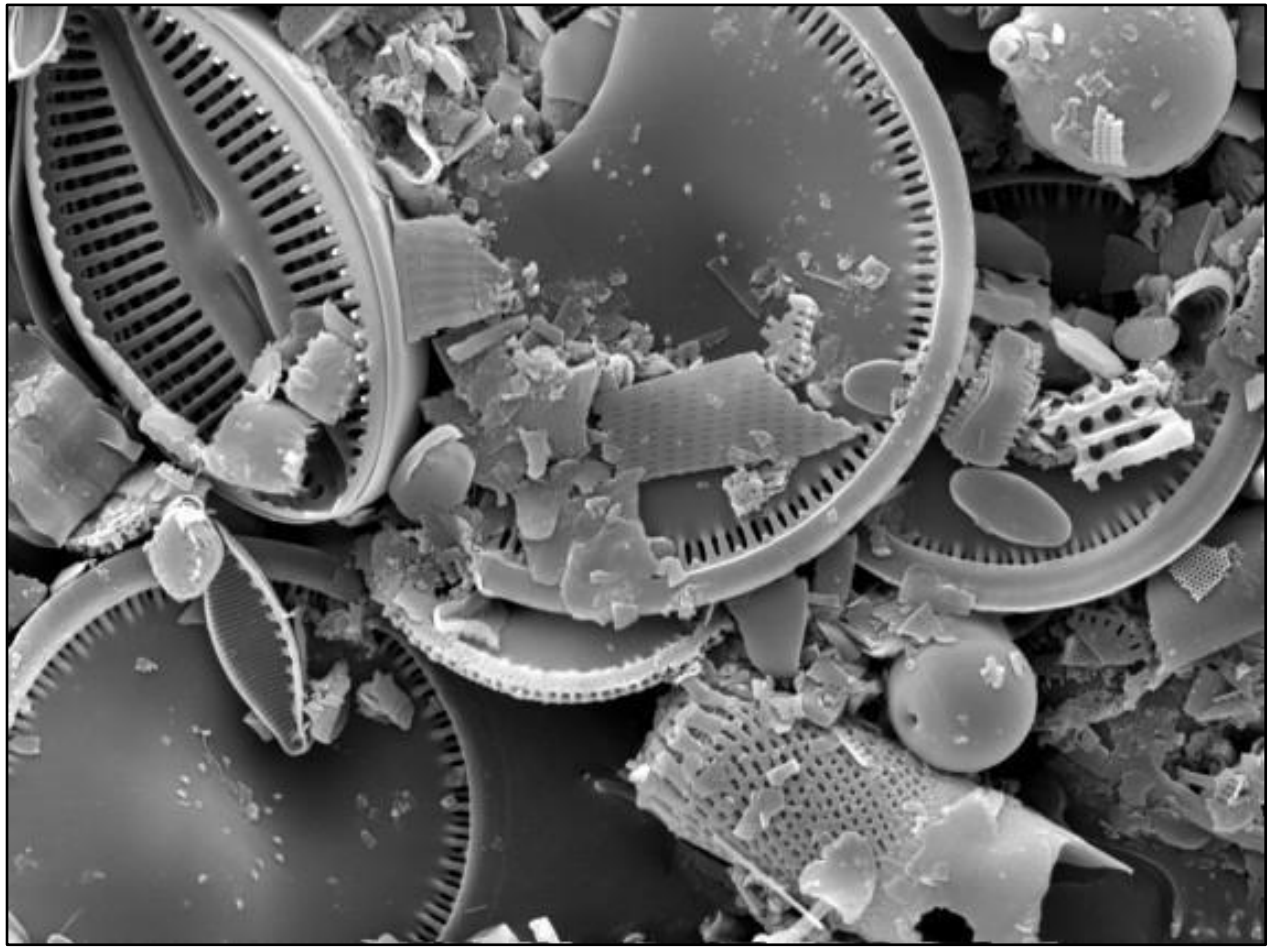
→ SE – topography

SE and BSE imaging

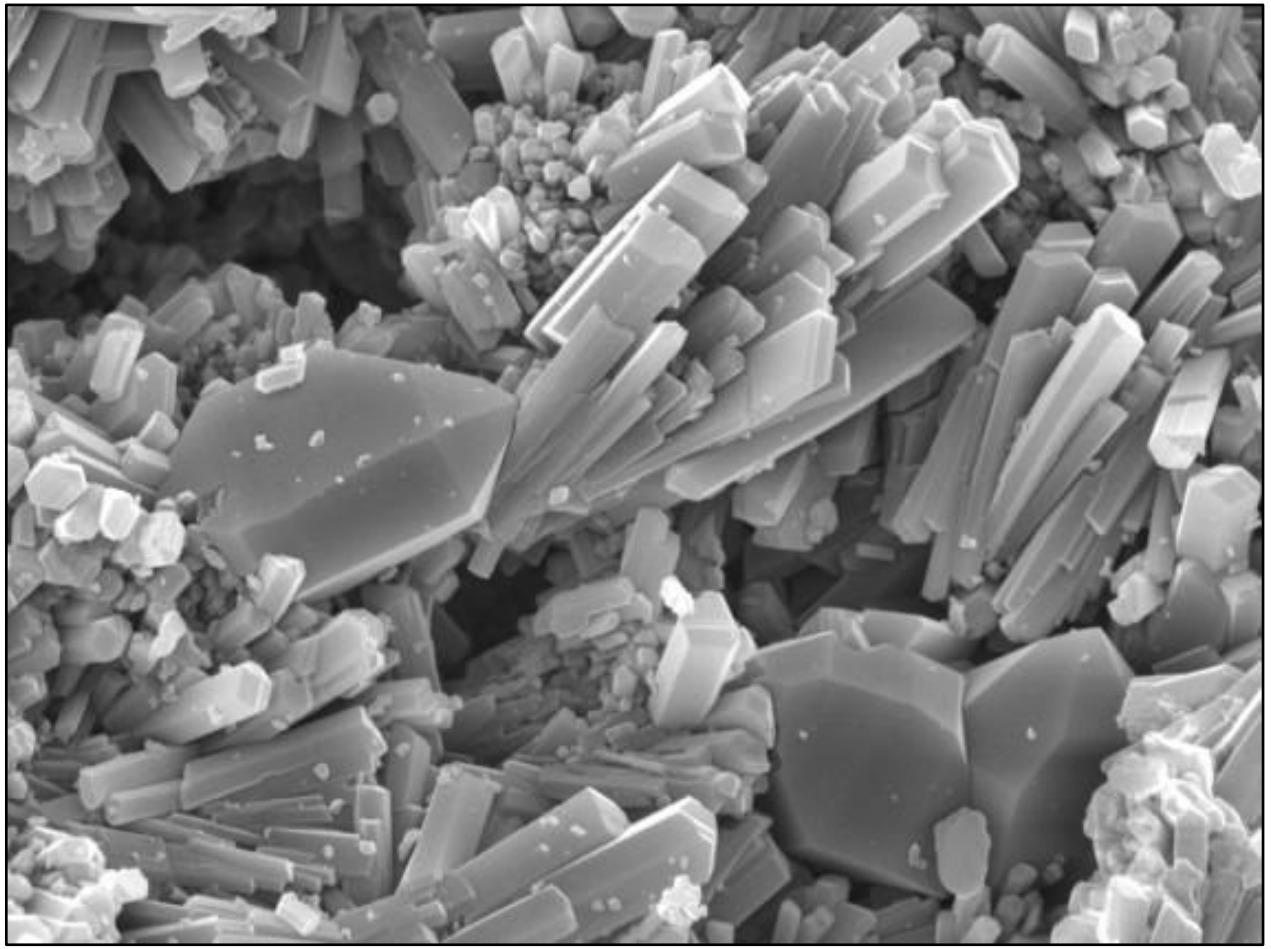


→ SE – topography
→ BSE – mineralogical and chemical information

Secondary electrons



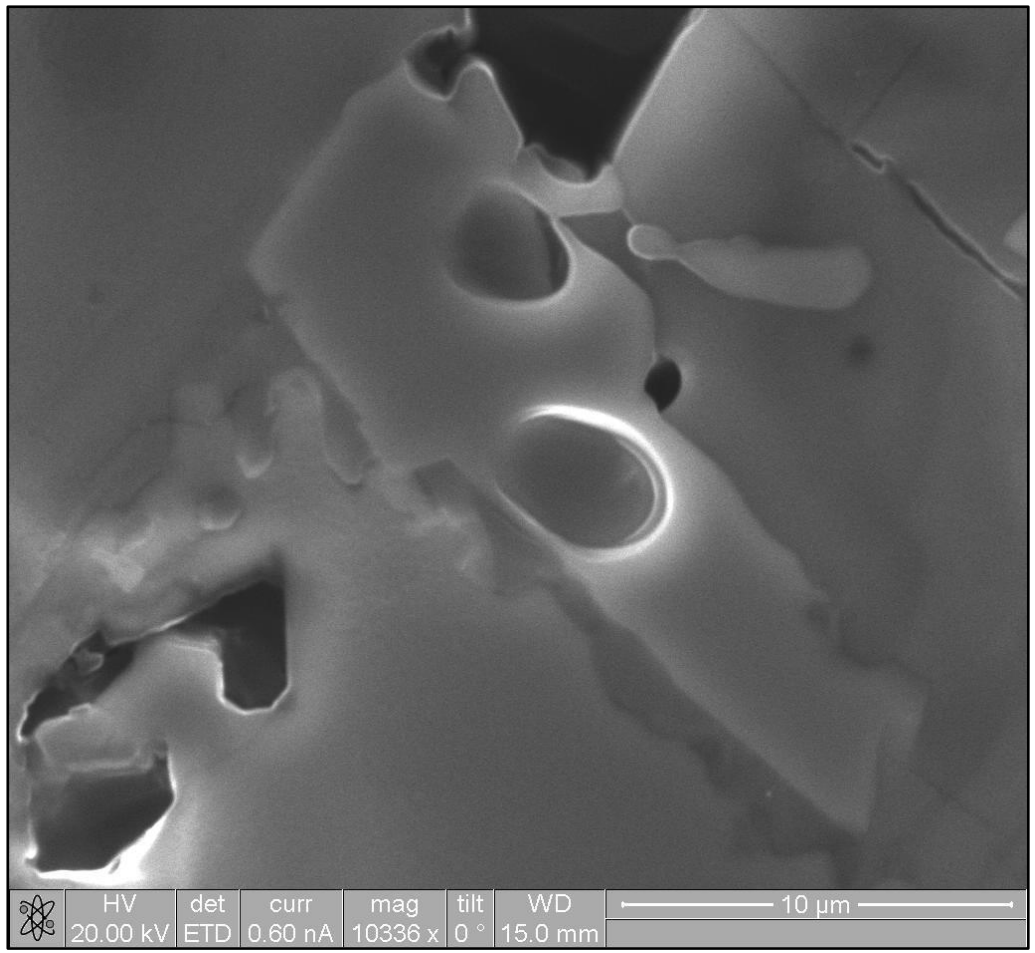
Diatoms from Lake Toskaljarvi, Finland (70 µm FOV)



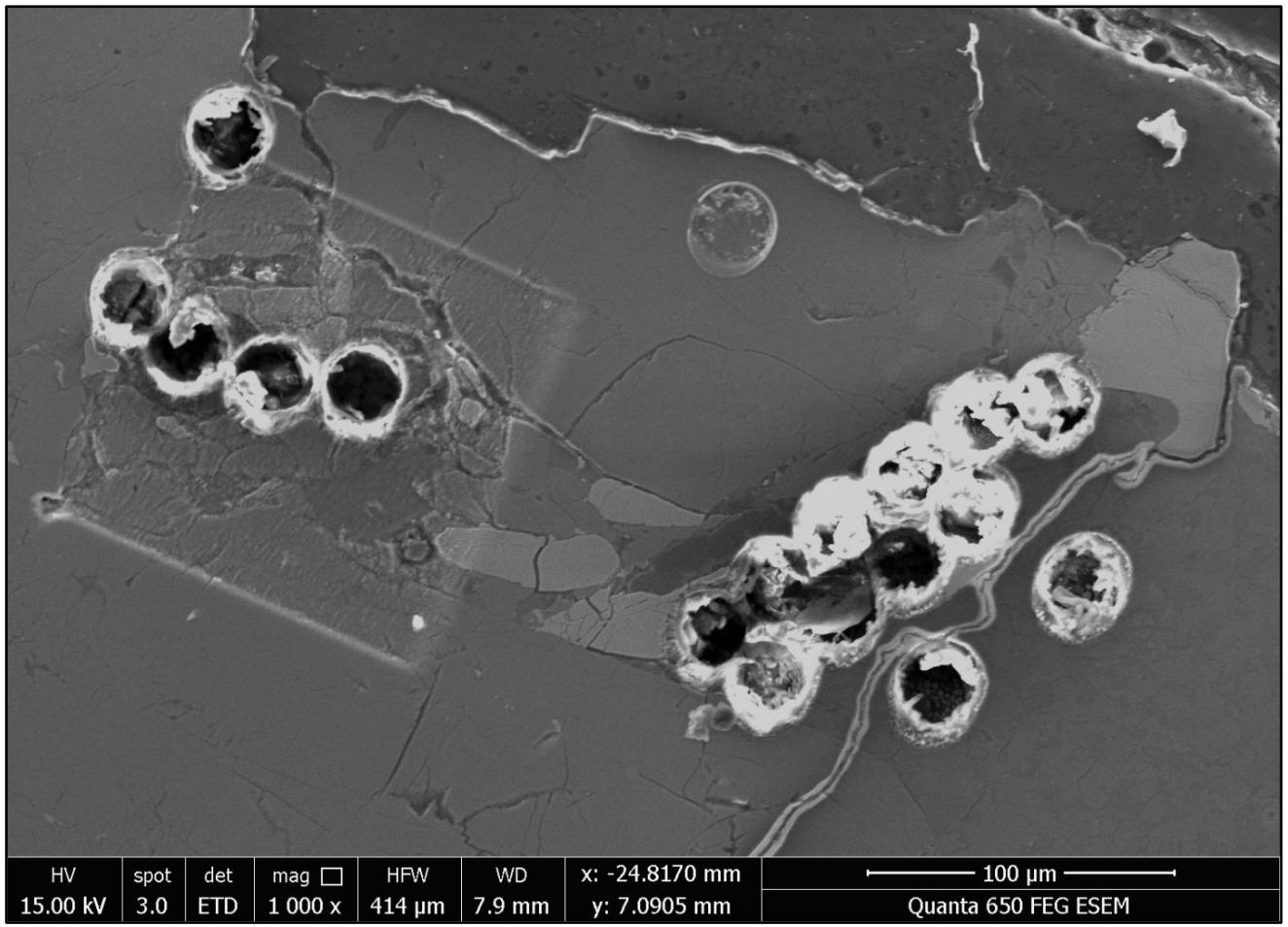
Experimentally grown plagioclase and quartz (50 µm FOV).

SE images are based on the characteristics of the **surface topography** (roughness) of the specimen, and are not dissimilar to images observed by eye.

Secondary electrons



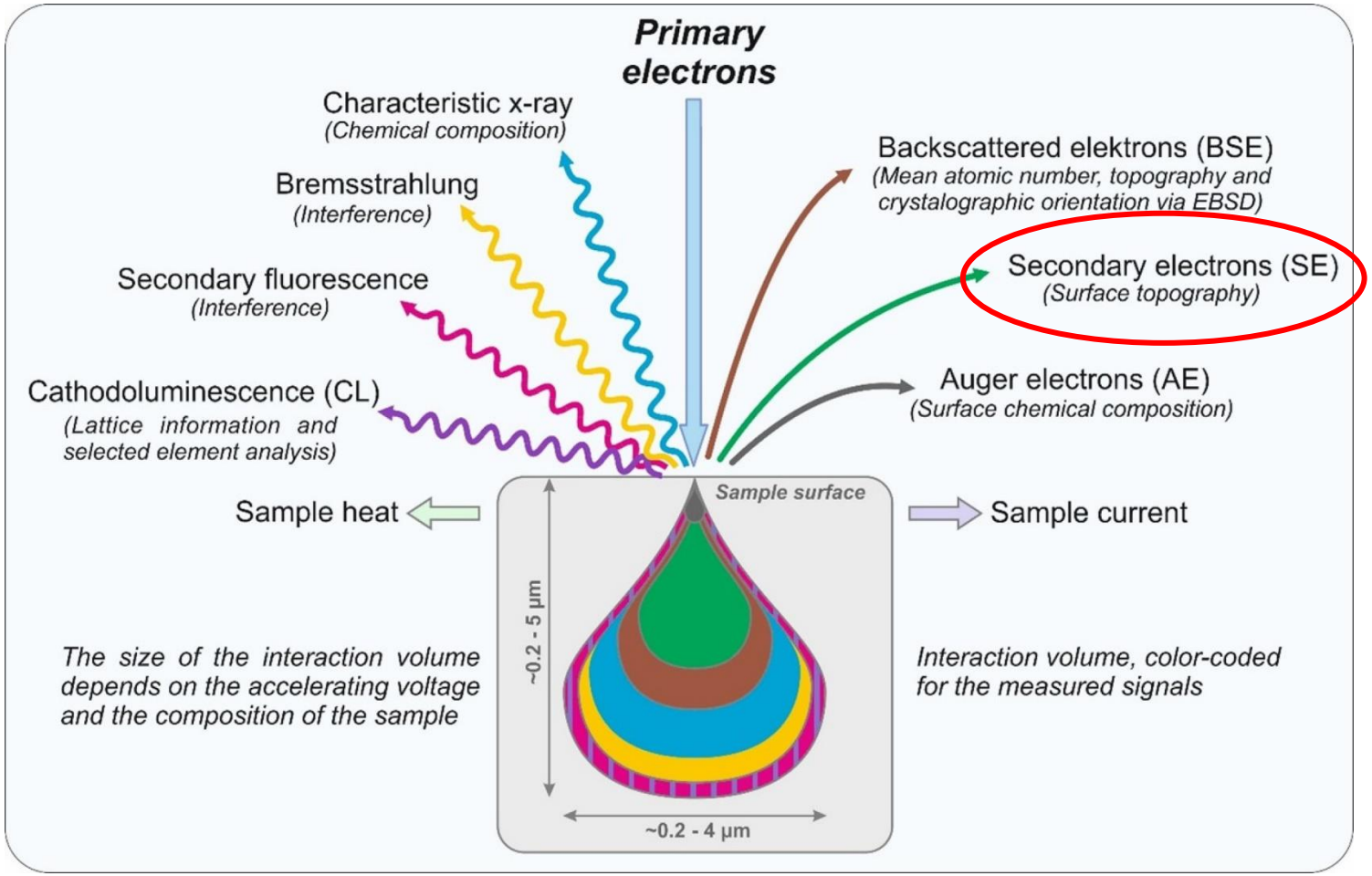
SIMS pits in tranquillityite in Apollo 17 basalt.



LA-ICP-MS pits in carbonates in martian meteorite ALH 84001.

SE images are based on the characteristics of the **surface topography** (roughness) of the specimen, and are not dissimilar to images observed by eye.

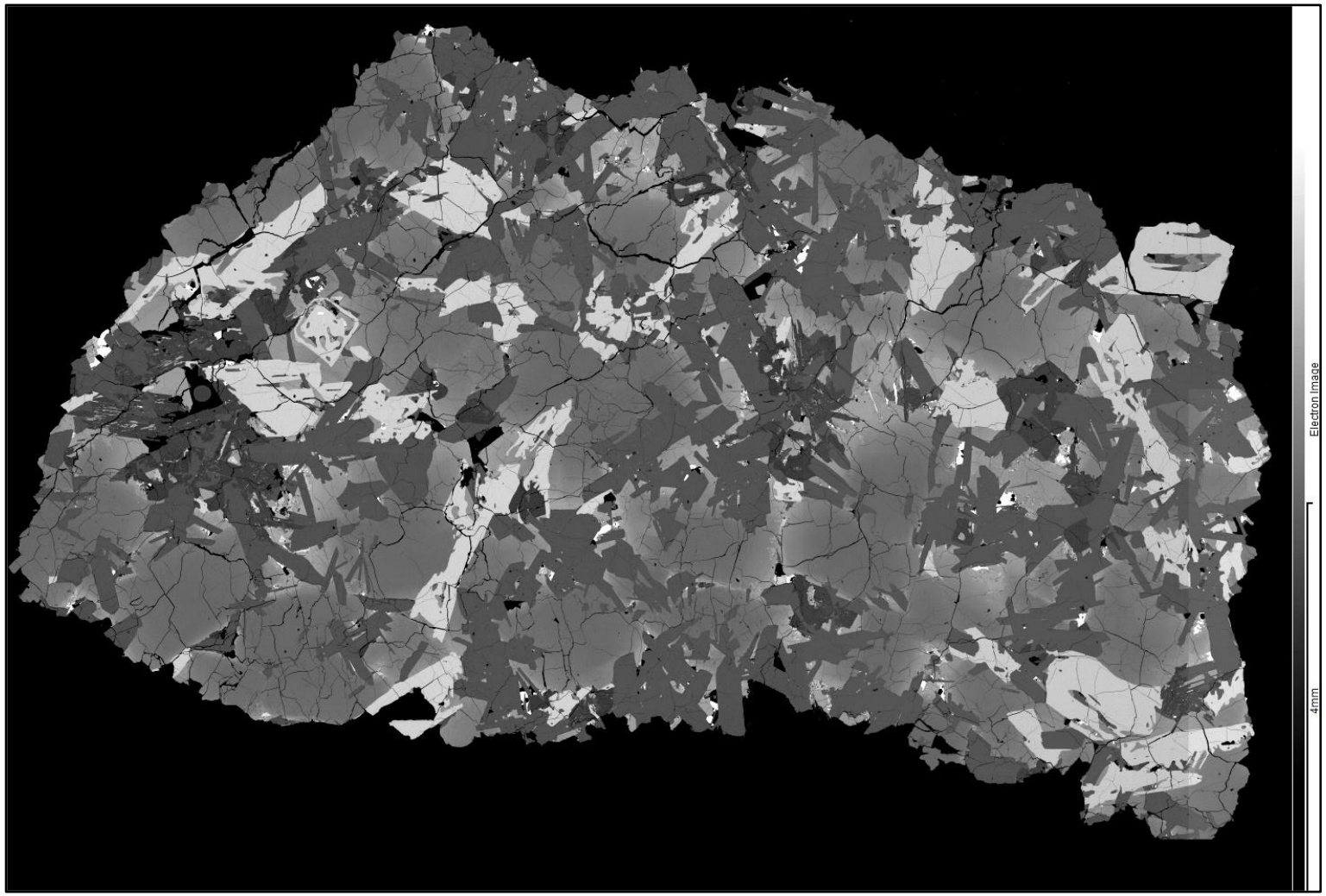
Secondary electrons



- SE are e^- with energies of 0 to ~30 eV, which originate from within the target specimen.
- **Target atomic number Z does not influence signal strength.**
- Most SE originate <100 nm from the specimen surface.
- Divergence of primary beam within target is thus almost zero → image resolution ~ incident beam diameter.

https://www.nhm-wien.ac.at/en/research/central_research_laboratories/analytical_electron_microscopy/theory

Backscattered electrons



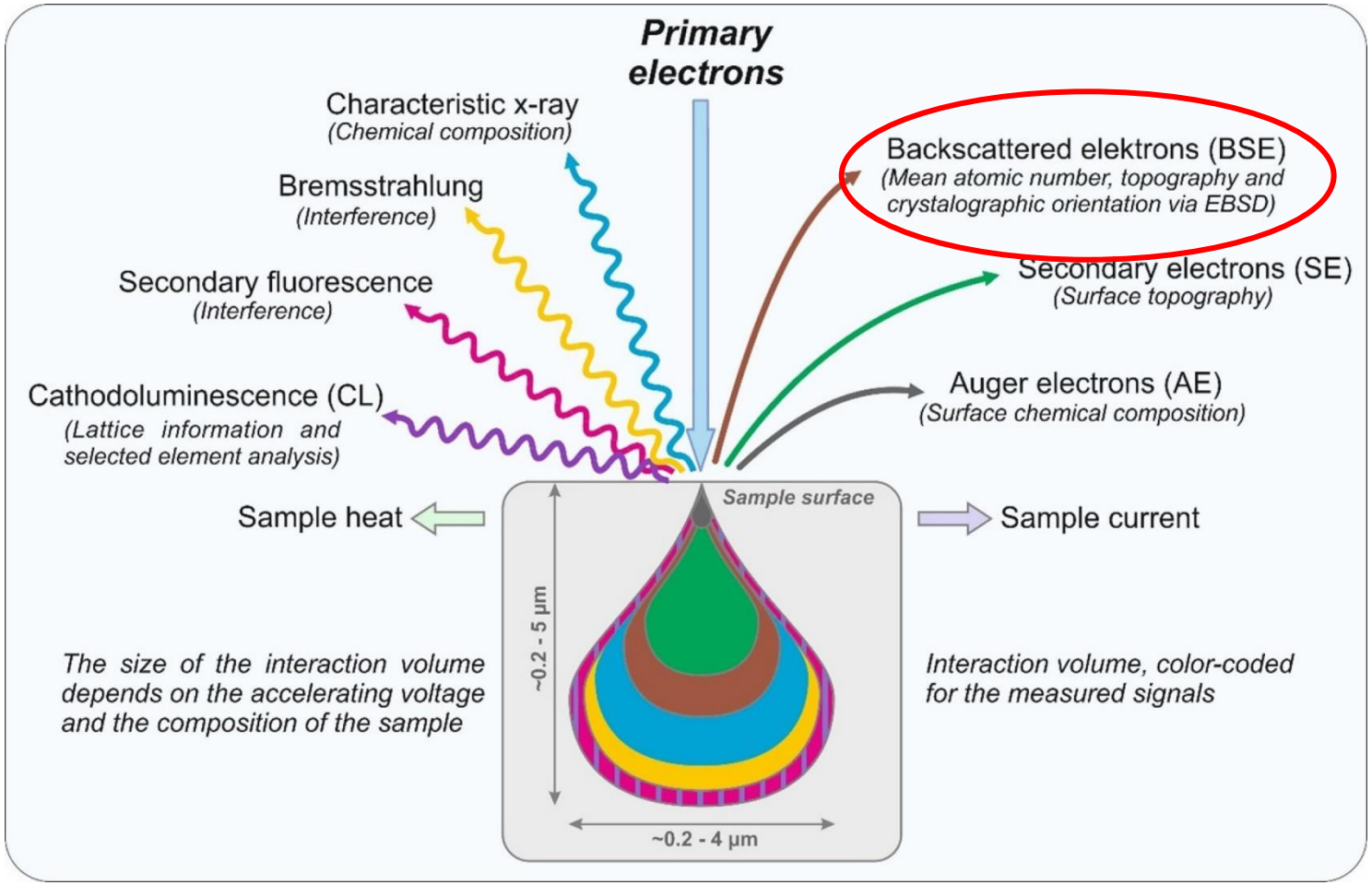
Apollo 11 basalt 10044

Different shades of grey in the BSE image of the whole sample correspond to different mineral phase:

- darkest grey: silica
- dark grey: plagioclase
- medium grey: clinopyroxene
- light grey: ilmenite
- white: sulfides

BSE originate from within the specimen. Information about the bulk properties of the target (composition, crystallography, etc).

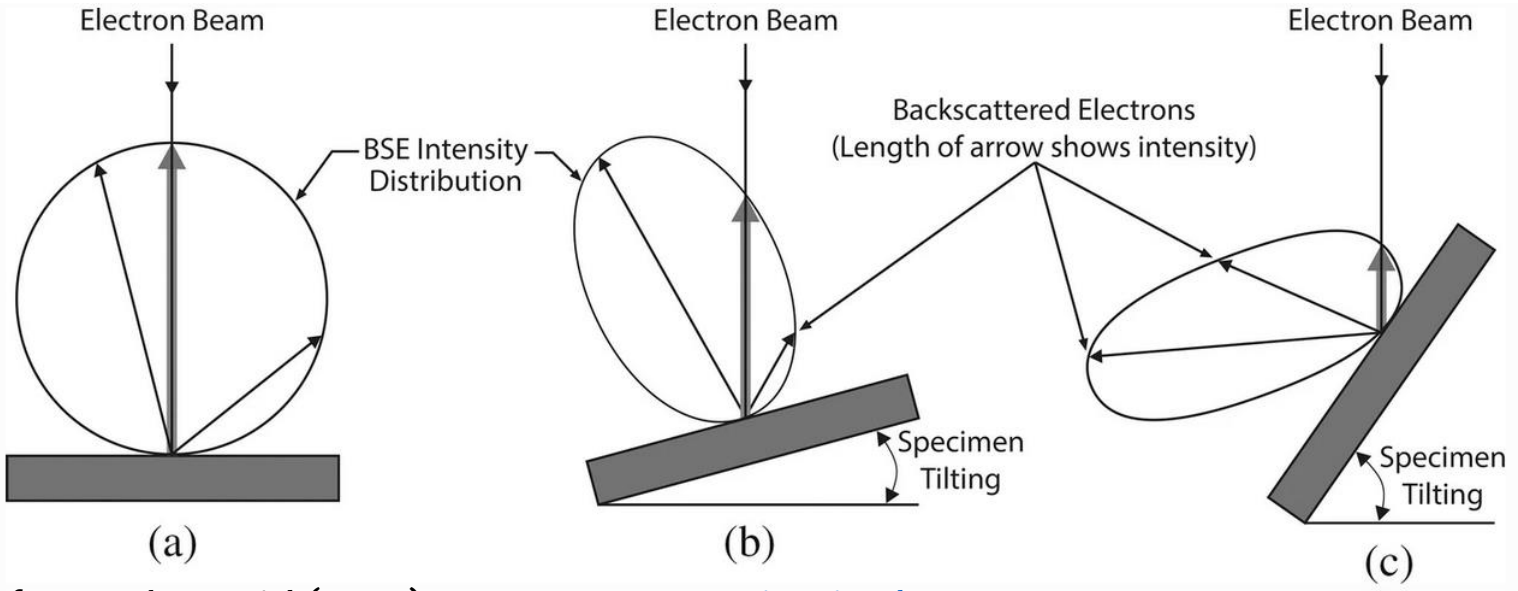
Backscattered electrons



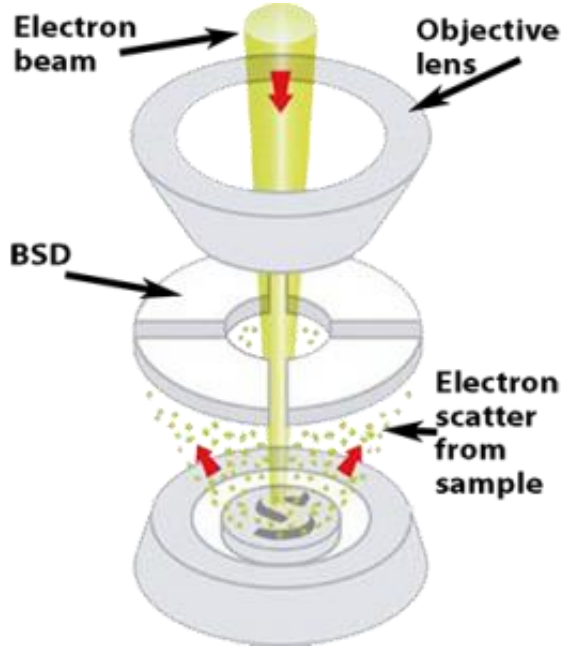
- BSE are e^- with energies from a few keV to energy of incident e^- (typically 15–30 keV), which originate from within the target specimen.
- **BSE signal strength increases with target atomic number Z.**
- BSE originate from ~0.5-2 µm of the specimen surface.

https://www.nhm-wien.ac.at/en/research/central_research_laboratories/analytical_electron_microscopy/theory

Backscattered electrons – detection

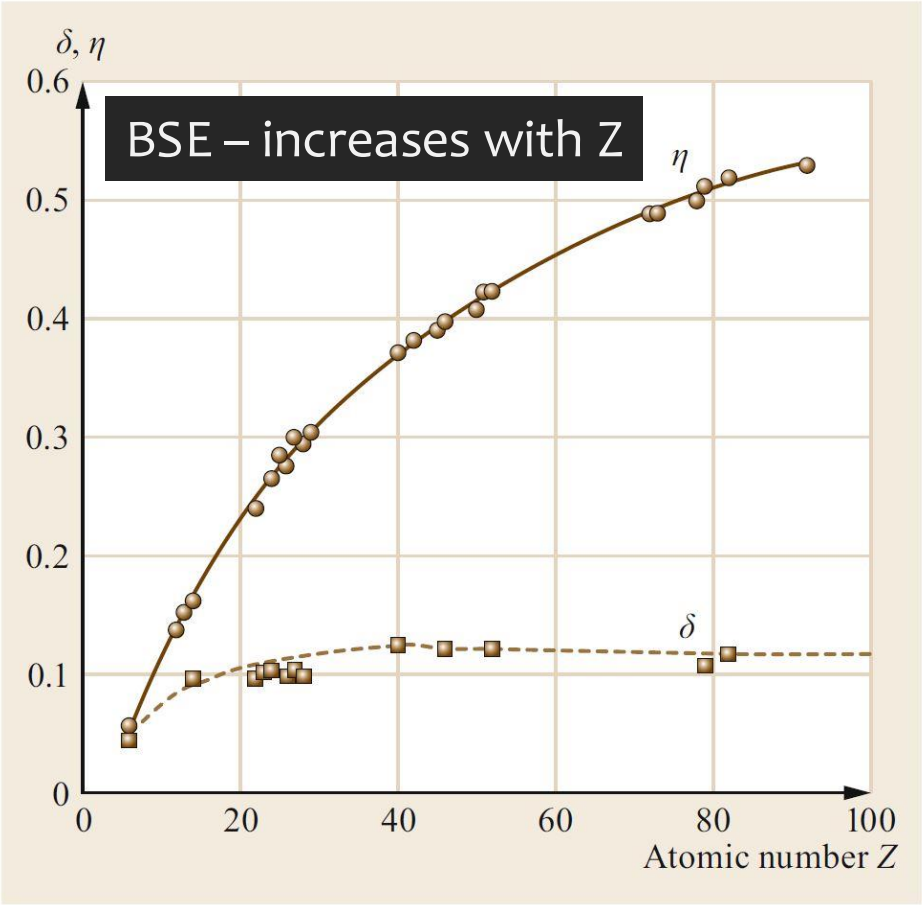


from Ul-Hamid (2018) Contrast Formation in the SEM.

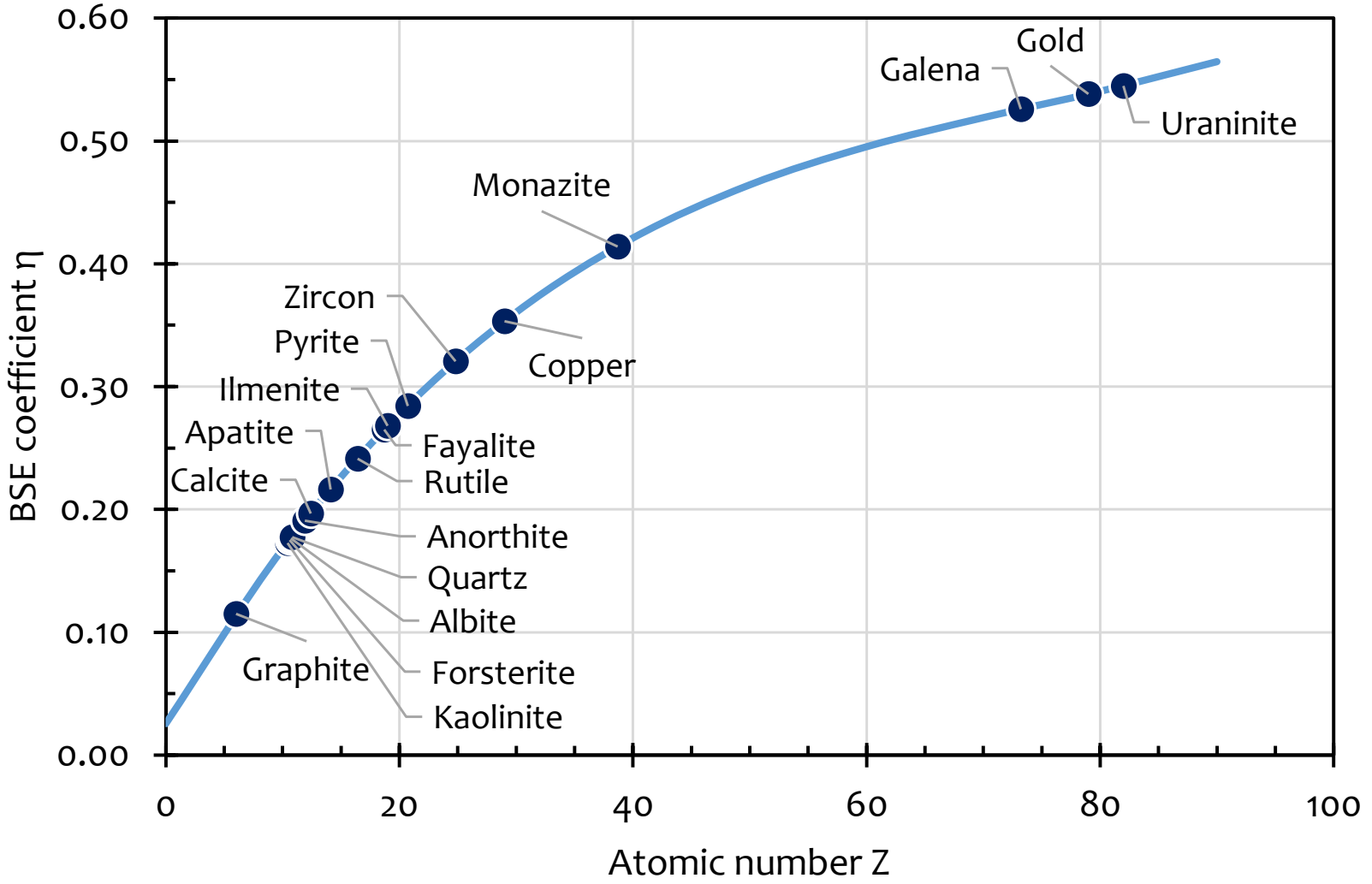


- BSE emission is highly directional → **flat specimen surfaces** are normally required to prevent uncontrolled scattering and BSE emission.
- BSE detectors must be positioned to form a large solid angle with the BSE emission distribution. In general, incident e^- beam has a normal incidence with the target → **BSE detectors usually positioned vertically above target**, with the e^- beam striking target through a central hole in the detector.

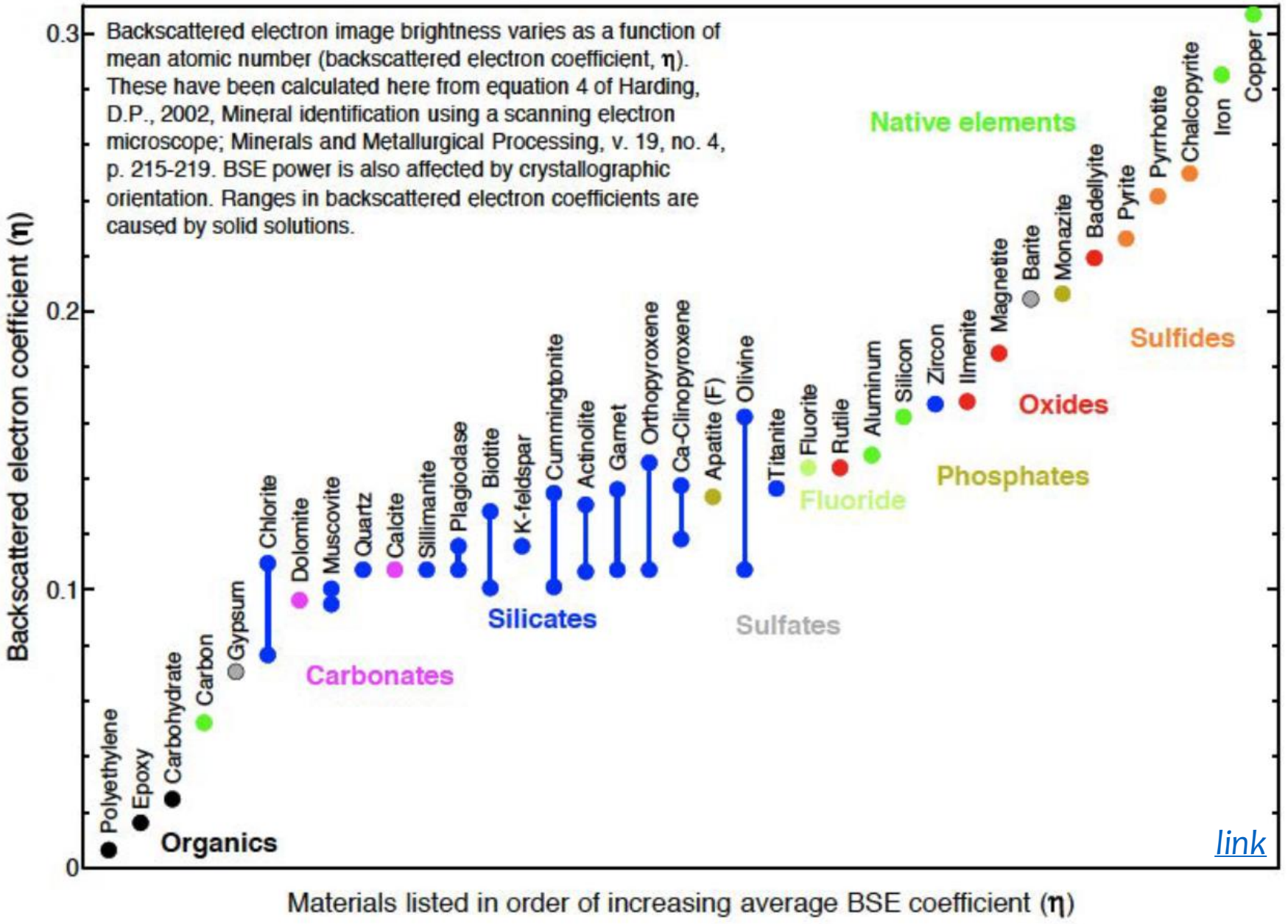
Backscattered electrons – mineralogy



from Herdman et al. (2019) – [link]



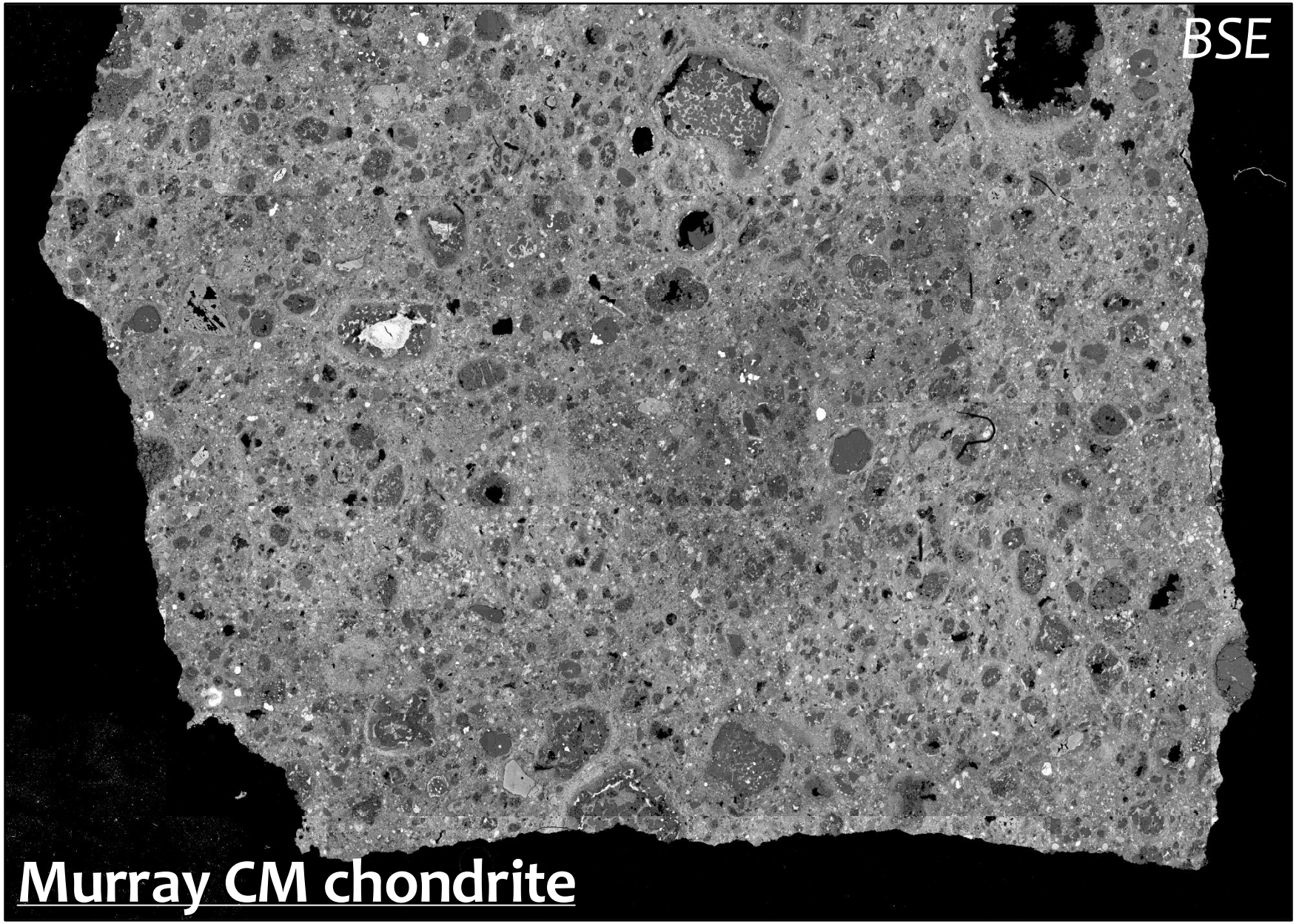
Backscattered electrons – mineralogy



link

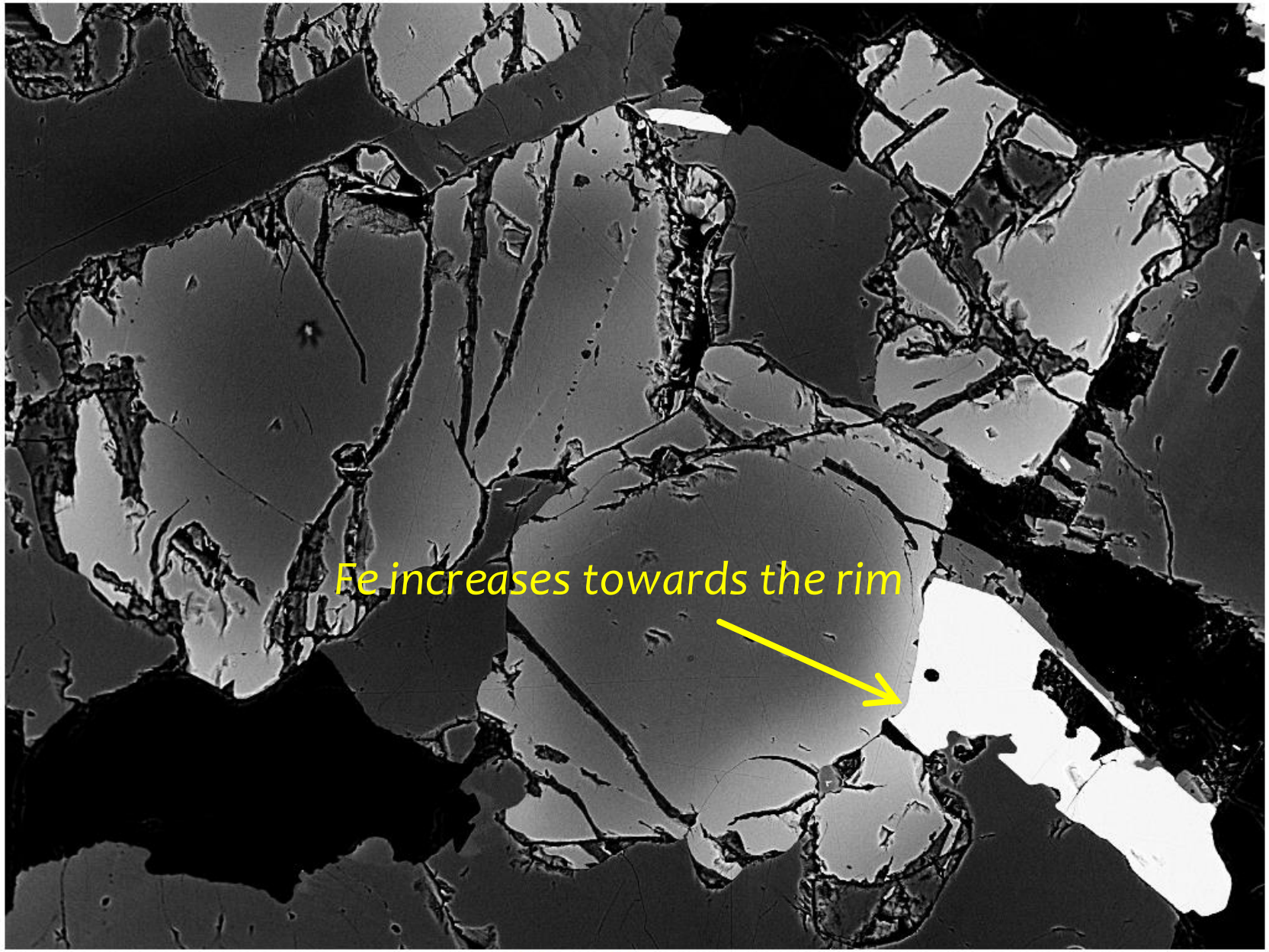
- BSE imaging of polyphased material allows visualising different phases (e.g., minerals in a rock).

Backscattered electrons – mineralogy



- BSE imaging of polyphased material allows visualising different phases (e.g., minerals in a rock).

Backscattered electrons – mineralogy



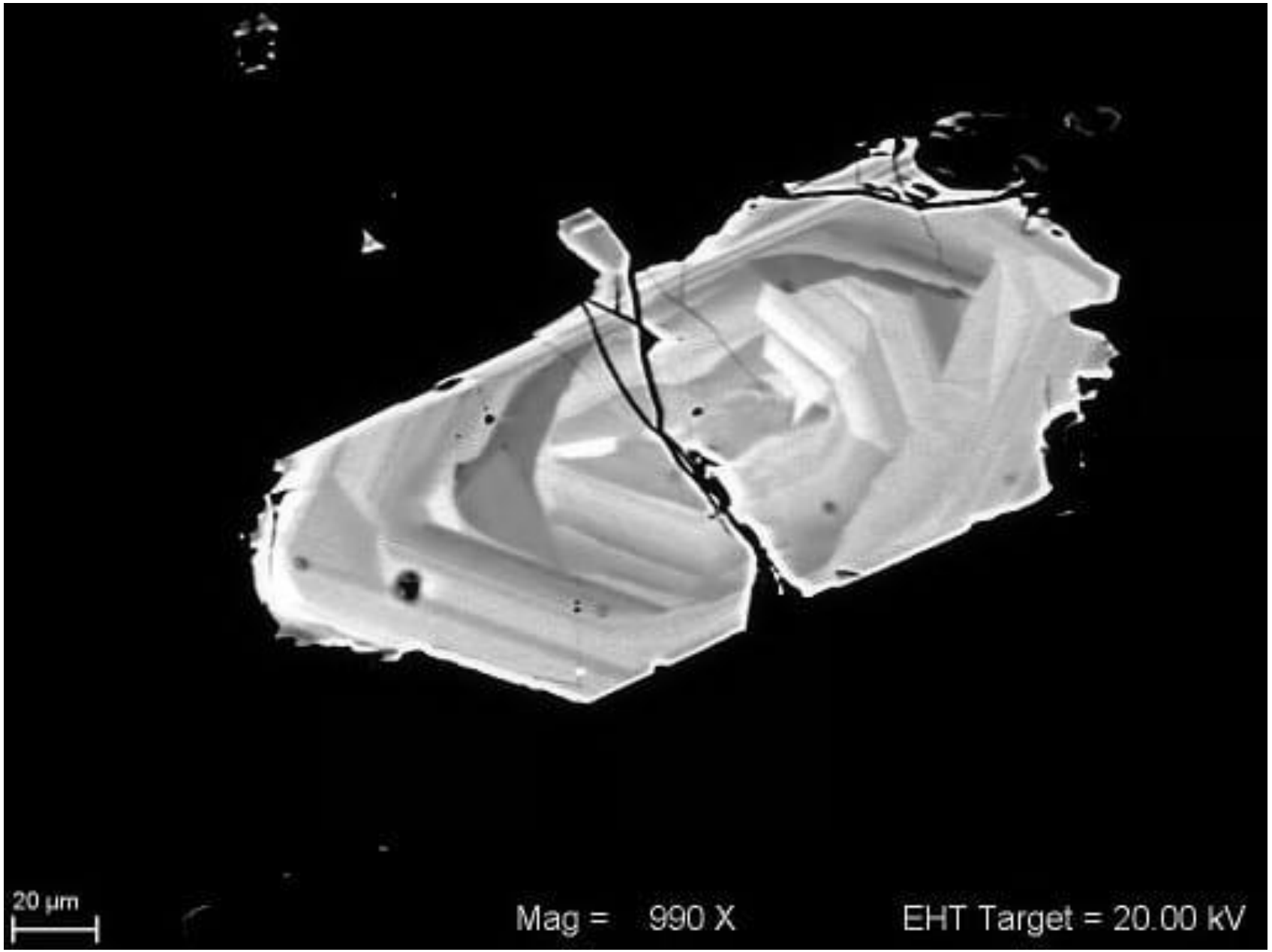


- BSE imaging of polyphased material allows visualising different phases (e.g., minerals in a rock).
- BSE imaging of single phases allows visualising chemical zoning:

  → *Mg-Fe in olivine*

https://mcf.tamu.edu/instruments/electron-microprobe/

Backscattered electrons – mineralogy

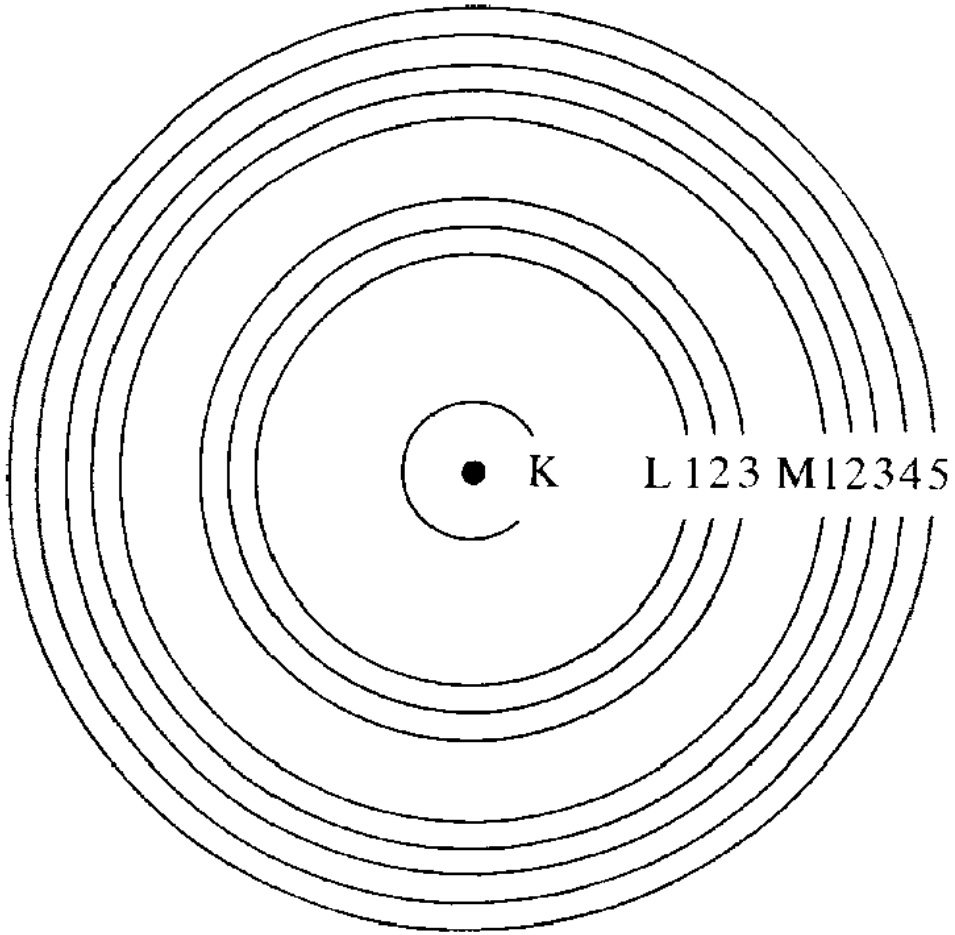


- BSE imaging of polyphased material allows visualising different phases (e.g., minerals in a rock).
- BSE imaging of single phases allows visualising chemical zoning:

  → *Mg-Fe in olivine*
  → *Trace elements (U, Th, REE) in zircon*

https://muse.union.edu/hollochk/kurt-hollocher/scanning-electron-microscope/backscattered-electron-bse-imaging/

Characteristic X-rays

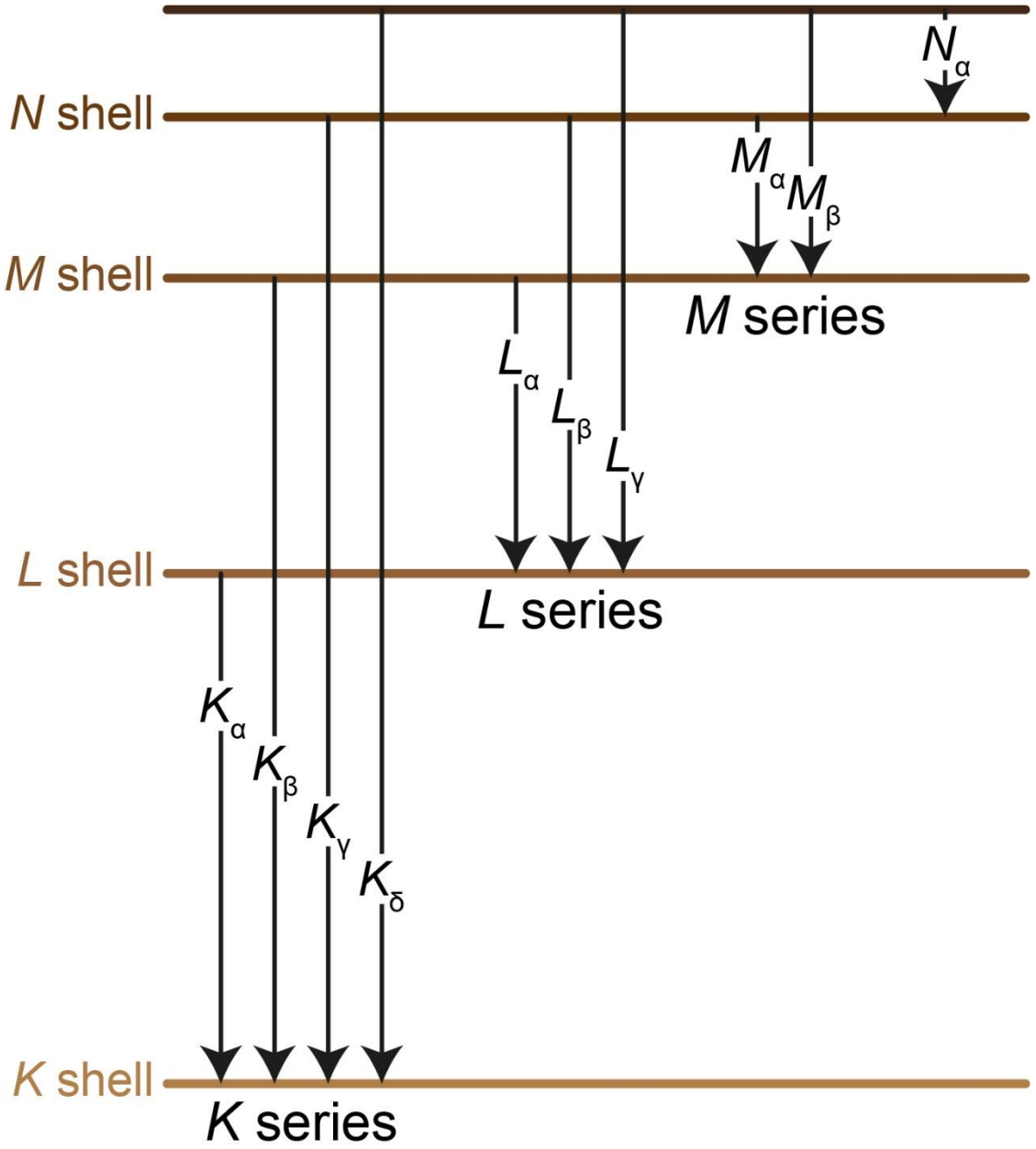


- **Characteristic X-ray** produced by electron transitions between bound electron orbits.
- Removal of an inner electron is necessary for production of characteristic X-ray.
- For characteristic X-ray line to be produced, **incident electron energy E_0 must exceed critical excitation energy E_C required to excite the relevant shell**.

Fig. 2.7. A schematic diagram of inner atomic electron shells; characteristic X-rays are produced by transitions between these shells.

from Reed (2005) – *Electron Microprobe Analysis and Scanning Electron Microscopy in Geology*

Characteristic X-rays



- K_α X-ray is produced due to removal of K shell electron, with L shell electron taking its place. K_β occurs when K shell electron is replaced by electron from the M shell.
- L_α X-ray is produced due to removal of L shell electron, replaced by M shell electron.
- M_α X-ray is produced due to removal of M shell electron, replaced by N shell electron.

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MATERIALS ANALYSIS DIVISION

	meters	dm	cm	mm	µm	nm	Å
1Å =	10^{-10}	10^{-9}	10^{-8}	10^{-7}	10^{-4}	10^{-1}	1
1nm =	10^{-9}	10^{-8}	10^{-7}	10^{-6}	10^{-3}	1	10
1µm =	10^{-6}	10^{-5}	10^{-4}	10^{-3}	1	10^{3}	10^{4}
1mm =	10^{-3}	10^{-2}	10^{-1}	1	10^{3}	10^{6}	10^{7}
1cm =	10^{-2}	10^{-1}	1	10	10^{4}	10^{7}	10^{8}
1dm =	10^{-1}	1	10	10^{2}	10^{5}	10^{8}	10^{9}
1m =	1	10	10^{2}	10^{3}	10^{6}	10^{9}	10^{10}

Crystal Structures

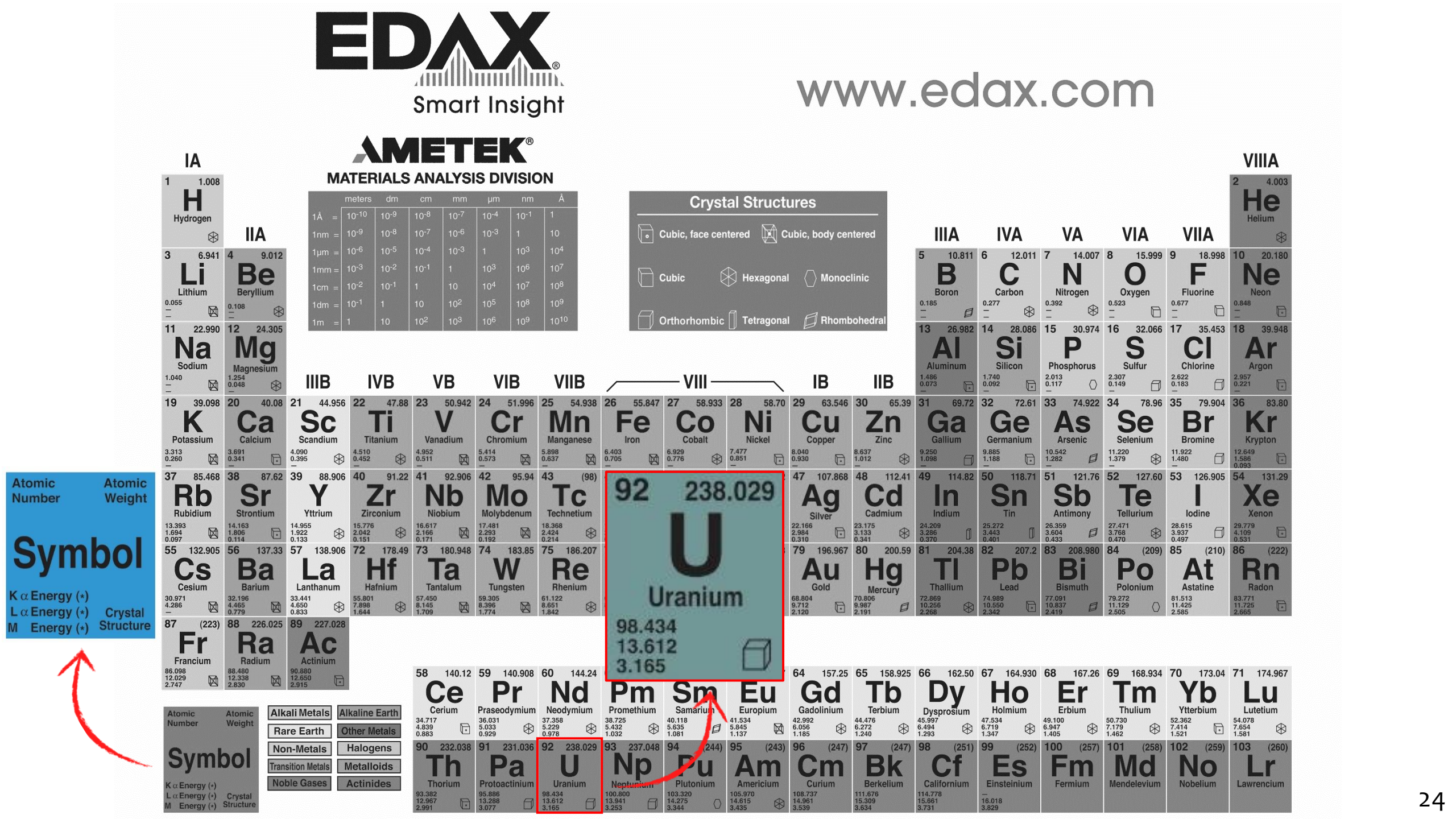
- Cubic, face centered
- Cubic, body centered
- Cubic
- Hexagonal
- Monoclinic
- Orthorhombic
- Tetragonal
- Rhombohedral

IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII	VIII	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 1.008 H Hydrogen																	2 4.003 He Helium
3 6.941 Li Lithium 0.055	4 9.012 Be Beryllium 0.108											5 10.811 B Boron 0.185	6 12.011 C Carbon 0.277	7 14.007 N Nitrogen 0.392	8 15.999 O Oxygen 0.523	9 18.998 F Fluorine 0.677	10 20.180 Ne Neon 0.848
11 22.990 Na Sodium 1.040	12 24.305 Mg Magnesium 1.254 0.048											13 26.982 Al Aluminum 1.486 0.073	14 28.086 Si Silicon 1.740 0.092	15 30.974 P Phosphorus 2.013 0.117	16 32.066 S Sulfur 2.307 0.149	17 35.453 Cl Chlorine 2.622 0.183	18 39.948 Ar Argon 2.957 0.221
19 39.098 K Potassium 3.313 0.260	20 40.08 Ca Calcium 3.691 0.341	21 44.956 Sc Scandium 4.090 0.395	22 47.88 Ti Titanium 4.510 0.452	23 50.942 V Vanadium 4.952 0.511	24 51.996 Cr Chromium 5.414 0.573	25 54.938 Mn Manganese 5.898 0.637	26 55.847 Fe Iron 6.403 0.705	27 58.933 Co Cobalt 6.929 0.776	28 58.70 Ni Nickel 7.477 0.851	29 63.546 Cu Copper 8.040 0.930	30 65.39 Zn Zinc 8.637 1.012	31 69.72 Ga Gallium 9.250 1.098	32 72.61 Ge Germanium 9.885 1.188	33 74.922 As Arsenic 10.542 1.282	34 78.96 Se Selenium 11.220 1.379	35 79.904 Br Bromine 11.922 1.480	36 83.80 Kr Krypton 12.649 1.586 0.093
37 85.468 Rb Rubidium 13.393 1.694 0.097	38 87.62 Sr Strontium 14.163 1.806 0.114	39 88.906 Y Yttrium 14.955 1.922 0.133	40 91.22 Zr Zirconium 15.776 2.042 0.151	41 92.906 Nb Niobium 16.617 2.166 0.171	42 95.94 Mo Molybdenum 17.481 2.293 0.192	43 (98) Tc Technetium 18.368 2.424 0.214	44 101.07 Ru Ruthenium 19.282 2.558 0.237	45 102.906 Rh Rhodium 20.217 2.696 0.260	46 106.42 Pd Palladium 21.180 2.838 0.282	47 107.868 Ag Silver 22.166 2.984 0.310	48 112.41 Cd Cadmium 23.175 3.133 0.341	49 114.82 In Indium 24.209 3.286 0.370	50 118.71 Sn Tin 25.272 3.443 0.401	51 121.76 Sb Antimony 26.359 3.604 0.433	52 127.60 Te Tellurium 27.471 3.768 0.470	53 126.905 I Iodine 28.615 3.937 0.497	54 131.29 Xe Xenon 29.779 4.109 0.531
55 132.905 Cs Cesium 30.971 4.286	56 137.33 Ba Barium 32.196 4.465 0.779	57 138.906 La Lanthanum 33.441 4.650 0.833	72 178.49 Hf Hafnium 55.801 7.898 1.644	73 180.948 Ta Tantalum 57.450 8.145 1.709	74 183.85 W Tungsten 59.305 8.396 1.774	75 186.207 Re Rhenium 61.122 8.651 1.842	76 190.2 Os Osmium 62.989 8.910 1.910	77 192.22 Ir Iridium 64.906 9.174 1.978	78 195.08 Pt Platinum 66.834 9.441 2.048	79 196.967 Au Gold 68.804 9.712 2.120	80 200.59 Hg Mercury 70.806 9.987 2.191	81 204.38 Tl Thallium 72.869 10.256 2.268	82 207.2 Pb Lead 74.989 10.550 2.342	83 208.980 Bi Bismuth 77.091 10.837 2.419	84 (209) Po Polonium 79.272 11.129 2.505	85 (210) At Astatine 81.513 11.425 2.585	86 (222) Rn Radon 83.771 11.725 2.665
87 (223) Fr Francium 86.098 12.029 2.747	88 226.025 Ra Radium 88.480 12.338 2.830	89 227.028 Ac Actinium 90.880 12.650 2.915															

58 140.12 Ce Cerium 34.717 4.839 0.883	59 140.908 Pr Praseodymium 36.031 5.033 0.929	60 144.24 Nd Neodymium 37.358 5.229 0.978	61 (145) Pm Promethium 38.725 5.432 1.032
62 150.36 Sm Samarium 40.118 5.635 1.081	63 151.97 Eu Europium 41.534 5.845 1.137	64 157.25 Gd Gadolinium 42.992 6.056 1.185	65 158.925 Tb Terbium 44.476 6.272 1.240
66 162.50 Dy Dysprosium 45.997 6.494 1.293	67 164.930 Ho Holmium 47.534 6.719 1.347	68 167.26 Er Erbium 49.100 6.947 1.405	69 168.934 Tm Thulium 50.730 7.179 1.462
70 173.04 Yb Ytterbium 52.362 7.414 1.521	71 174.967 Lu Lutetium 54.078 7.654 1.581		
90 232.038 Th Thorium 93.382 12.967 2.991	91 231.036 Pa Protoactinium 95.886 13.288 3.077	92 238.029 U Uranium 98.434 13.612 3.165	93 237.048 Np Neptunium 100.800 13.941 3.253
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98 (251) Cf Californium 114.778 15.661 3.731	99 (252) Es Einsteinium – 16.018 3.829	100 (257) Fm Fermium	101 (258) Md Mendelevium
102 (259) No Nobelium	103 (260) Lr Lawrencium		

Atomic Number / Atomic Weight
Symbol
K α Energy (•)
L α Energy (•)
M Energy (•)
Crystal Structure

Alkali Metals	Alkaline Earth
Rare Earth	Other Metals
Non-Metals	Halogens
Transition Metals	Metalloids
Noble Gases	Actinides


EDAX
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AMETEK
MATERIALS ANALYSIS DIVISION
meters dm cm mm µm nm Å
1Å = 10^-10 10^-9 10^-8 10^-7 10^-4 10^-1 1
1nm = 10^-9 10^-8 10^-7 10^-6 10^-3 1 10
1µm = 10^-6 10^-5 10^-4 10^-3 1 10^3 10^4
1mm = 10^-3 10^-2 10^-1 1 10^3 10^6 10^7
1cm = 10^-2 10^-1 1 10 10^4 10^7 10^8
1dm = 10^-1 1 10 10^2 10^5 10^8 10^9
1m = 1 10 10^2 10^3 10^6 10^9 10^10
Crystal Structures
Cubic, face centered
Cubic, body centered
Cubic
Hexagonal
Monoclinic
Orthorhombic
Tetragonal
Rhombohedral
IA IIA IIIB IVB VB VIB VIIB VIII IB IIB IIIA IVA VA VIA VIIA VIIIA
1 1.008 H Hydrogen
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3 6.941 Li Lithium 0.055
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Atomic Weight
Symbol
K α Energy (•)
L α Energy (•)
M Energy (•)
Crystal Structure
Alkali Metals
Alkaline Earth
Rare Earth
Other Metals
Non-Metals
Halogens
Transition Metals
Metalloids
Noble Gases
Actinides

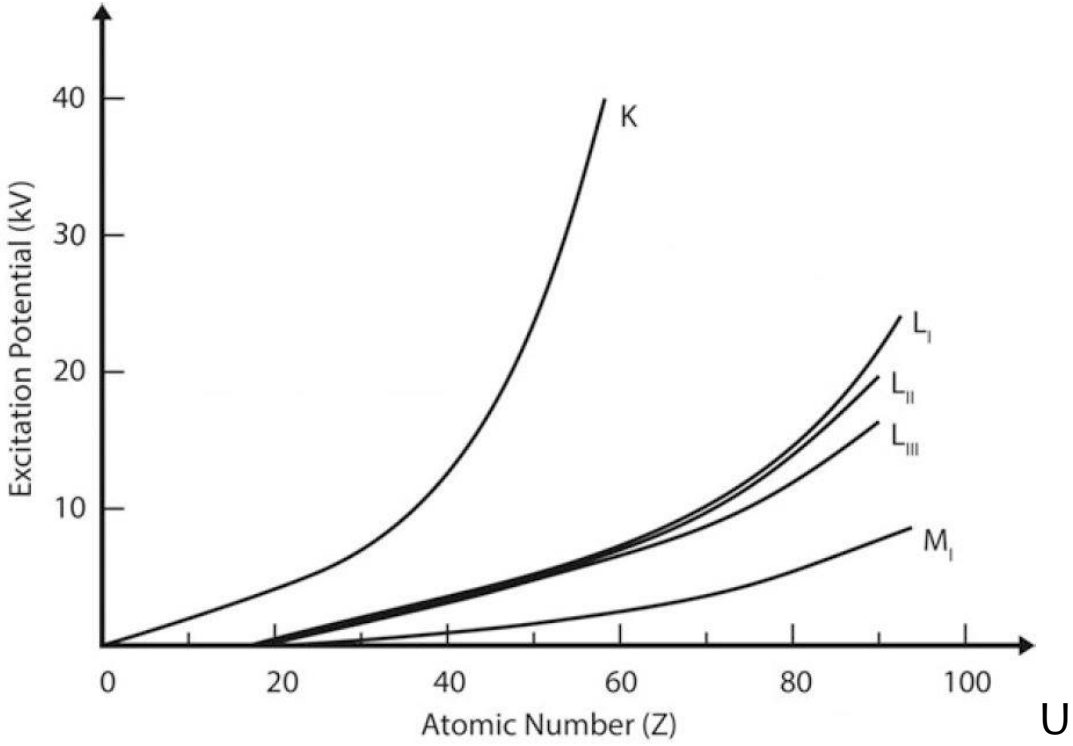
Characteristic X-rays – Critical excitation energy

Atom	Z	E_c K_α	K_α	L_α	M
Al	13	1.56	1.486	0.073	-
Si	14	1.83	1.740	0.092	-
Fe	26	7.11	6.403	0.705	-
Cu	29	8.98	8.040	0.930	-
Au	79	80.72	68.804	9.712	2.120
U	92	115.60	98.434	13.712	3.165

- **Critical excitation energy** E_C = min. energy eject electron from atomic shell.
- Also known as *excitation potential* and *X-ray absorption edge energy.*

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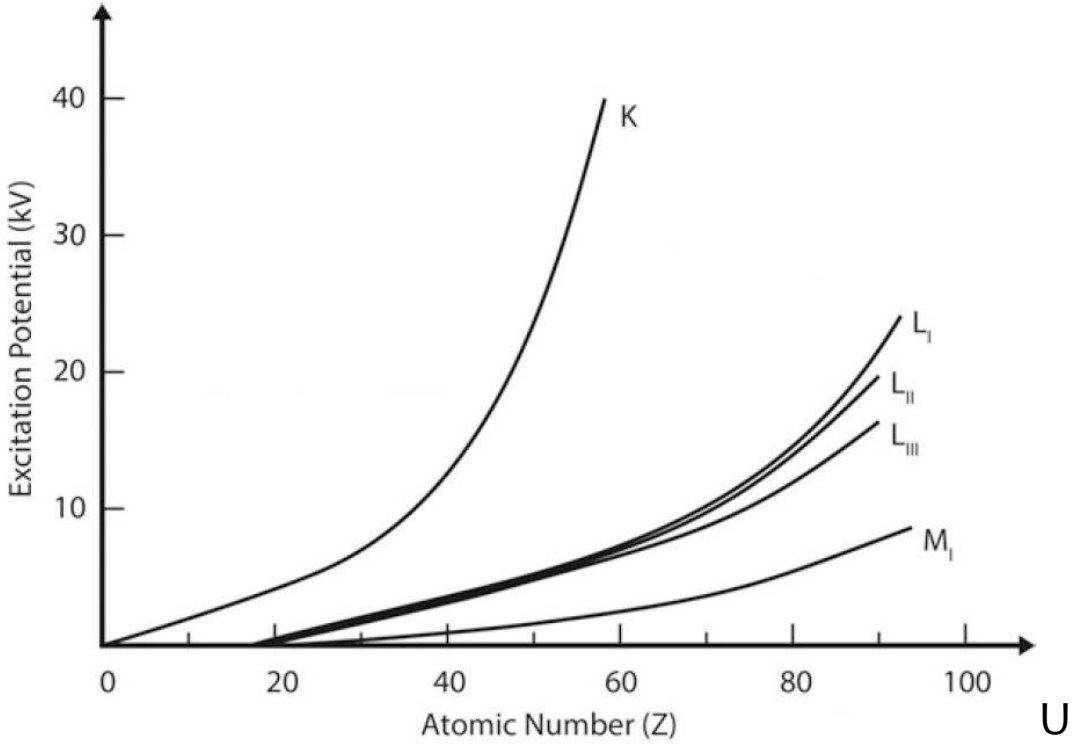


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- With increasing atom size (increasing Z), E_C required to excite any particular transition line also increases.

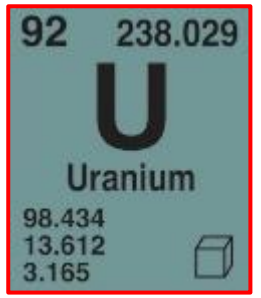
Ul-Hamid (2018) – *A Beginners' Guide to Scanning Electron Microscopy*.

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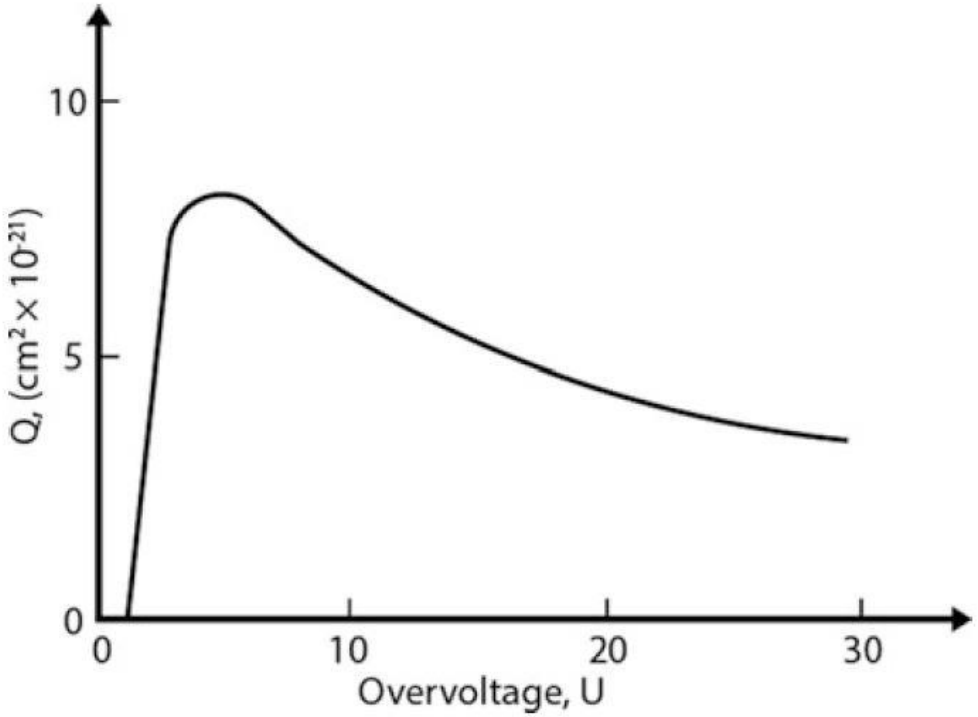
- **<u>Critical excitation energy</u>** E_C = min. energy eject electron from atomic shell.
- Also known as *excitation potential* and *X-ray absorption edge energy.*
- With increasing atom size (increasing Z), E_C required to excite any particular transition line also increases.
- E_C is higher than associated *characteristic X-ray energy.*
- E_C K_α ~ Σ(line energies) ~ $K_\alpha + L_\alpha + M_\alpha$



Ul-Hamid (2018) – <u>A Beginners' Guide to Scanning Electron Microscopy</u>.

Condition for ionisation – Overvoltage

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Si	14	1.83	1.740	0.092	-
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- **Overvoltage** $U = E_0/E_C$

  → ratio of accelerating voltage (gun) to critical excitation energy for selected line.
- Max. efficiency at ~2-3 times E_C.
- **Fe**: min. accelerating voltage needed to excite K_α ~ 15 kV.
- **U**: min. accelerating voltage needed to excite L_α ~ 30 kV.

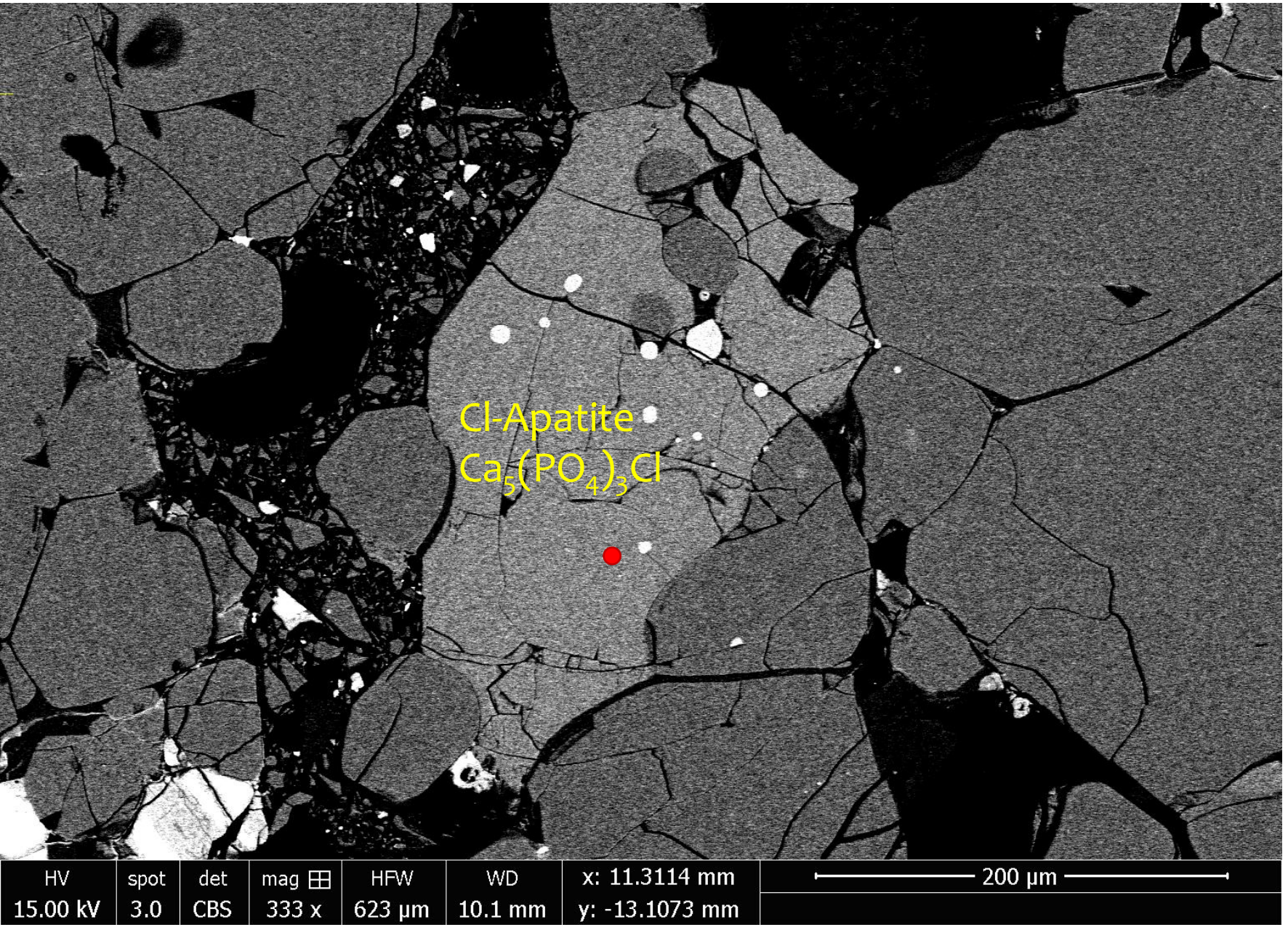
Energy dispersive spectrometer (EDS)

- Solid-state detector (semiconductor).
- As each X-ray photon hits detector, small current produced.
- Records X-ray of all energies simultaneously.
- Low energy resolution, typically 80-150 eV wide.

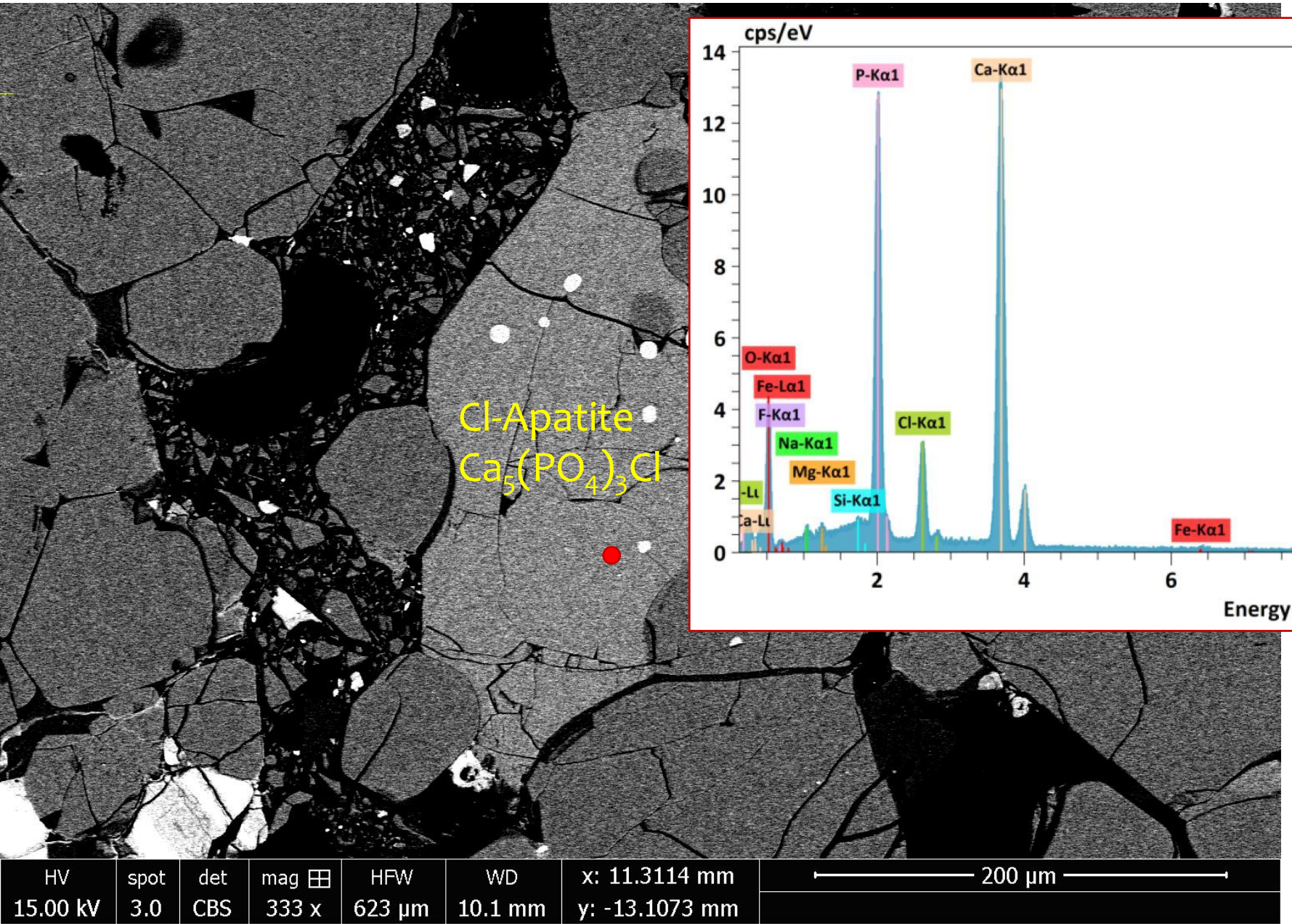
→ good for broad qualitative characterisation of chemistry of samples investigated.

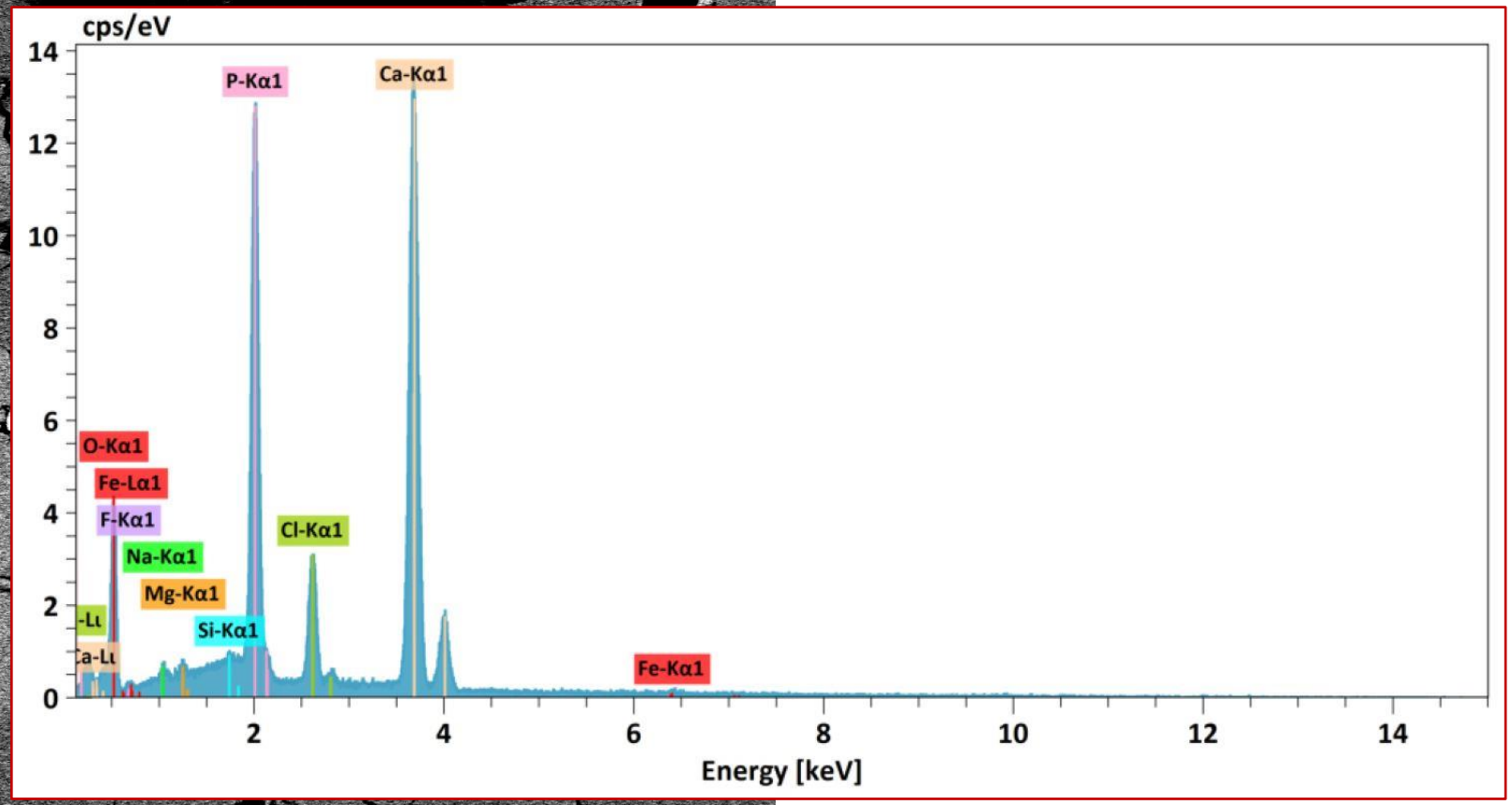


Energy dispersive spectrometer (EDS)

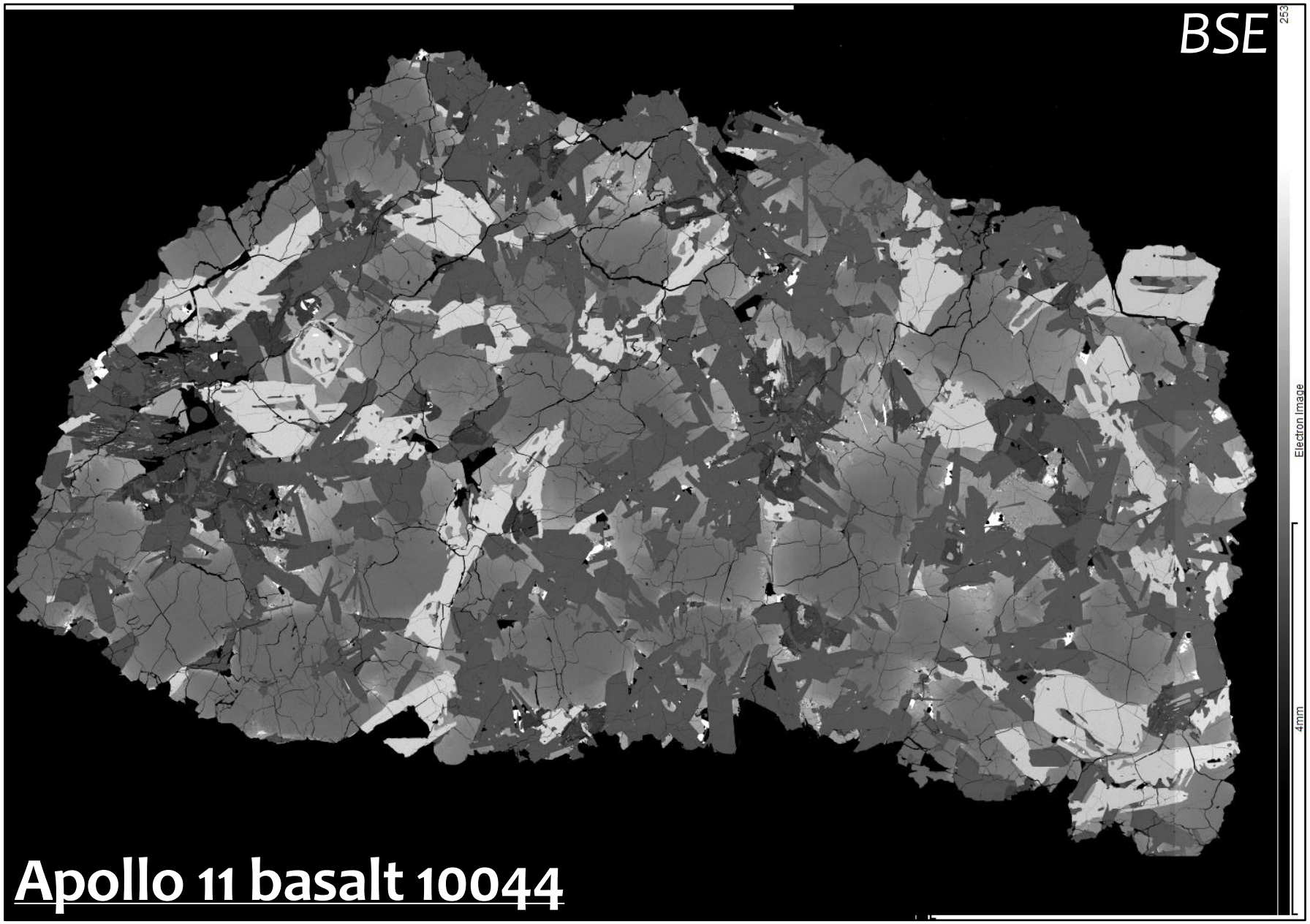


Energy dispersive spectrometer (EDS)



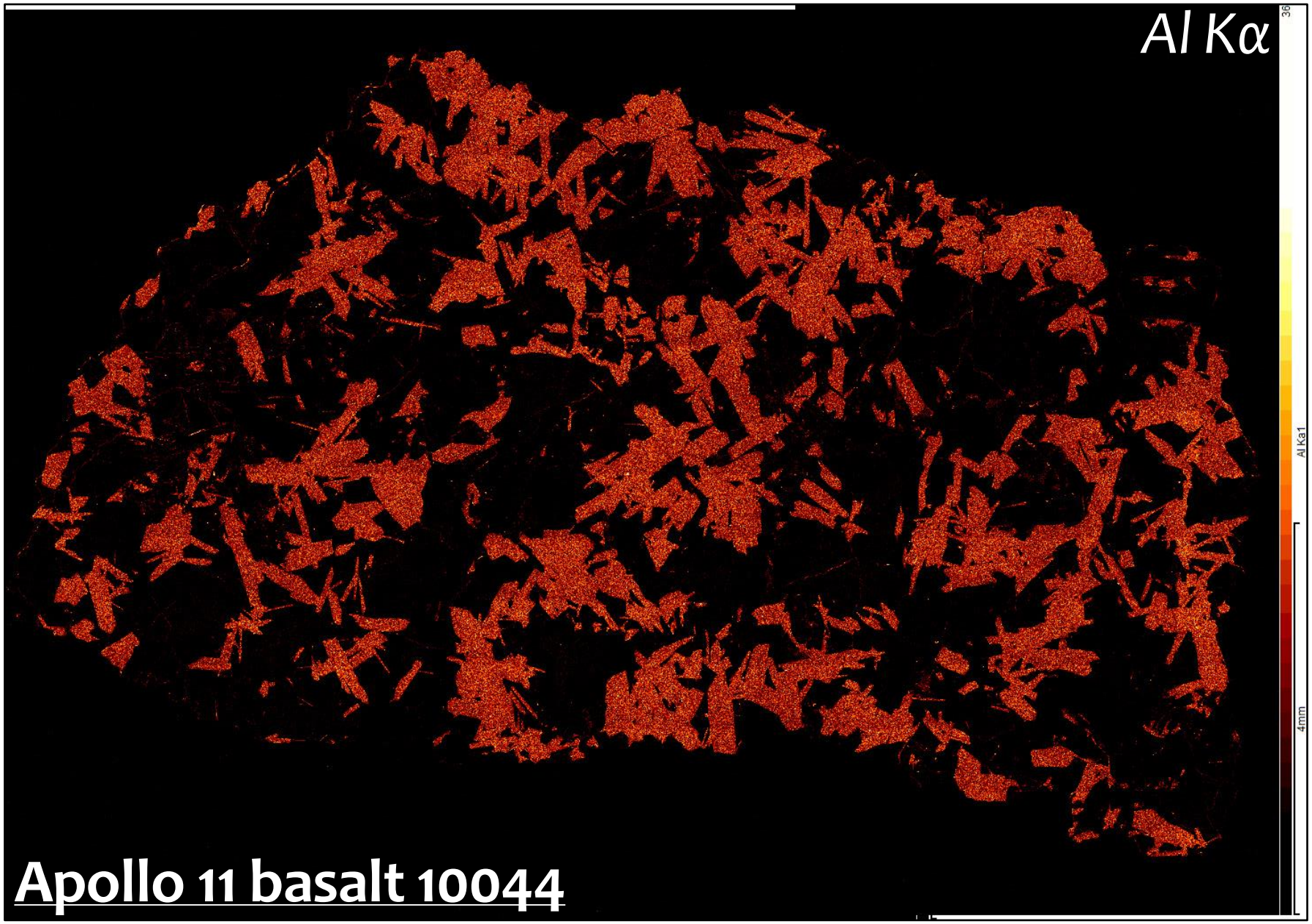


Energy dispersive spectrometer (EDS)



Apollo 11 basalt 10044

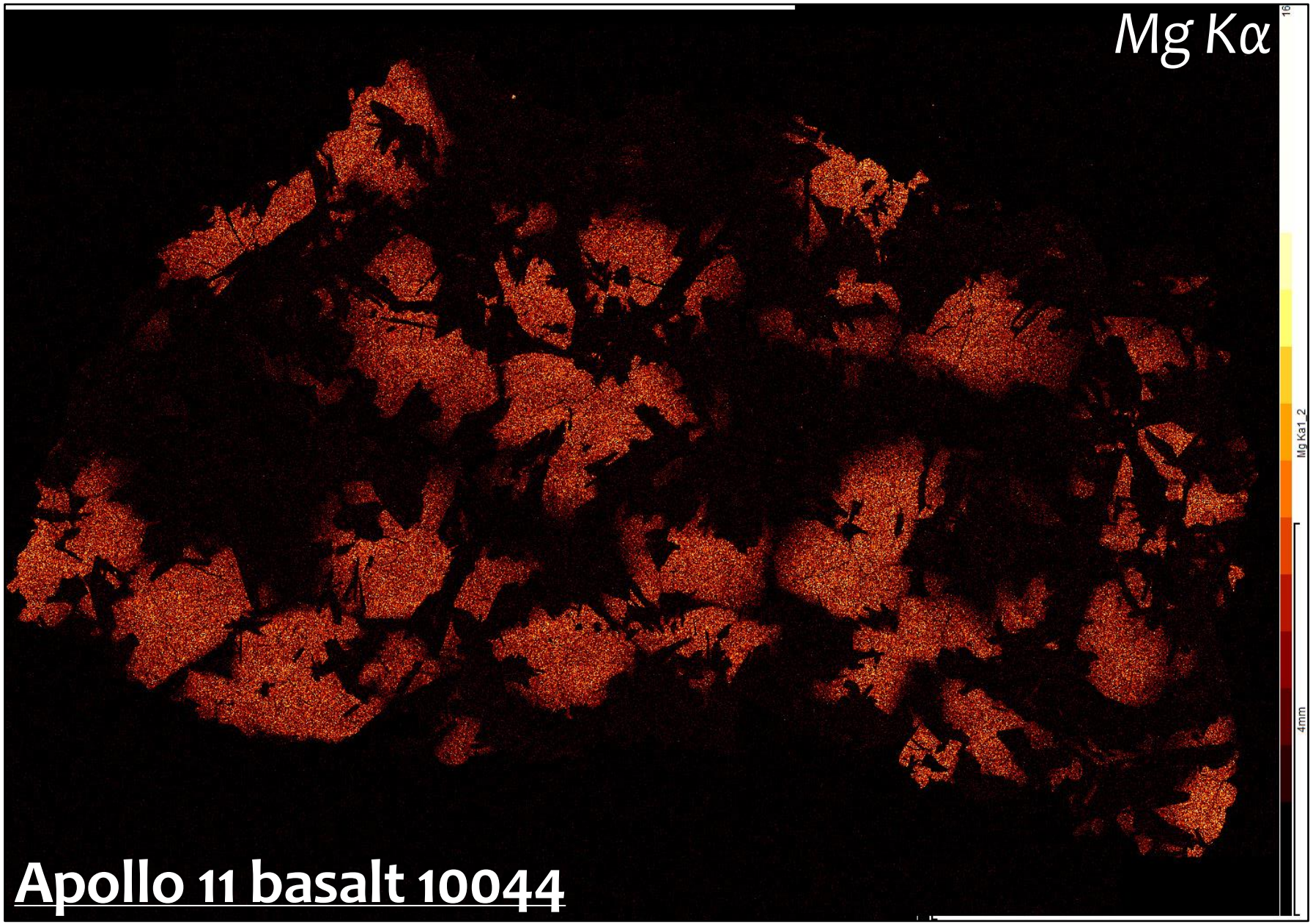
Energy dispersive spectrometer (EDS)



Apollo 11 basalt 10044

EDS allows producing **X-ray element maps** of whole sample → very useful for looking at their mineralogy.

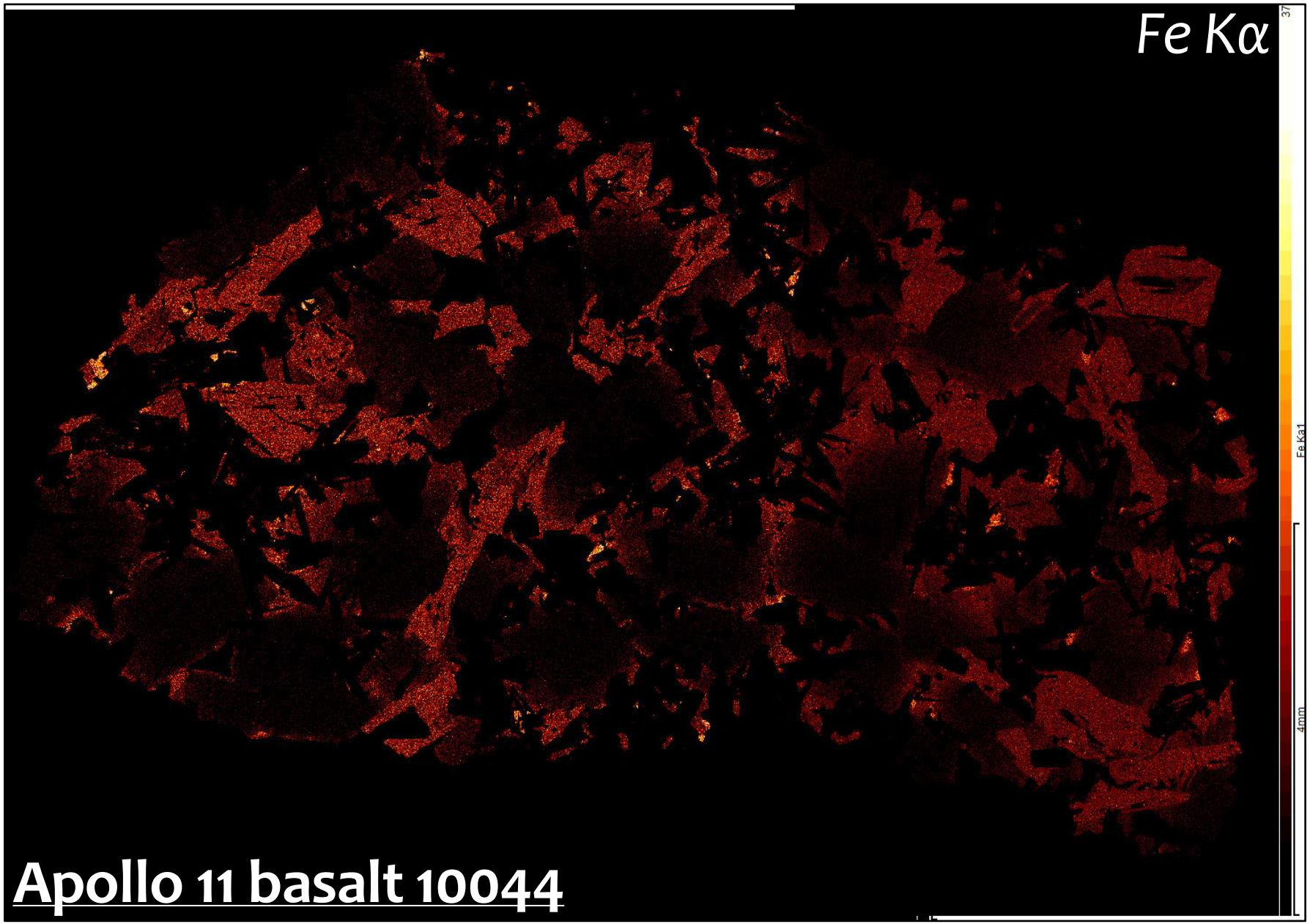
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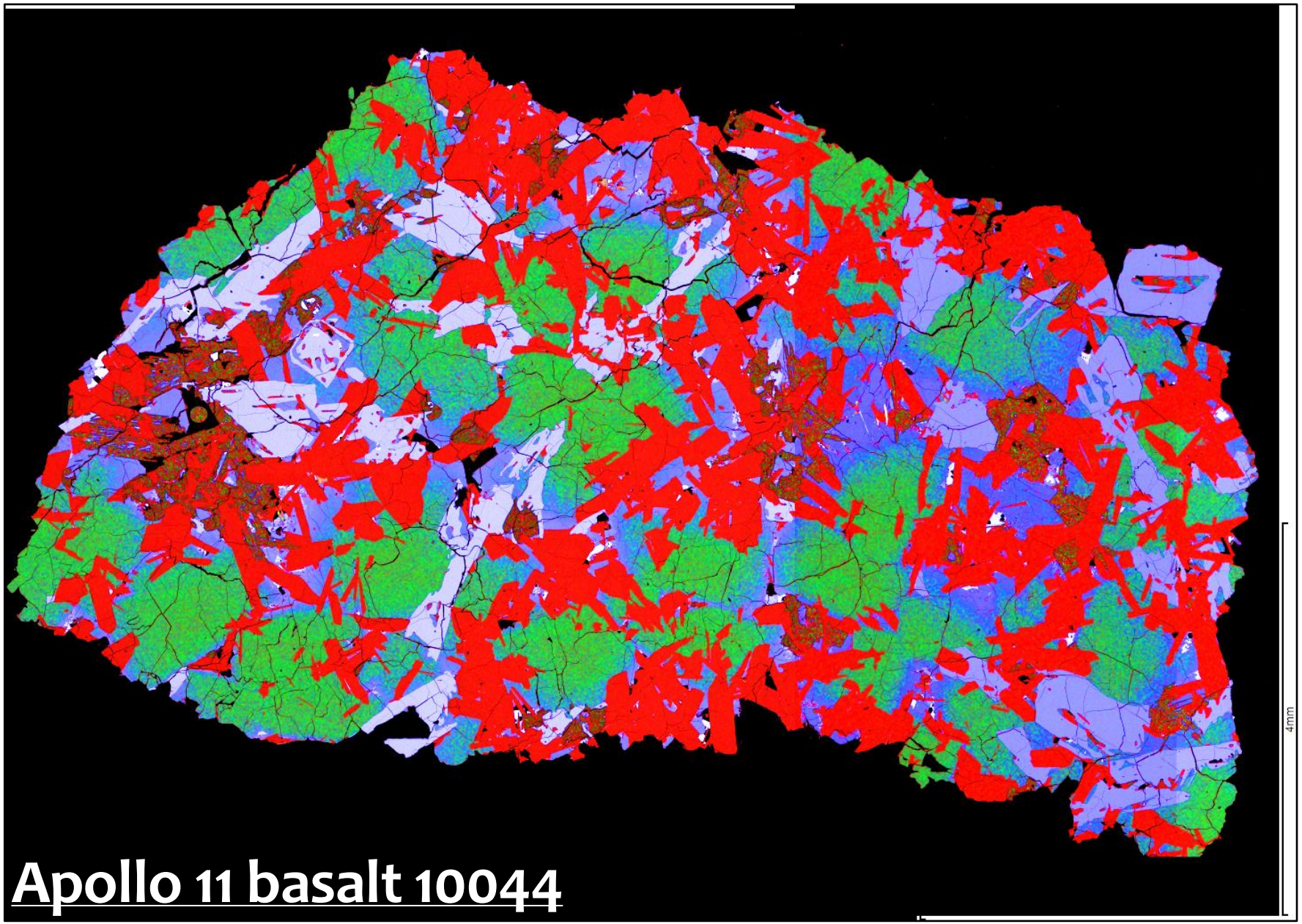
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Energy dispersive spectrometer (EDS)



Apollo 11 basalt 10044

EDS allows producing **X-ray element maps** of whole sample → very useful for looking at their mineralogy.

Can combine several elements into composite X-ray map, here red = Al, green = Mg, and blue = Fe.

Sample preparation – same for SEM and EPMA

Secondary electrons

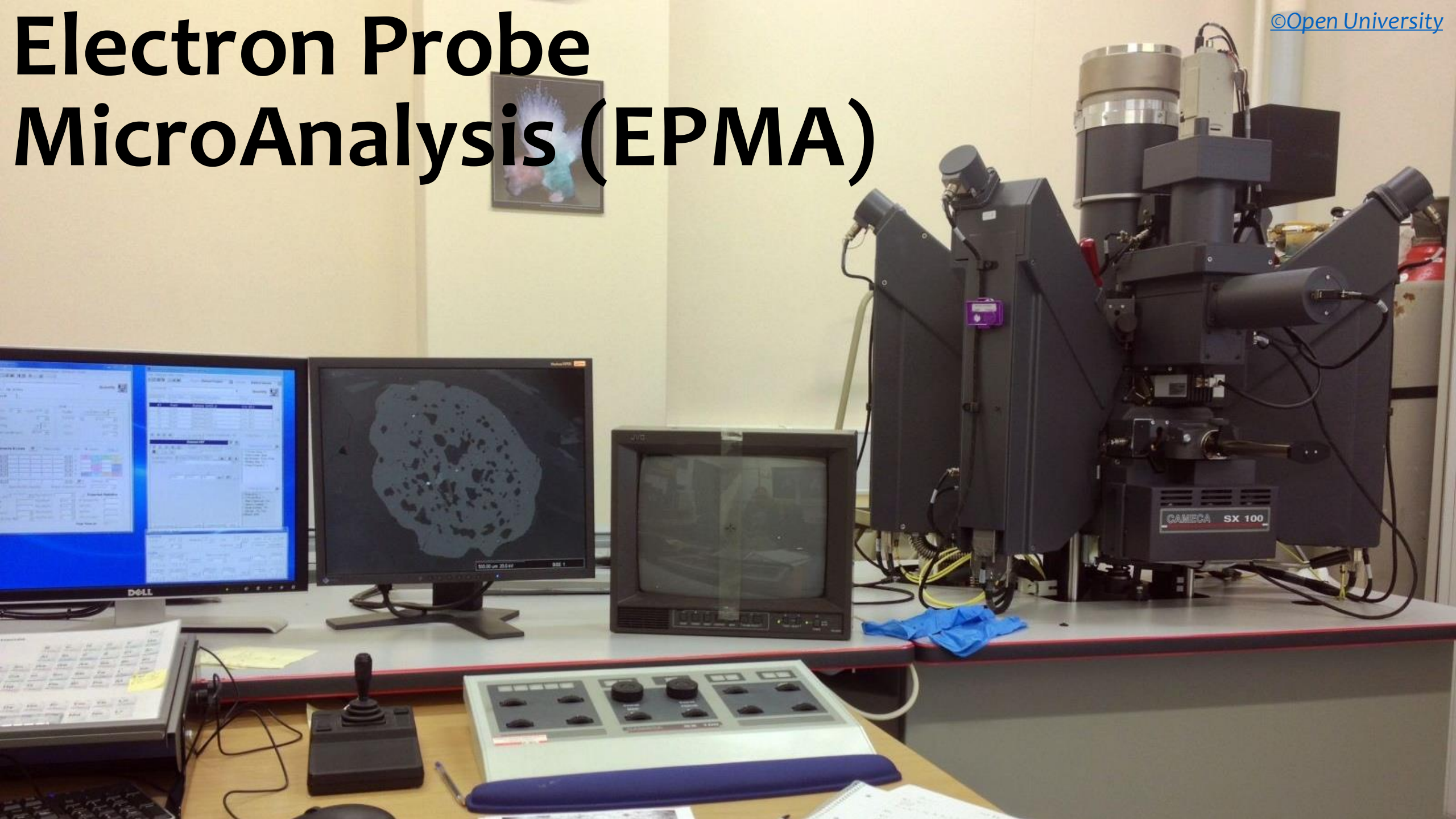
- Information on topography – no need to be flat.
- May need fixing to stabilise friable components.
- Sample typically glued to a stub.

Backscatter electrons and X-rays

- Need flat surface nicely polished to remove topographic effects.
- May need impregnating as for thin section preparation.

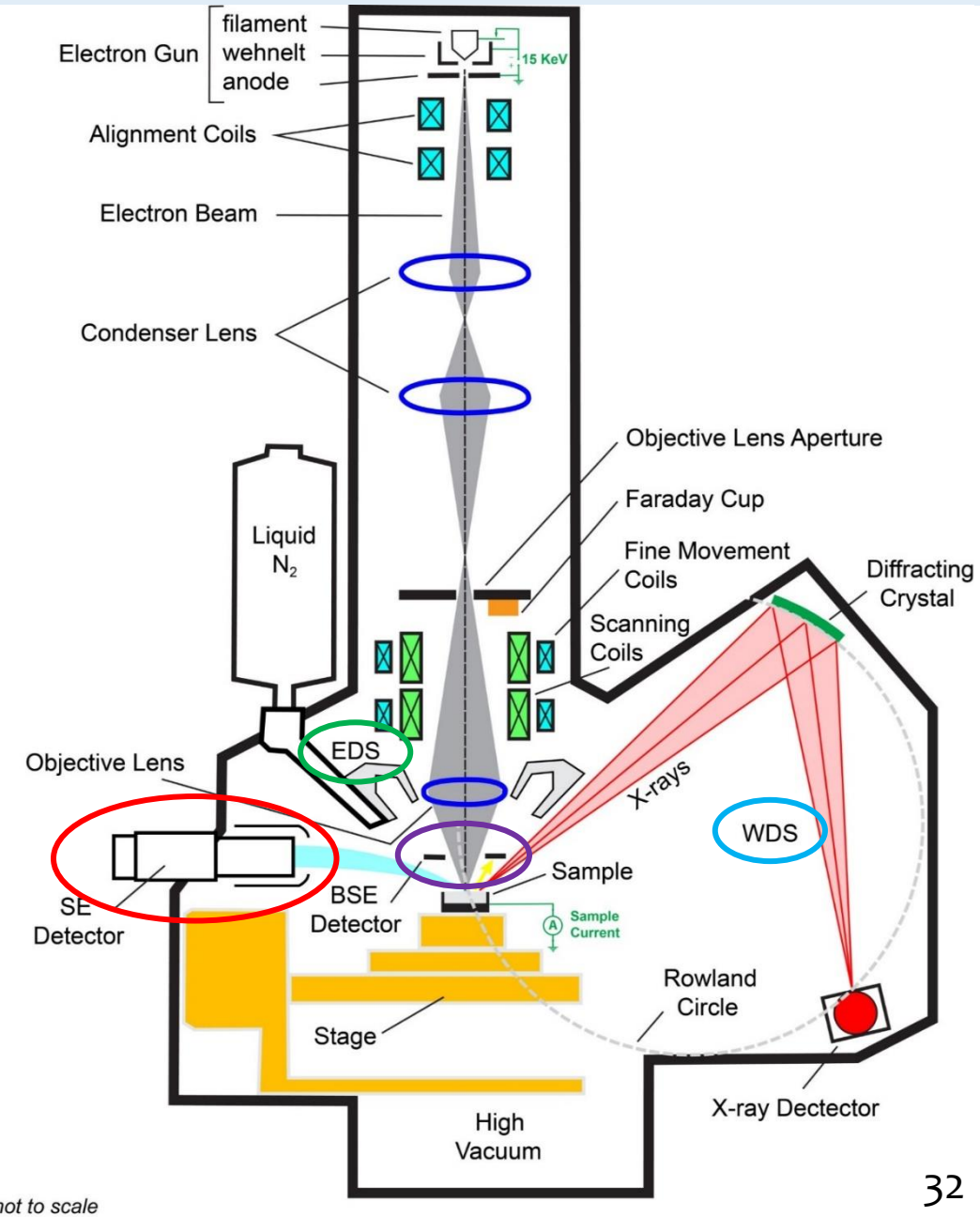
Interaction between the incident e^- beam and target results in electric current developing on specimen surface. Unless target is a conductor (e.g., metal), electrical charges build up → apply thin layer (10's nm) of conductive coating over the sample, typically C.

Electron Probe MicroAnalysis (EPMA)



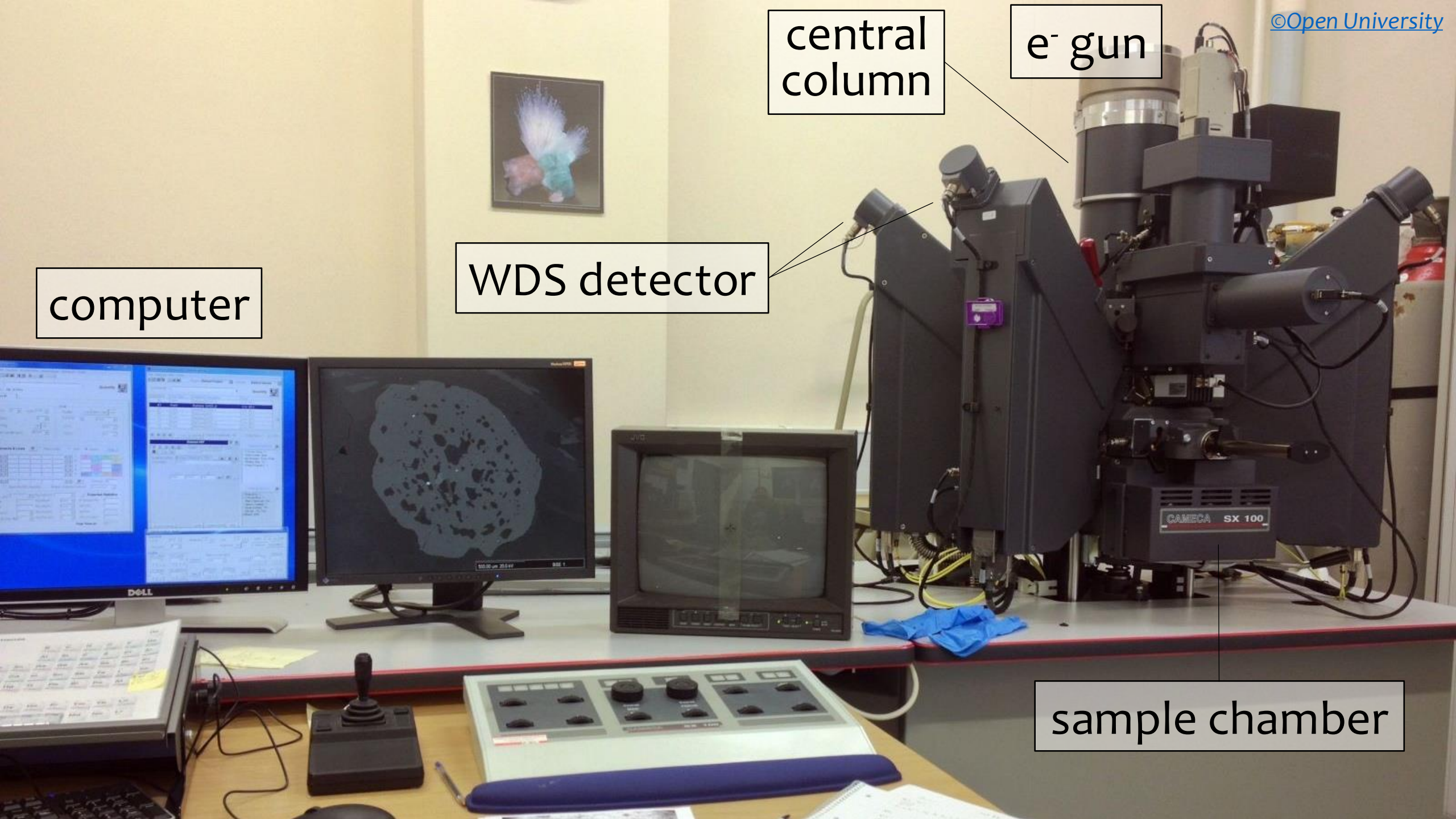
System overview

- **Electron source/gun** produces beam of e^-. Typically W filament or FEG.
- Electron beam focused by coils, condenser and objective lenses, and apertures → beam typically ~1 μm Φ.
- **Scanning coil** used to deflect the beam in X-Y directions → raster scan over the sample surface.
- Detectors for secondary electron (SE) and back-scattered electron (BSE), and X-ray energy dispersive (EDS) and wavelength dispersive spectrometers (WDS).
- High vacuum to limit interaction with gas molecules (~10^{-4}-10^{-5} Pa).

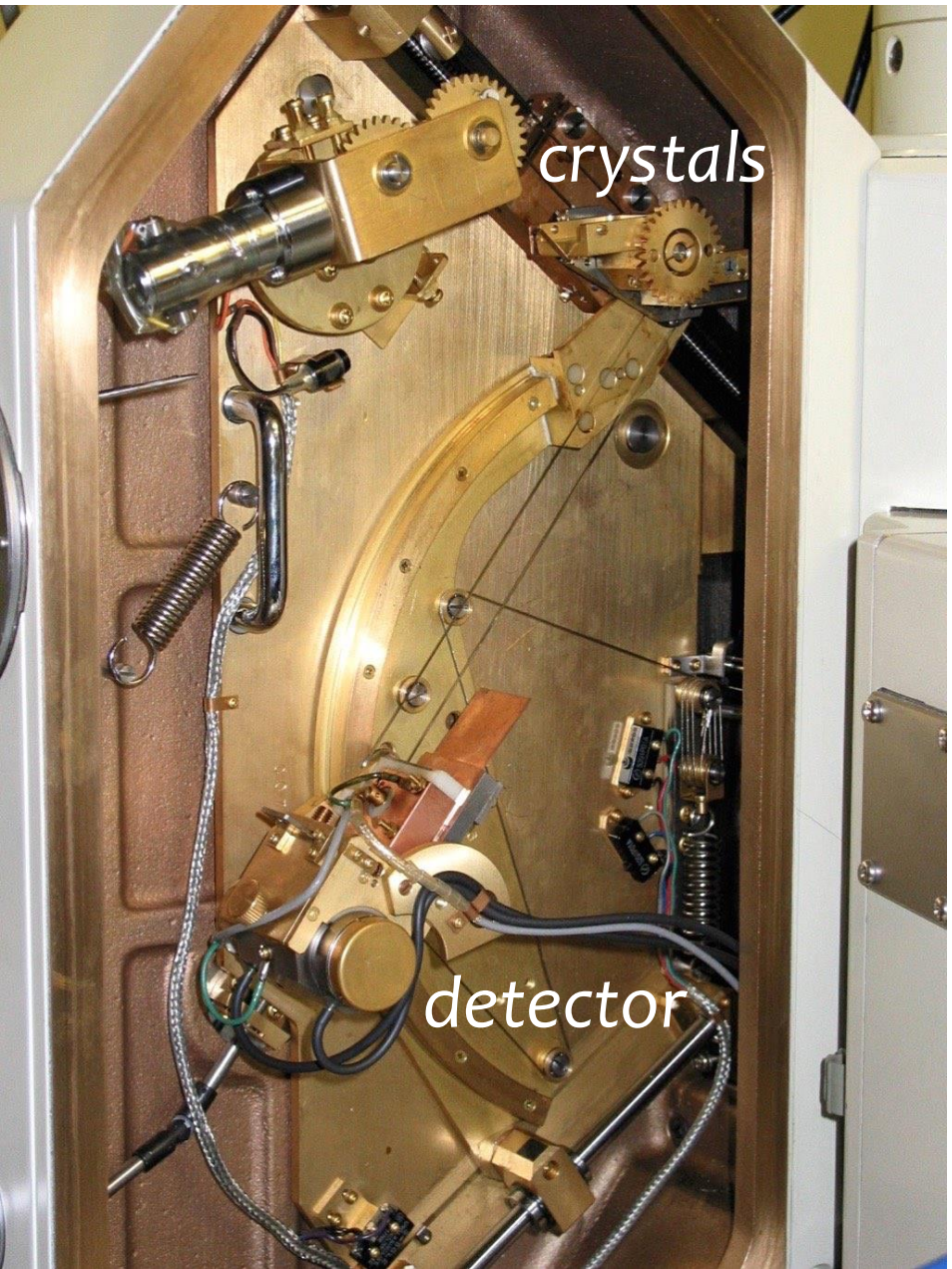


https://e-probe.epss.ucla.edu/?page_id=519

not to scale


central column
e- gun
©Open University
WDS detector
computer
CAMECA SX 100
sample chamber

Wavelength dispersive spectrometer (WDS)

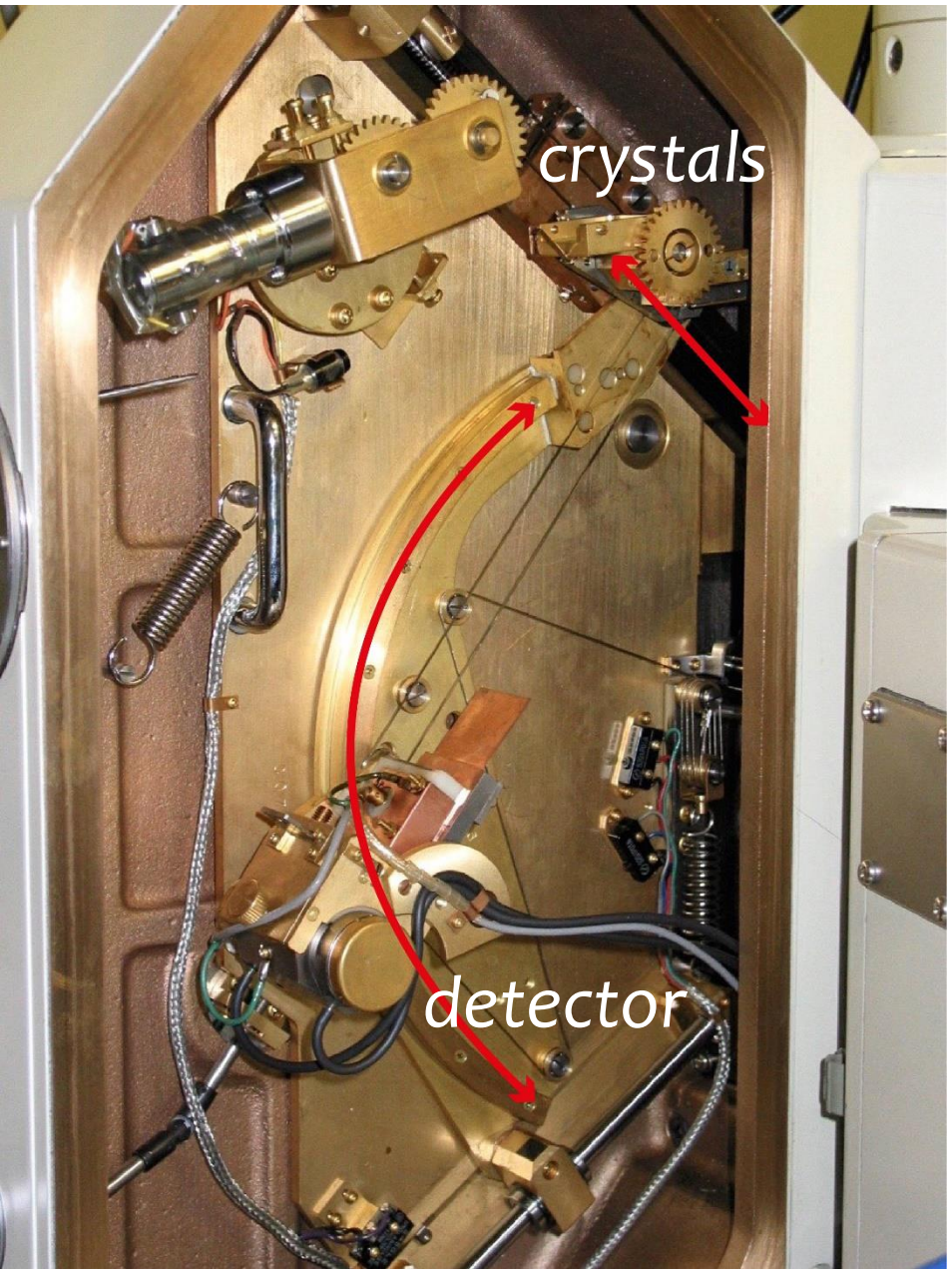


- Typically 4-5 WDS spectrometers with 1-4 diffracting crystals.
- Incoming X-ray dispersed by crystals, with a single wavelength diffracted to detector.
- Each detector records one wavelength at a time.
- Detector is gas-filled tube where gas is ionised by X-rays, yielding large multiplication factor.
- High spectral resolution, typically 5-20 eV.

→ high precision quantitative chemical analysis of chemistry of samples investigated.

http://www.jepspectro.com/htm/jeol733-repair.htm

Wavelength dispersive spectrometer (WDS)

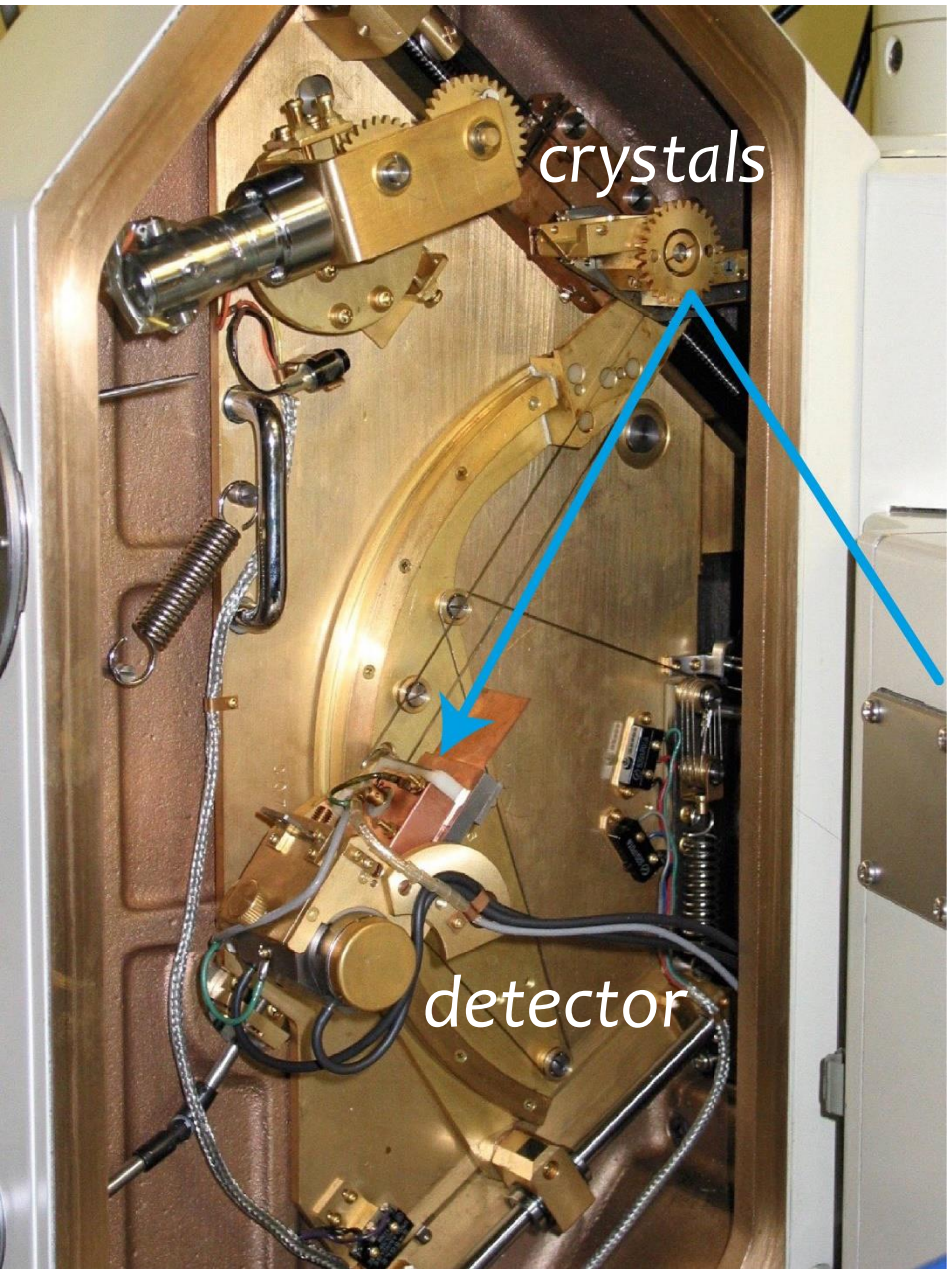


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- High spectral resolution, typically 5-20 eV.

→ high precision quantitative chemical analysis of chemistry of samples investigated.

http://www.jepspectro.com/htm/jeol733-repair.htm

Wavelength dispersive spectrometer (WDS)

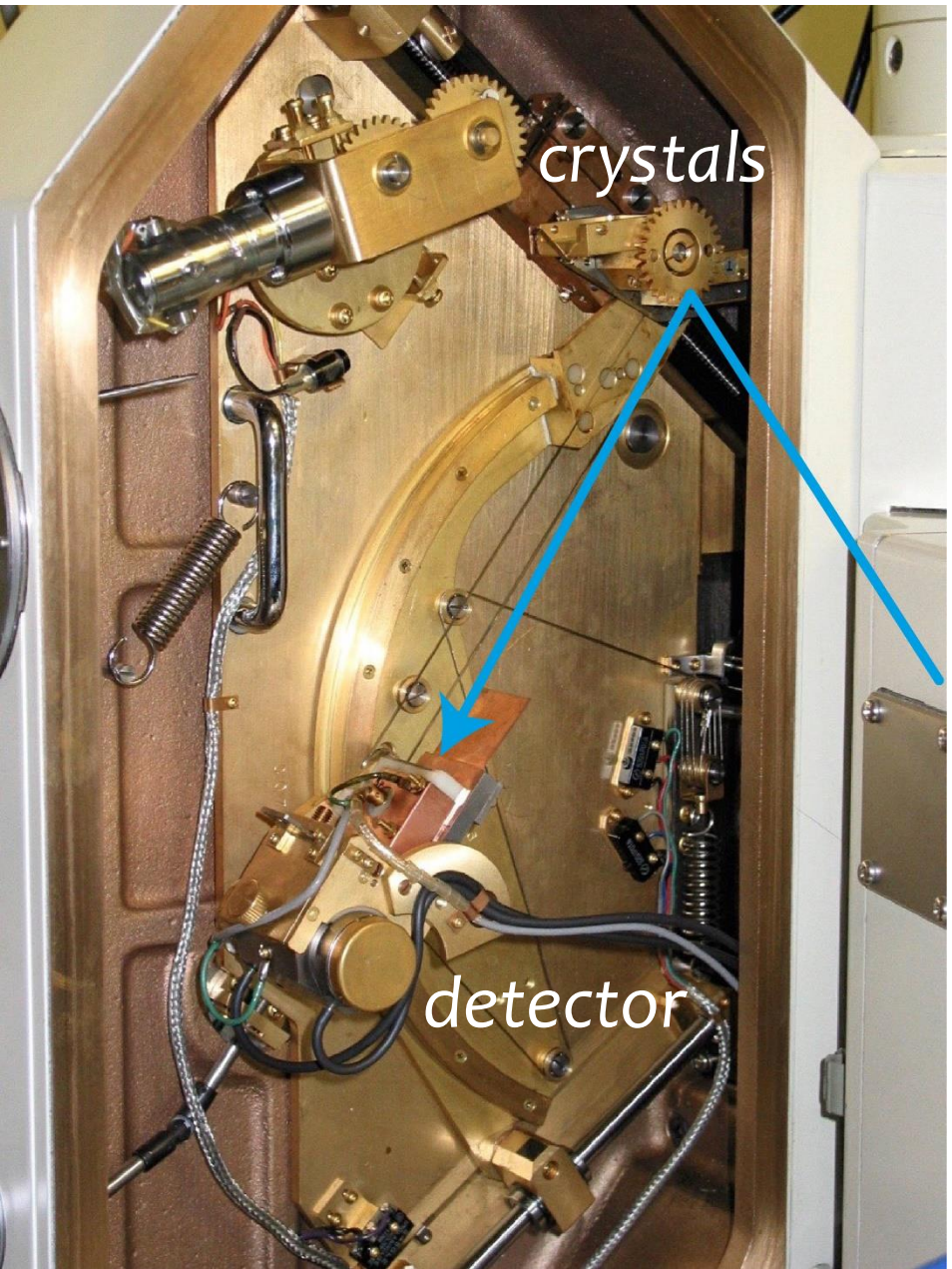


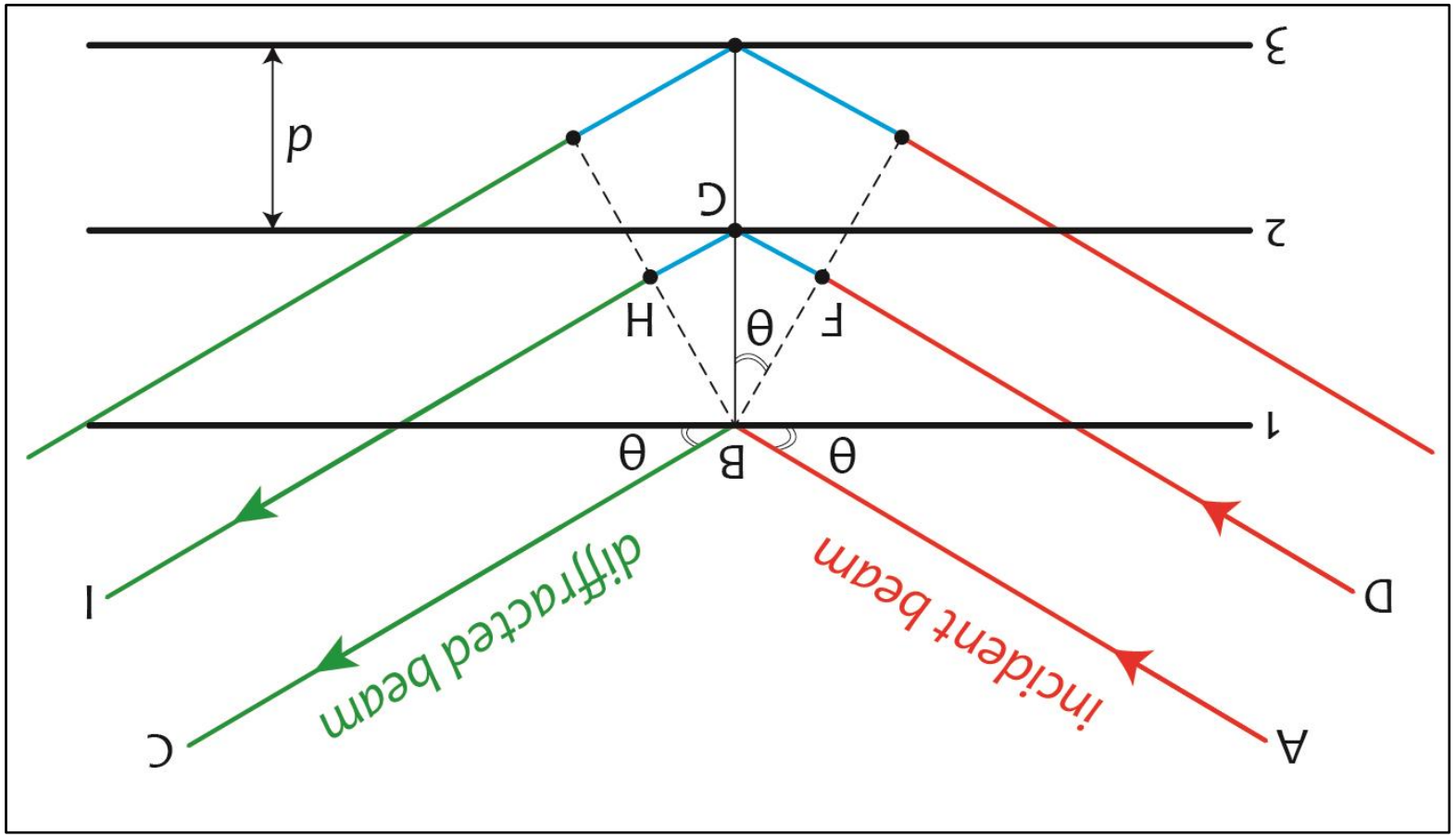
- Typically 4-5 WDS spectrometers with 1-4 diffracting crystals.
- Incoming X-ray dispersed by crystals, with a single wavelength diffracted to detector.
- Each detector records one wavelength at a time.
- Detector is gas-filled tube where gas is ionised by X-rays, yielding large multiplication factor.
- High spectral resolution, typically 5-20 eV.

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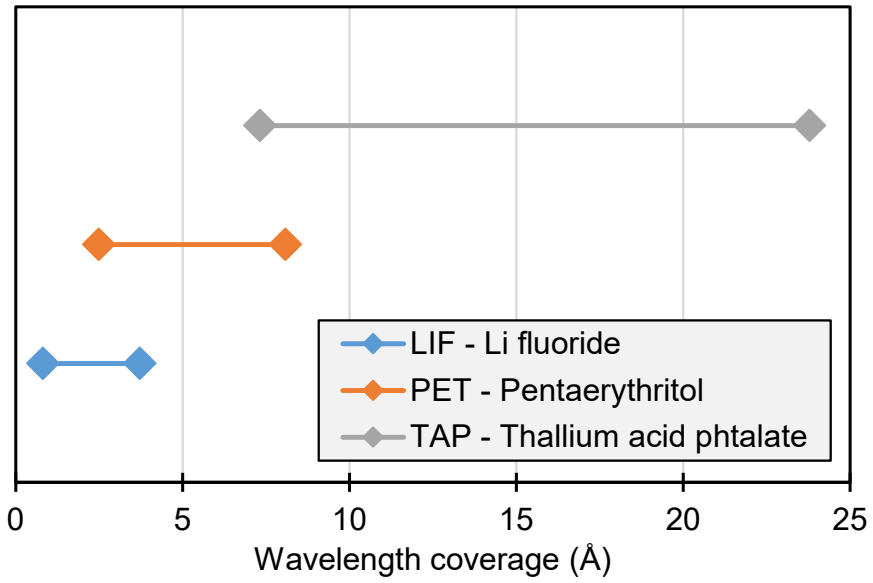




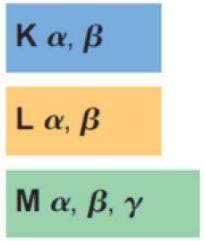
$n\lambda = 2d.\sin(\theta) \rightarrow$ Braggs' law

http://www.jepspectro.com/htm/jeol733-repair.htm

Wavelength dispersive spectrometer (WDS)



Analytical ranges of crystals

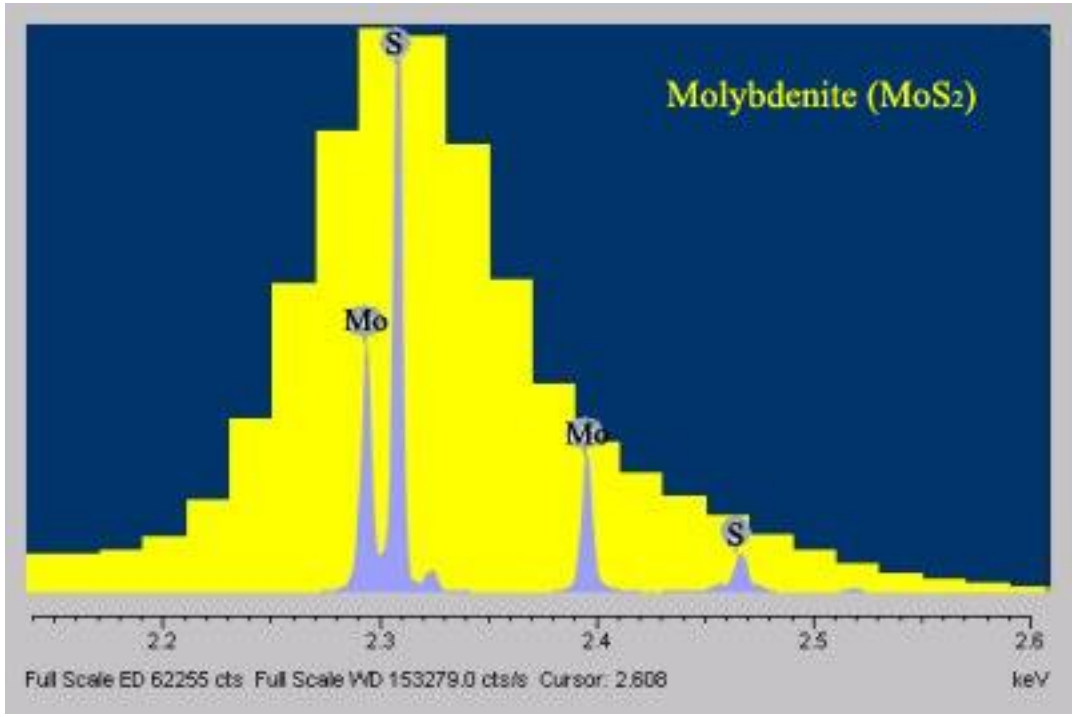


	2d (nm)	6 C	14 Si	22 Ti	30 Zn	38 Sr	46 Pd	54 Xe	62 Sm	70 Yb	78 Pt	86 Rn
TAP	2.576	8O–15P		24Cr–41Nb			46Pd–79Au					
TAPH	2.576	9F–13Al		24Cr–35Br			47Ag–70Yb					
PET	0.8742	13Al–25Mn		36Kr–65Tb						70Yb		
PETL	0.8742	14Si–24Cr		36Kr–62Sm						71Lu		
PETH	0.8742	14Si–22Ti		37Rb–56Ba						72Hf		
LIF	0.4027	19K–37Rb					48Cd					
LIFL	0.4027	20Ca–35Br					50Sn–87Fr					
LIFH	0.4027	20Ca–31Ga					50Sn–79Au					

- LIF, PET & TAP most common crystals.
- H-crystals: smaller circle radius, more limited range, but higher count rates.
- L-crystals: large analysing crystals for trace element analyses.

link

EDS vs. WDS



https://serc.carleton.edu/research_education/geochemsheets/wds.html

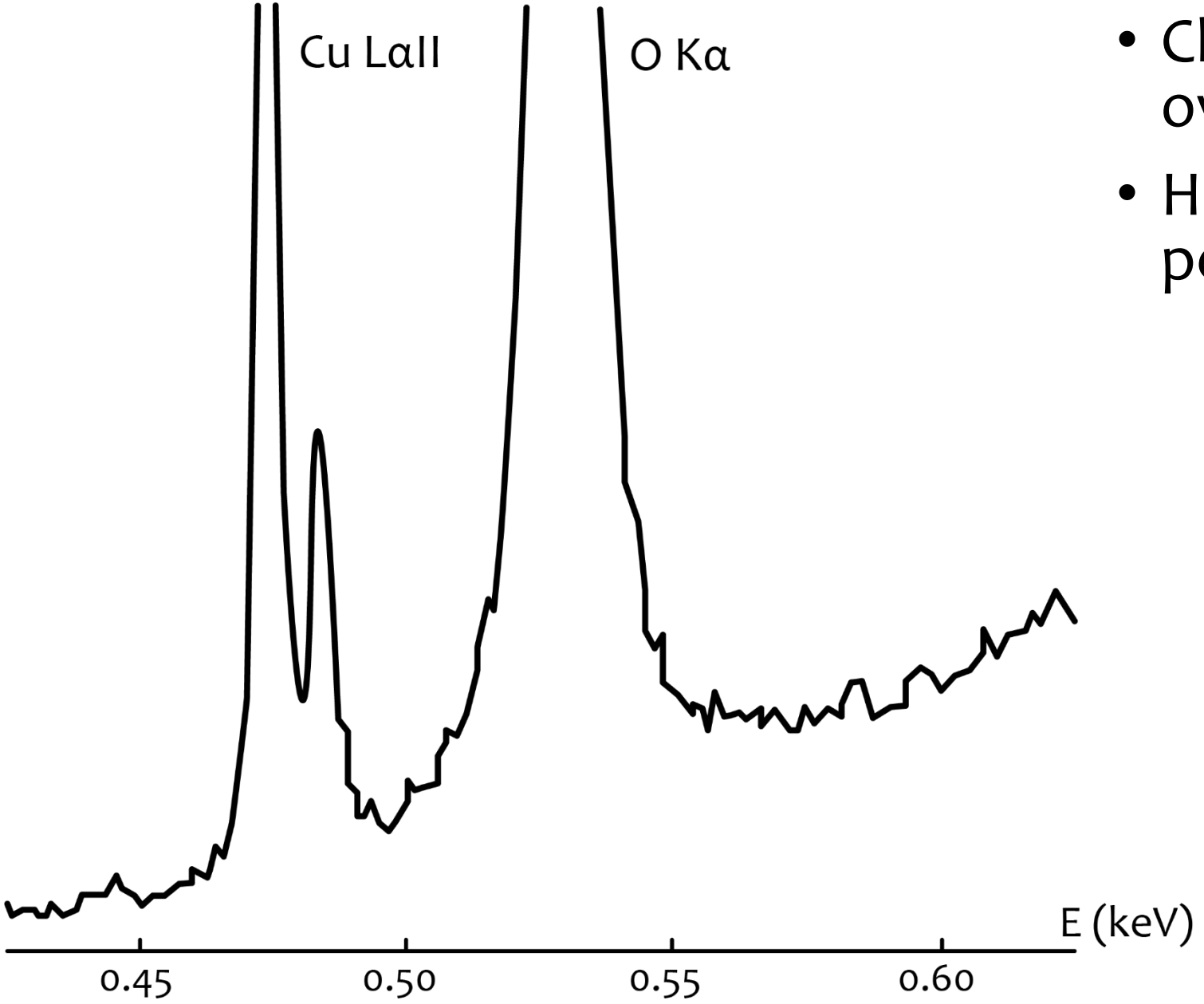
- Molybdenite (MoS_2) EDS (yellow) and WDS (purple) spectra.
- In EDS spectrum, Mo Lα line @ 2.293 keV severely overlapped by S Kα @ 2.307 keV.
- These lines are clearly resolved in WDS spectrum. Also resolves S Kβ and Mo Lβ.

EDS vs. WDS

	WDS	EDS
Z range	^{3}Li – ^{92}U	^{4}Be – ^{92}U
Typical resolution (eV)	~5-20	~80-150
Detection limit (ppm)	~10-100	~500-2000
# elements analysed simultaneously	1	entire spectrum
Time to acquire full spectrum (min)	~20-30	~0.5

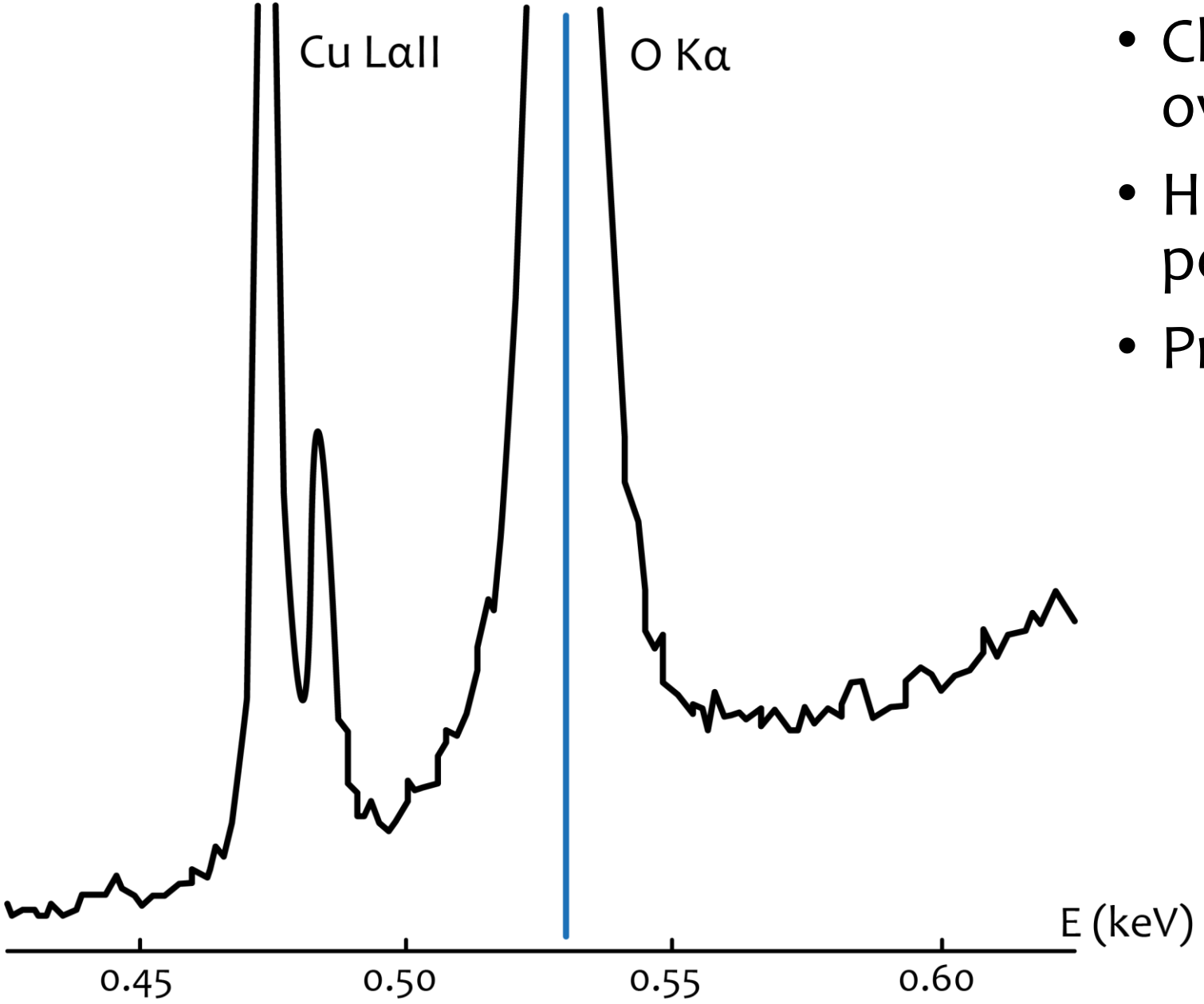
- EDS is cheaper, quicker, but some interferences impossible to resolve, and detection limits are fairly high → ***qualitative overview***.
- WDS is more expensive and time consuming, but much better spectral resolution and detection limits → ***quantitative chemical analysis***.

From WDS spectra to quantitative chemical data



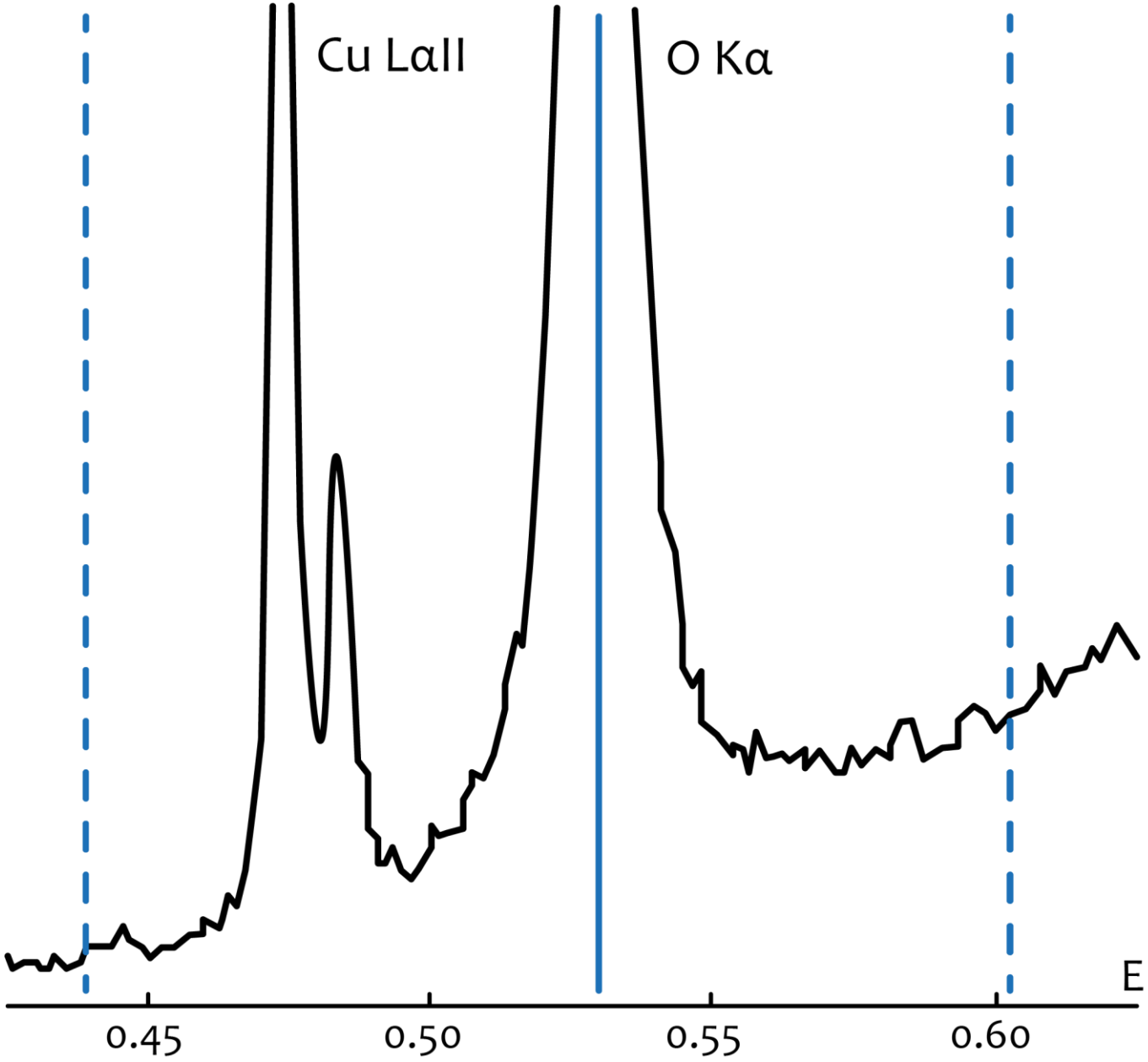
- Check for possible peak overlaps.
- Here want to quantify O Kα peak.

From WDS spectra to quantitative chemical data



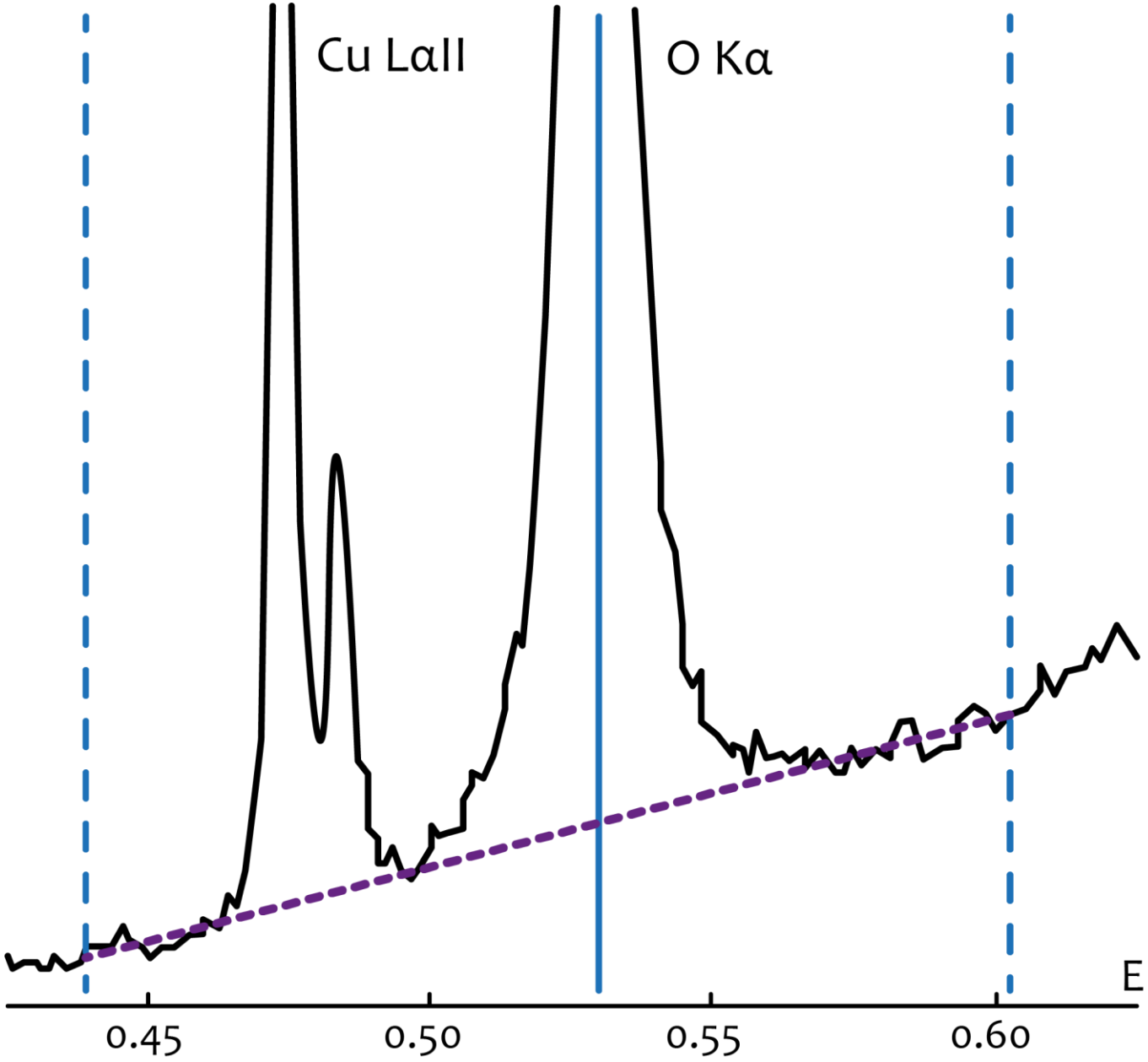
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- Here want to quantify O Kα peak.
- Precisely position peak centre.

From WDS spectra to quantitative chemical data



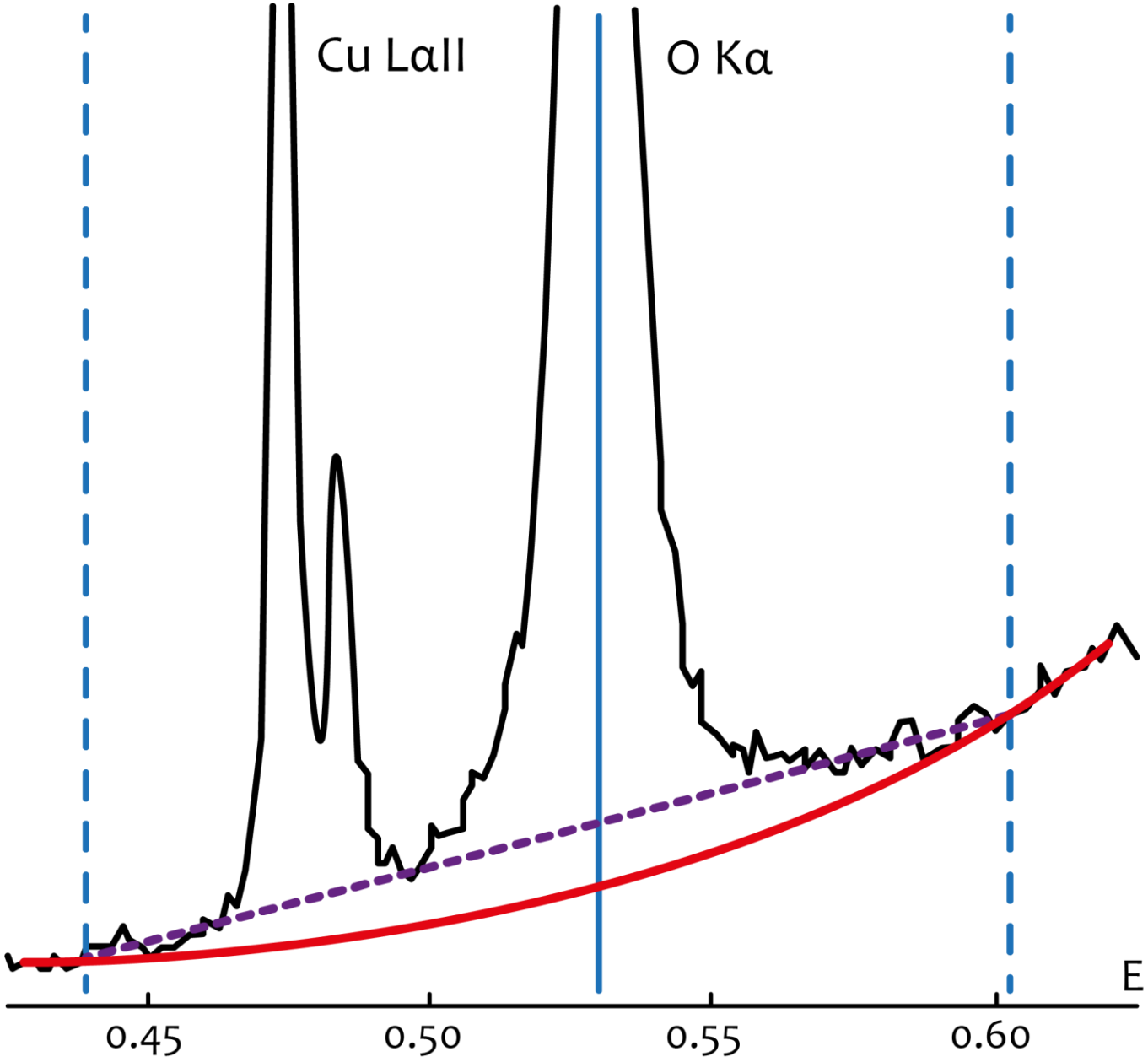
- Check for possible peak overlaps.
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- Precisely position peak centre.
- Position background analysis positions on both side. *Caution with interfering peaks*.

From WDS spectra to quantitative chemical data



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- Decide on appropriate model to fit background – linear?

From WDS spectra to quantitative chemical data



- Check for possible peak overlaps.
- Here want to quantify O Kα peak.
- Precisely position peak centre.
- Position background analysis positions on both side. *Caution with interfering peaks*.
- Decide on appropriate model to fit background – linear?
- Polynomial fit better here.

Standardisation and empirical corrections

$$C'_{unk} = C_{std} \times \frac{I_{unk}}{I_{std}}$$

Apparent concentration in unknown is equal to concentration in standard multiplied by intensity ratio of unknown/standard.

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Real concentration in unknown is equal to apparent concentration in unknown multiplied by correction factor ratio of unknown/standard.

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Apparent concentration in unknown is equal to concentration in standard multiplied by intensity ratio of unknown/standard.

$$C_{unk} = C'_{unk} \times \frac{F_{unk}}{F_{std}}$$

Real concentration in unknown is equal to apparent concentration in unknown multiplied by correction factor ratio of unknown/standard.

$$F = F_Z \times F_A \times F_F$$

Correction factor is function of corrections resulting from backscattering (Z), absorption (A), and fluorescence (F). Known as matrix correction ZAF.

Standardisation and empirical corrections

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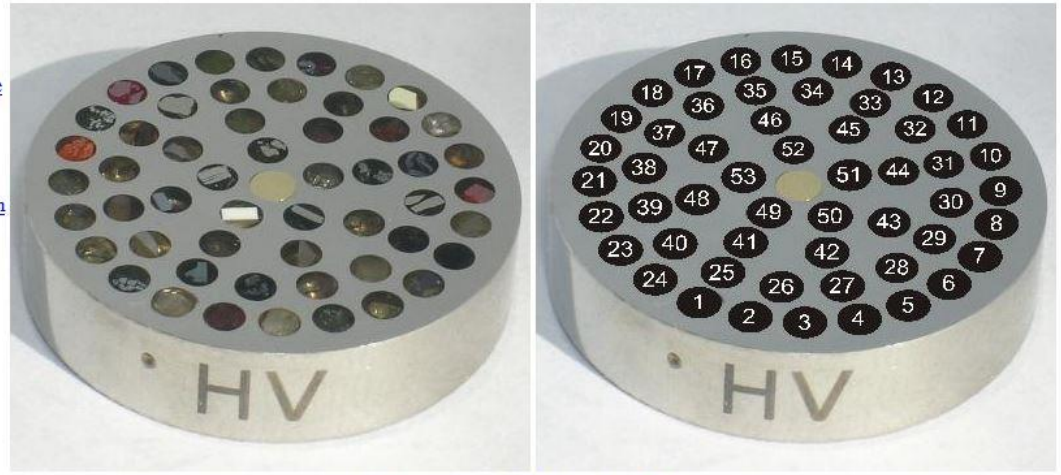
We choose standards that are as similar as possible to the unknown to minimise the correction factor extrapolation $\rightarrow F_{unk}/F_{std} \sim 1$.

Standards

C. Astimex 53 Minerals Mount MINM25-53

Go to:

- Astimex Home Page
- Package Mount Layouts
- Ordering Information



53 Minerals Standard Mount MINM25-53

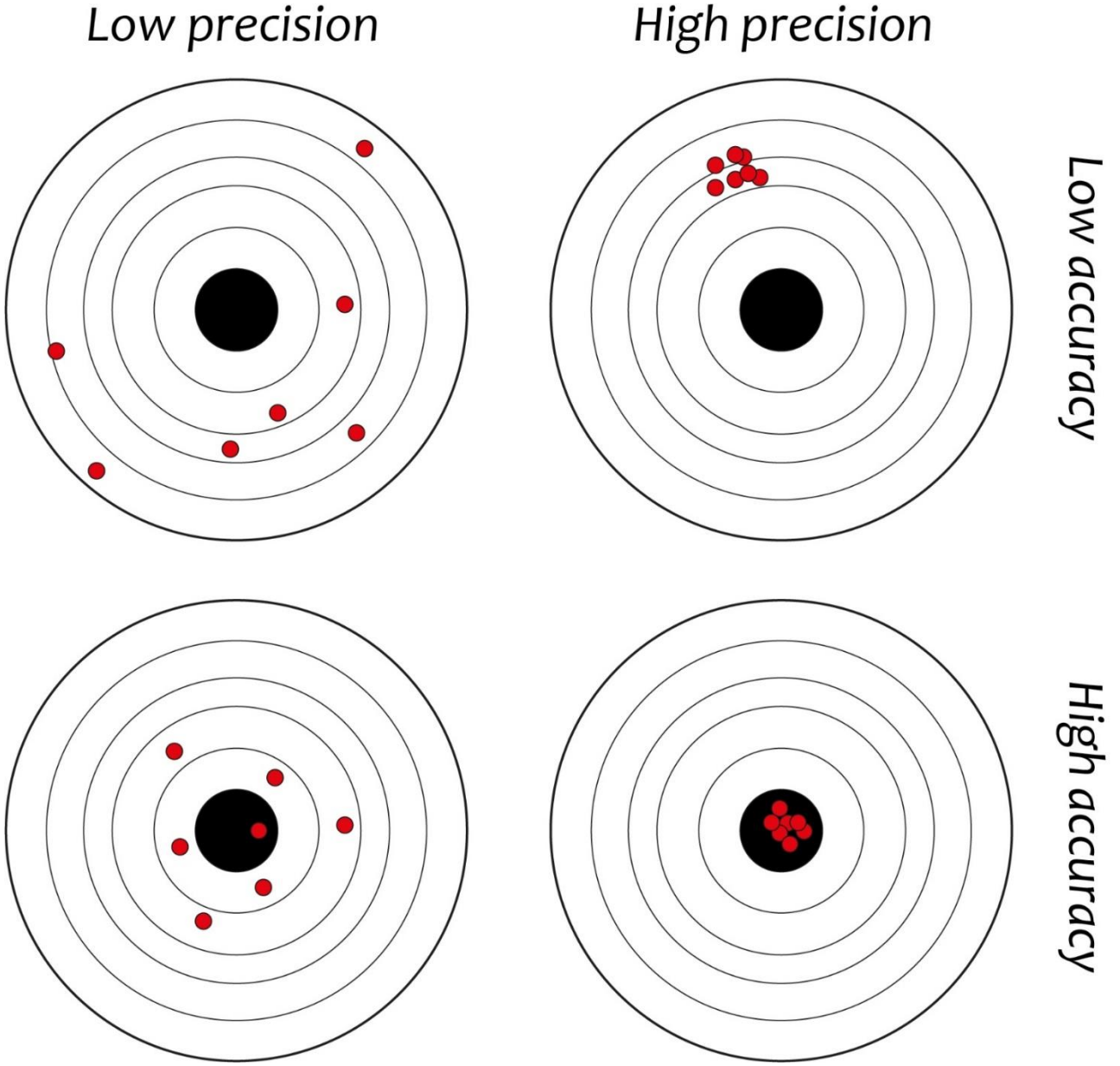
1.	Albite	$NaAlSi_3O_8$	28.	Jadeite	$NaAlSi_2O_6$
2.	Almandine Garnet	$Fe_3Al_2Si_3O_{12}$	29.	Kaersutite	$Ca_2Na(MgFe)_4Ti_1Si_6Al_2O_{22}(OH)_2$
3.	Anhydrite	$CaSO_4$	30.	Marcasite	FeS_2
4.	Apatite	$Ca_5(PO_4)_3F$	31.	Molybdenite	MoS_2
5.	Barite	$BaSO_4$	32.	Monazite	$(Ce,La,Y,Th)PO_4$
6.	Benitoite	$BaTiSi_3O_9$	33.	Obsidian	Na,K,Al,Fe silicate glass
7.	Biotite	$K(Fe,Mg)_3AlSi_3O_{10}(OH)_2$	34.	Olivine	$(MgFe)_2SiO_4$
8.	Pollucite	$CsSi_2AlO_6$	35.	Plagioclase	$(Ca,Na)Al(Al,Si)Si_2O_8$
17.	Chromium Oxide	Cr_2O_3	44.	Stibnite	Sb_2S_3
18.	Cinnabar	HgS	45.	Tugtupite	$Na_4AlBeSi_4O_{12}Cl$
19.	Arsenopyrite	FeAsS	46.	Willemite	$(ZnMn)_2SiO_4$
20.	Crocoite	$PbCrO_4$	47.	Cubic Zirconia	ZrO_2
21.	Diopside	$MgCaSi_2O_6$	48.	Gallium Arsenide	GaAs
22.	Dolomite	$MgCa(CO_3)_2$	49.	Boron Nitride	BN
23.	Fluorite	CaF_2	50.	Bismuth Selenide	Bi_2Se_3
24.	Galena	PbS	51.	Yttrium Al Garnet	$Y_3Al_5O_{12}$
25.	Hematite	Fe_2O_3	52.	Nickel Silicide	Ni_2Si
26.	Magnetite	Fe_3O_4	53.	Antimony Telluride	Sb_2Te_3
27.	Periclase	MgO			

Critical to analyse a range of standards that match material being analysed.

https://astimex.com/com/catalog/min.html

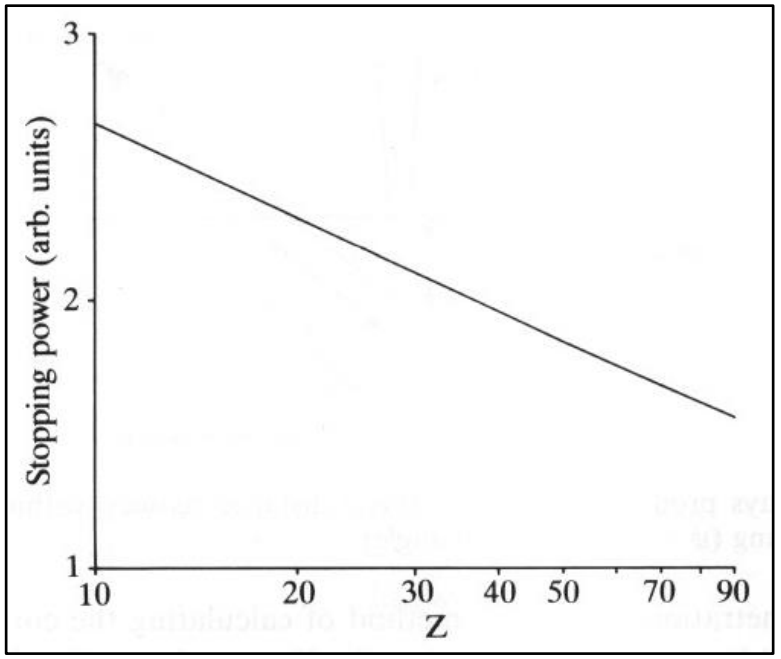
What's a good analysis?

- **Totals**: Having sum of all major elements analysed close to 100 wt.% suggests analysis is ok. Typically in range 98.5-100.5 wt.% for silicates and glasses.
- **Stoichiometry**: calculated mineral stoichiometry from analyses makes sense (*we'll come back to this in practical*).

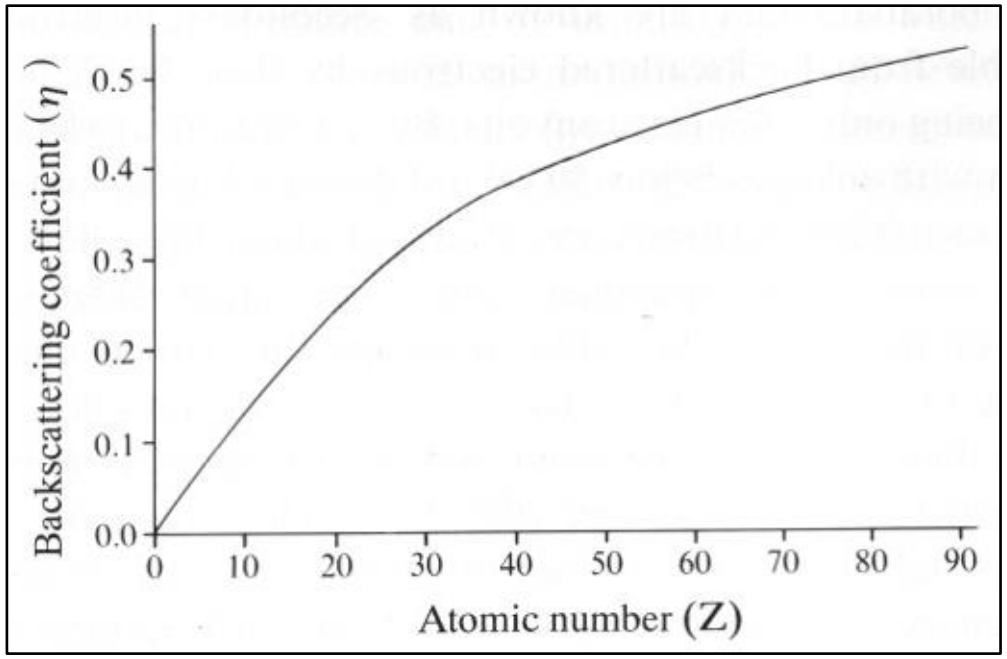


ZAF correction: Z correction

from Reed (2005) – *Electron Microprobe Analysis and Scanning Electron Microscopy in Geology*



+



Stopping power correction

- Incident electrons lose energy by interacting with inner shell electrons.
- Stopping power (= energy lost by electrons) drops with increasing Z.

→ *higher number of X-rays produced in higher Z targets.*

Backscatter correction

- Fraction of incident electrons backscattered increase with Z.
- Less incident electrons penetrate into targets with higher Z.

→ *smaller number of X-rays produced in higher Z targets.*

ZAF correction: Absorption (A) correction

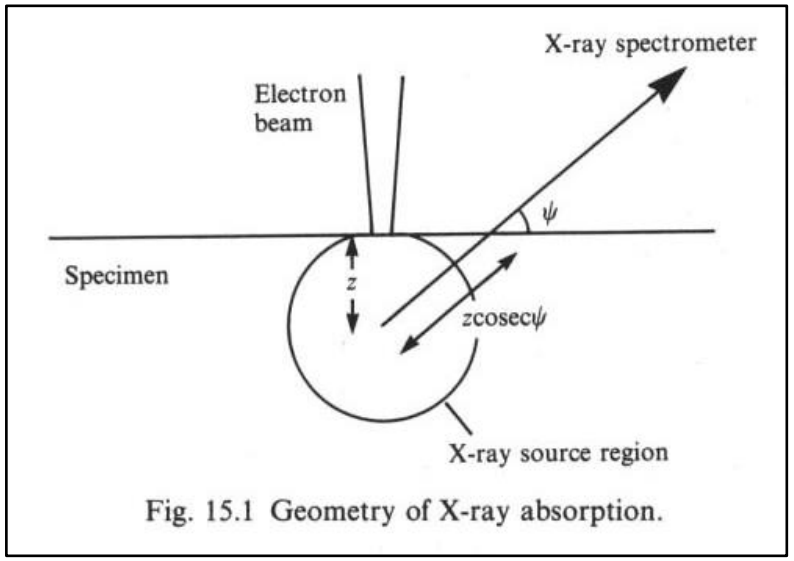


Fig. 15.1 Geometry of X-ray absorption.

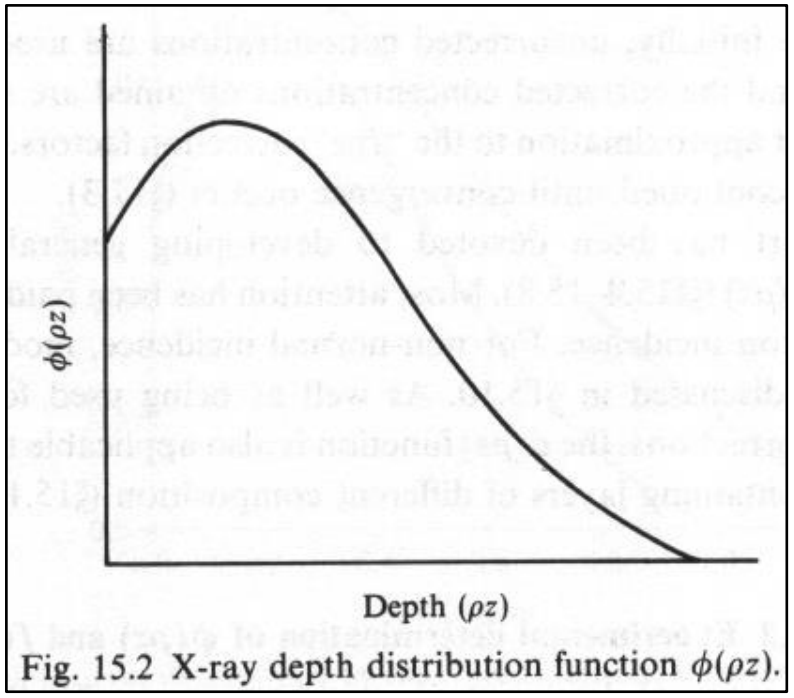


Fig. 15.2 X-ray depth distribution function $\phi(\rho z)$.

- X-ray produced in a material propagate in all directions.
- Some of these X-ray are **absorbed** before escaping the specimen surface → part of the intensity lost.
- Not linear impact – the absorbed ratio depends on X-ray energy and chemical composition of the sample.
- To correct for this, need to know how X-ray production varies with depth in material, known as the $\Phi(\rho z)$ function [phi-rho-z].
- $\Phi(\rho z)$ varies with accelerating voltage E_0, critical excitation energy of a specific element line E_C, and mean Z of material.

from Reed (1993) – *Electron Microprobe Analysis*

ZAF correction: Fluorescence (F) correction

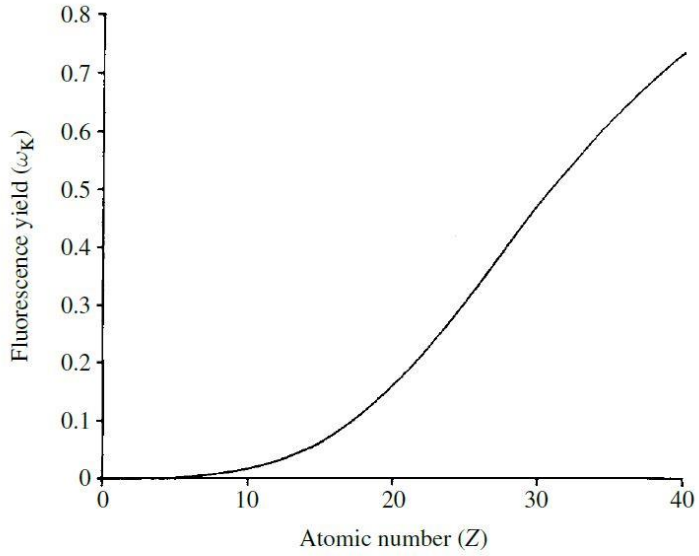


Fig. 2.13. The atomic-number dependence of the 'fluorescence yield' of the K shell (ω_K), which is the probability of K-shell ionisation being followed by characteristic X-ray emission (rather than Auger electron emission).

from Reed (2005) – *Electron Microprobe Analysis and Scanning Electron Microscopy in Geology*

- X-ray produced in a material have potential to produce a 2nd generation of X-ray → **fluorescence**.
- Occurs when characteristic X-ray has energy greater than critical excitation energy of another element present in material.
- Typically, for atom of atomic number Z it promotes secondary fluorescence for atoms with atomic numbers (Z – 1) and (Z – 2) for same series line (e.g., K vs. K, L vs. L, etc).
- For example, K_α (6.4 keV) of Fe (Z = 26) in steel produce Mn (Z = 25) and Cr (Z = 24) ionisation.
- Fluorescence yield increases with Z.